
Calculus I

Explanations, Problems & Solutions (Solutions)

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2026-07-21

Preface

Welcome to **Calculus I**, a rigorous journey into the mathematics of change and motion that forms the backbone of modern scientific analysis.

1 About This Textbook

This textbook has been carefully structured to guide you through the essential pillars of single variable calculus, summarizing the core concepts and techniques of the course:

Limits and Continuity : Establishing the precise mathematical foundation required to analyze functions at their boundaries, defining the critical concepts of instantaneous velocity and continuous motion.

The Derivative and Its Applications : Mastering the tools to measure instantaneous rates of change, using differentiation to solve complex optimization problems, approximate linear behaviors, and sketch the precise shapes of curves.

Integration : Exploring the accumulation of quantities to determine net change, calculating complex areas and volumes, and bridging the gap between slopes and areas through the Fundamental Theorem of Calculus.

Transcendental Functions and Advanced Techniques : Extending calculus to exponential, logarithmic, and inverse trigonometric functions, and developing sophisticated strategies such as integration by parts and trigonometric substitution to solve intricate problems.

2 How to Use This Book

To facilitate a deep and structured understanding of the material, this book utilizes specific block types to organize information:

Definition blocks clarify fundamental concepts, ensuring you grasp the precise language of calculus.

Example blocks demonstrate standard problems derived from the syllabus, illustrating how to apply theoretical concepts.

Solution blocks offer detailed, step-by-step walkthroughs to model the logical flow required for your own proofs and calculations.

Theorem blocks highlight the pivotal mathematical truths, such as the Mean Value Theorem and the Fundamental Theorem of Calculus.

Proof blocks guide you through the logical derivations of theorems, fostering the analytical reasoning skills emphasized in this course.

Note blocks provide crucial insights, common calculation pitfalls, and connections between differentiation and integration.

3 A Note on Learning

Calculus is more than a set of rules for manipulating symbols; it is a way of thinking about the world. True mastery requires **active engagement** with the material. We encourage you to:

- Attempt to solve the examples yourself before revealing the solution blocks.
- Focus on the geometric and physical intuition behind the formulas.
- Understand not just **how** to differentiate or integrate, but **when** and **why** these tools are used.
- Collaborate with peers to discuss concepts and refine your logical argumentation.

The notes provided here are designed to support your learning journey, clarifying the lecture material and helping you build the confidence to solve real-world problems mathematically.

4 Looking Ahead

The concepts you master in **Calculus I** (limits, derivatives, and integrals) are the indispensable tools of the future. They are the language used to describe the motion of planets, the flow of fluids, the growth of populations, and the fluctuations of markets. Whether you pursue physics, engineering, economics, or computer science, the analytical framework you build here will serve as the foundation for your advanced studies.

We hope this book serves as a clear and supportive guide as you learn to see the world through the lens of calculus.

Sihoo Lee
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Chapter 00

Preview of Calculus

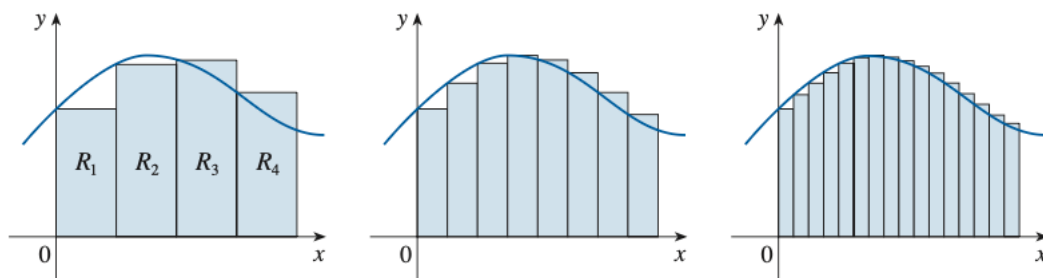
This section bridges the gap between static pre-calculus mathematics and the dynamic nature of calculus by addressing two fundamental problems: determining the slope of a tangent line to a curve and calculating the instantaneous velocity of a moving object. Both problems are solved using the same limiting process, where approximations—such as the slopes of secant lines or average velocities over shrinking time intervals—converge to a single, precise value. This concept of the "limit" serves as the foundation for the entire course, establishing the mathematical tools necessary to analyze instantaneous rates of change.

Chapter 00.00

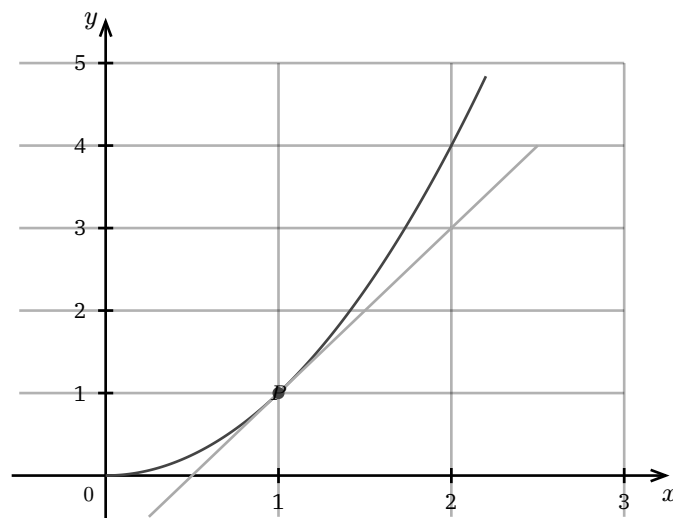
Preview of Calculus

1 What is calculus?

- We have two main problems that we deal in calculus: The area problem & the tangent problem.
 - The area problem is about finding the area under a curve.
 - The tangent problem is about finding the slope of a curve at a given point.



- Approximation of an area using rectangles.



NOTATION | The Area Problem

We can define the area under the curve as

$$A = \lim_{n \rightarrow \infty} A_n$$

Here, A_i represents sum of each area block equally dividing the given range into n parts.

NOTATION | The Tangent Problem

The slope of a curve at a given point a can be defined as

$$m_a = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Here, m represents the slope of the tangent line to the curve at point x . Or, alternatively, we can write it as :

$$m_a = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

NOTE | So what's the relation?

Both problems involve limits. The area problem uses a limit to find the exact area under a curve. The tangent problem uses a limit to find the exact slope of a tangent line. In the end, solving one of them leads to solving the other. We call this the **Fundamental Theorem of Calculus**.

Chapter 01

Functions and Limits

This chapter formalizes the concept of the limit, moving from intuitive approximations to rigorous mathematical definitions including the precise Epsilon-Delta definition. It provides the algebraic rules (Limit Laws) necessary to calculate limits without graphing, methods for handling indeterminate forms, and defines continuity to describe functions without breaks or holes. These concepts culminate in the Intermediate Value Theorem, a powerful tool used to prove the existence of roots and solutions within specific intervals, setting the stage for the definition of the derivative.

The Tangent & Speed Problem

1 The Tangent Problem

- What does tangent mean in calculus?

DEFINITION | Tangency

A tangent line to a curve at a given point is a straight line that just “touches” the curve at that point. In calculus, the tangent line represents the instantaneous rate of change of the function at that point, which is given by the derivative of the function.

EXAMPLE | Tangency Basics

Say we have the function $f(x) = x^2$. Let us define the base point of our tangent slope calculation as $x = 1$. Here, our slope calculation equation would become :

$$m_{pq} = \frac{p_x^2 - q_x^2}{p_x - q_x}$$

where $q_x = 1$. If we plot in values from $p_x = 2$ to $p_x \approx 1$, we get the following table:

p_x	m_pq
2	3
1.5	2.5
1.1	2.1
1.01	2.01
1.001	2.001

As we can see from the table, as p_x approaches 1, the slope m_{pq} approaches 2. Therefore, the slope of the tangent line to the curve at the point where $x = 1$ is 2. Thus, after a bit of calculation, we can find that the equation of the tangent line at the point $(1, f(1)) = (1, 1)$ is:

$$y - 1 = 2(x - 1)$$

or simplified,

$$y = 2x - 1$$

NOTE | Approach from L/R

Note from the example above that there are two directions a line can approach a curve: from the left side (as p_x approaches 1 from values less than 1) and from the right side (as p_x approaches 1 from values greater than 1).

- Normally, this doesn't pose that much of a problem, but if the function is **severed** at a point, the left-hand limit and right-hand limit may not be equal, leading to different tangent slopes from each side.

2 The Velocity Problem

- Take a car. How do we define its **speed**?
- We know from middle school knowledge that speed is defined by the following equation :

$$v = \frac{\text{total distance travelled}}{\text{total time spent}}$$

- Note here that this is the **exact same form** as the slope equation we used in the tangent problem!

DEFINITION | Average Velocity

Average velocity over a time interval is defined as the total displacement divided by the total time taken. Mathematically, if a car moves from position $s(a)$ at time a to position $s(b)$ at time b , the average velocity v_{avg} over the interval $[a, b]$ is given by:

$$v_{\text{avg}} = \frac{s(b) - s(a)}{b - a}$$

DEFINITION | Instantaneous Velocity

A bit off course, but we can define instantaneous velocity using the later-defined idea of the limit.

Instantaneous velocity at a specific time is the limit of the average velocity as the time interval approaches zero. It represents the velocity of the car at a precise moment in time. Mathematically, the instantaneous velocity $v(t)$ at time t is given by:

$$v(t) = \lim_{\Delta t \rightarrow 0} \frac{s(t + \Delta t) - s(t)}{\Delta t}$$

Chapter 01.05

The Limit of a Function

1 The limit

- In the previous section, whilst discussing tangents and velocity, we kept encountering a problem: how do we define the slope of a curve at a single point, or the velocity of a car at a precise moment in time?
- Both of these problems can be solved using the concept of **limits**.

DEFINITION | Intuitive Definition of the Limit

Suppose $f(x)$ is defined whilst x is near a . Then we write

$$\lim_{x \rightarrow a} f(x) = L$$

We read this as “the limit of $f(x)$ as x approaches a is L ”. This claim must be preceded by the statement that as $f(x)$ gets arbitrarily closer and closer to L as x gets closer and closer to a .

EXAMPLE | Limit Basics

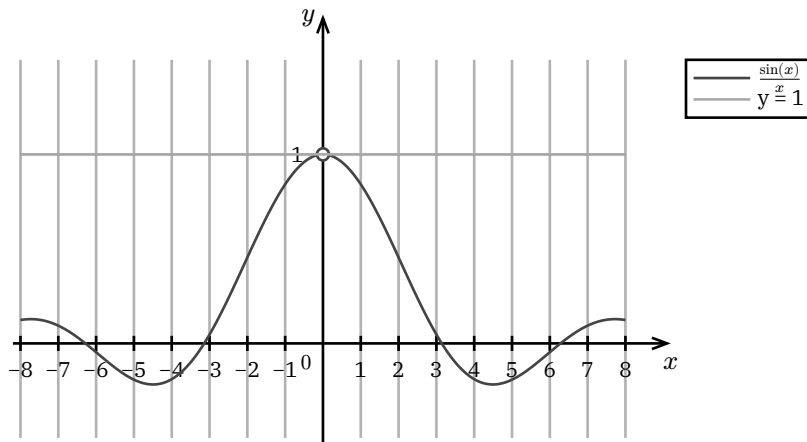
Consider the function $f(x) = \frac{\sin(x)}{x}$. Evaluate $f(x)$ at values of x that get closer and closer to 0:

Solution 1 |

x	$f(x)$
1	0.84147098
0.1	0.99833417
0.01	0.99998333
0.001	0.99999983
0.0001	0.99999998

By the intuitive definition of the limit, we can see that as x approaches 0, $f(x)$ approaches 1. Therefore, we can conclude that:

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$$



- Above, we briefly mentioned that limits may differ when approached from different sides. This can be expanded to the fact that limits may not exist at all from some sides in some cases.

DEFINITION | Intuitive Definition of One-Sided Limits

Suppose $f(x)$ is defined whilst x is near a . Then we write

$$\lim_{x \rightarrow a^+} f(x) = L_1$$

and

$$\lim_{x \rightarrow a^-} f(x) = L_2$$

We read these as “the limit of $f(x)$ as x approaches a from the right is L_1 ” and “the limit of $f(x)$ as x approaches a from the left is L_2 ”. This claim must be preceded by the statement that as $f(x)$ gets arbitrarily closer and closer to L_1 as x gets closer and closer to a from the right, and similarly for L_2 from the left. Here, we call L_1 as the **right-hand limit** and L_2 as the **left-hand limit**.

EXAMPLE | The Heaveside Function

Consider the Heaveside function $H(x)$ defined as follows:

$$H(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$$

Evaluate $H(x)$ at values of x that get closer and closer to 0 from both sides:

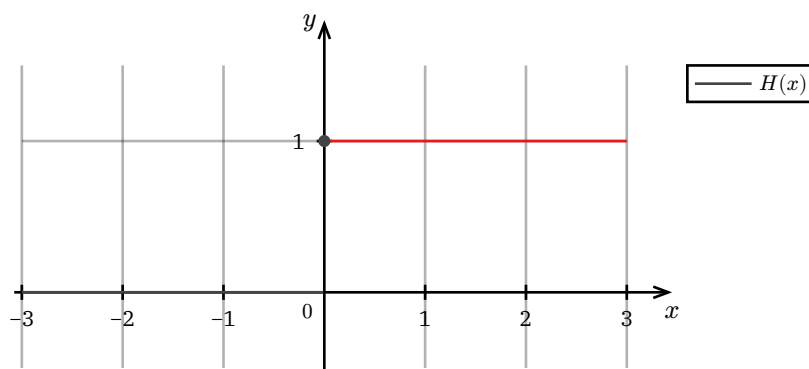
Solution 1 | More Evaluation...

x	$H(x)$
-1	0
-0.1	0
-0.01	0
0.01	1
0.1	1
1	1

By the intuitive definition of one-sided limits, we can see that as x approaches 0 from the left, $H(x)$ approaches 0, and as x approaches 0 from the right, $H(x)$ approaches 1. Therefore, we can conclude that:

$$\lim_{x \rightarrow 0^-} H(x) = 0$$

$$\lim_{x \rightarrow 0^+} H(x) = 1$$



- With the basic idea, we can now consider the bigger problem : can limits **fail** to exist?

EXAMPLE | Uh Oh

Investigate the following :

$$\lim_{x \rightarrow 0} \sin\left(\frac{\pi}{x}\right)$$

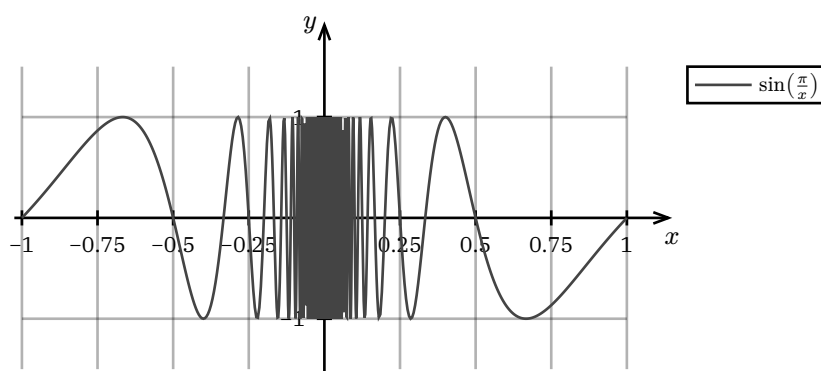
Solution 1 |

x	$\sin(\frac{\pi}{x})$
0.6666666666666666	-1
0.5	0
0.4	1
0.25	0
0.2222222222222222	1
0.18181818181818182	-1

By evaluating $\sin(\frac{\pi}{x})$ at values of x that get closer and closer to 0, we can see that the function does not approach a single value. Instead, it oscillates between -1 and 1 . Therefore, we can conclude that:

$$\lim_{x \rightarrow 0} \sin\left(\frac{\pi}{x}\right)$$

does not exist.



EXAMPLE | Uh Oh 2

Find

$$\lim_{x \rightarrow 0} \left(\frac{1}{x^2} \right)$$

if it exists

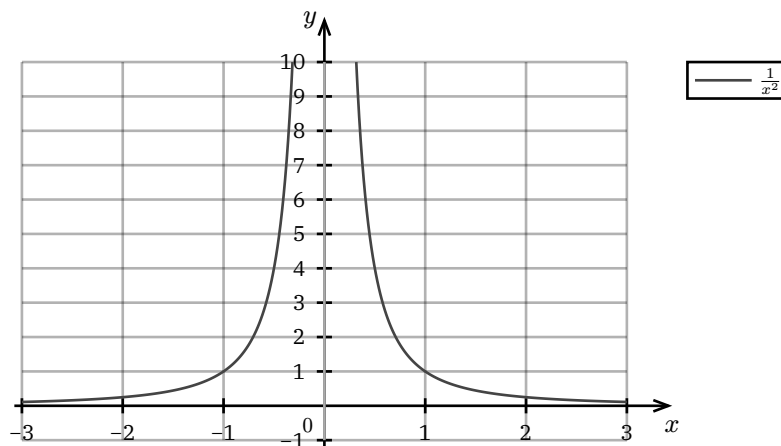
Solution 1 |

x	$\frac{1}{x^2}$
1	1
0.1	100
0.01	10000
0.001	1000000
0.0001	100000000

By evaluating $\frac{1}{x^2}$ at values of x that get closer and closer to 0, we can see that the function grows without bound. Therefore, we can conclude that:

$$\lim_{x \rightarrow 0} \left(\frac{1}{x^2} \right) = \infty$$

(which mathematically means that the limit does not exist in the real number system)



DEFINITION | Intuitive Definition of an Infinite Limit

Suppose $f(x)$ is defined whilst x is near a on both sides, except at a itself. Then we write

$$\lim_{x \rightarrow a} f(x) = \infty$$

which means that the values of $f(x)$ can be made arbitrarily large by taking x sufficiently close but not equal to a .

- Of course, we can do the same for negative infinity.

DEFINITION | Intuitive Definition of a Negative Infinity Limit

Suppose $f(x)$ is defined whilst x is near a on both sides, except at a itself. Then we write

$$\lim_{x \rightarrow a} f(x) = -\infty$$

which means that the values of $f(x)$ can be made arbitrarily small (negatively large) by taking x sufficiently close but not equal to a .

- At these points, we can define a **vertical asymptote**.

DEFINITION | Vertical Asymptote

A vertical asymptote is a vertical line $x = a$ when the function $f(x)$ suffices at least one of the 6 conditions :

- $\lim_{x \rightarrow a^+} f(x) = \infty$
- $\lim_{x \rightarrow a^+} f(x) = -\infty$
- $\lim_{x \rightarrow a^-} f(x) = \infty$
- $\lim_{x \rightarrow a^-} f(x) = -\infty$
- $\lim_{x \rightarrow a} f(x) = \infty$
- $\lim_{x \rightarrow a} f(x) = -\infty$

In simpler terms, a vertical asymptote is a vertical line where the function grows without bound as it approaches the line from at least one side.

EXAMPLE | Vertical Asymptote Example

Investigate if the function $y = \frac{2x}{x-3}$ have a vertical asymptote.

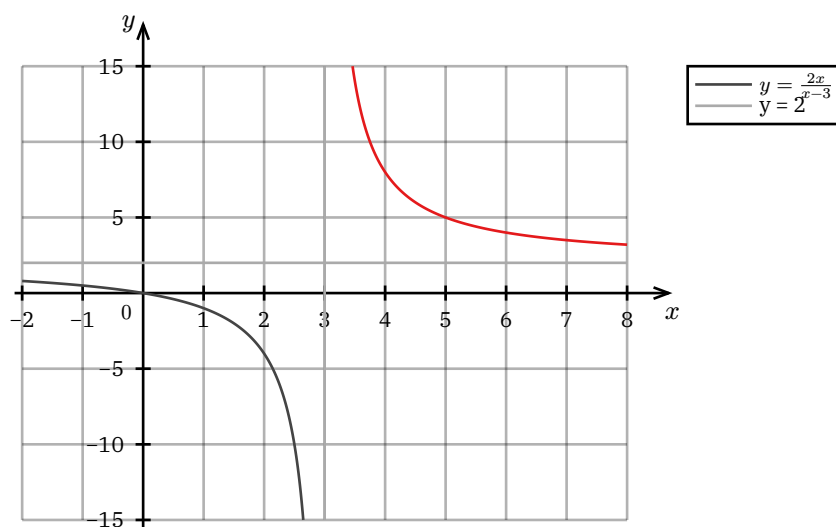
Solution 1 |

To find vertical asymptotes, we need to look for values of x that make the denominator equal to 0. Here, the denominator $x - 3$ equals 0 when $x = 3$. Next, we need to evaluate the limits as x approaches 3 from both sides:

$$\lim_{x \rightarrow 3^+} \frac{2x}{x - 3} = \infty$$

$$\lim_{x \rightarrow 3^-} \frac{2x}{x - 3} = -\infty$$

Since both one-sided limits approach infinity (one positive, one negative), we can conclude that the function has a vertical asymptote at $x = 3$.



Chapter 01.06

Calculating Limits using the Limit Laws

- Limits Have some rules that we must comply to.

THEOREM | The Laws of Limits

Suppose that c is a constant and the limits $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist. Then, the following properties hold:

1. $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$
2. $\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$
3. $\lim_{x \rightarrow a} [c * f(x)] = c * \lim_{x \rightarrow a} f(x)$
4. $\lim_{x \rightarrow a} [f(x) * g(x)] = \lim_{x \rightarrow a} f(x) * \lim_{x \rightarrow a} g(x)$
5. $\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$, provided that $\lim_{x \rightarrow a} g(x) \neq 0$

Using these laws, we can derive a few more :

THEOREM | The Power Law of Limits

Suppose that $\lim_{x \rightarrow a} f(x) = L$ and n is a positive integer. Then, $\lim_{x \rightarrow a} [f(x)]^n = [\lim_{x \rightarrow a} f(x)]^n = L^n$

THEOREM | The Root Law of Limits

Suppose that $\lim_{x \rightarrow a} f(x) = L$ and n is a positive integer. Then, $\lim_{x \rightarrow a} [f(x)]^{\frac{1}{n}} = [\lim_{x \rightarrow a} f(x)]^{\frac{1}{n}} = L^{\frac{1}{n}}$, provided that if n is even, then $L \geq 0$

also, some special limits :

THEOREM | The Constant Function Law

For any constant c , $\lim_{x \rightarrow a} c = c$

Written differently, $\lim_{x \rightarrow a} x = a$

EXAMPLE | Limit Law Basics

Evaluate $\lim_{x \rightarrow 5} (2x^2 - 3x + 4)$.

Solution 1 |

$$\lim_{x \rightarrow 5} (2x^2 - 3x + 4) = \lim_{x \rightarrow 5} 2x^2 - \lim_{x \rightarrow 5} 3x + \lim_{x \rightarrow 5} 4$$

($\because$ the Sum and Difference Law)

$$= 2 * \lim_{x \rightarrow 5} x^2 - 3 * \lim_{x \rightarrow 5} x + 4$$

($\because$ the Constant Multiple Law)

$$= 2 * 5^2 - 3 * 5 + 4$$

($\because$ the Power Law and Constant Function Law)

$$= 50 - 15 + 4 = 39$$

Above, we have discussed that $\lim_{x \rightarrow a} x = a$. What if we generalize this into any function $f(x)$?

THEOREM | The Direct Substitution Property

If f is a polynomial/rational function and a is in the domain of f , at $x = a$, then

$$\lim_{x \rightarrow a} f(x) = f(a)$$

EXAMPLE | Direct Substitution Example

Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 4x + 1}{x^2 + 3}$.

Solution 1 |

Since the function is a rational function and 2 is in the domain of the function, we can use the Direct Substitution Property.

$$\lim_{x \rightarrow 2} \frac{x^3 - 4x + 1}{x^2 + 3} = \frac{2^3 - 4*2 + 1}{2^2 + 3} = \frac{8 - 8 + 1}{4 + 3} = \frac{1}{7}$$

EXAMPLE | Laws of Limits General

Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$.

Solution 1 |

We cannot directly substitute 1 into the function, since it would create a division by zero. However, we can simplify the expression first.

$$\frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1 \because x \neq 1$$

.

Now we can find the limit:

$$\lim_{x \rightarrow 1} (x + 1) = 1 + 1 = 2$$

From the intuitive definition of left/right limits, we can know the following :

THEOREM | Relationship between One-Sided Limits and Two-Sided Limits

Let $f(x)$ be defined whilst x is near a . Then,

$$\lim_{x \rightarrow a} f(x) = L$$

if and only if both

$$\lim_{x \rightarrow a^+} f(x) = L$$

and

$$\lim_{x \rightarrow a^-} f(x) = L$$

1 The Squeeze Theorem

- Sometimes, it is hard to find the limit of a function directly. However, if we can find two other functions that “squeeze” the function we are interested in, we can use the Squeeze Theorem to find the limit.

THEOREM | The Squeeze Theorem

Let $f(x)$, $g(x)$, and $h(x)$ be functions defined on an open interval containing a , except possibly at a itself. If for every x in this interval, except possibly at a , we have

$$g(x) \leq f(x) \leq h(x)$$

and if

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L$$

then,

$$\lim_{x \rightarrow a} f(x) = L$$

EXAMPLE | Squeeze Theorem Example

Evaluate

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$$

Solution 1 |

We know that

$$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$$

for all $x \neq 0$. Multiplying all parts of the inequality by x^2 (which is always non-negative), we get :

$$-x^2 \leq x^2 * \sin\left(\frac{1}{x}\right) \leq x^2$$

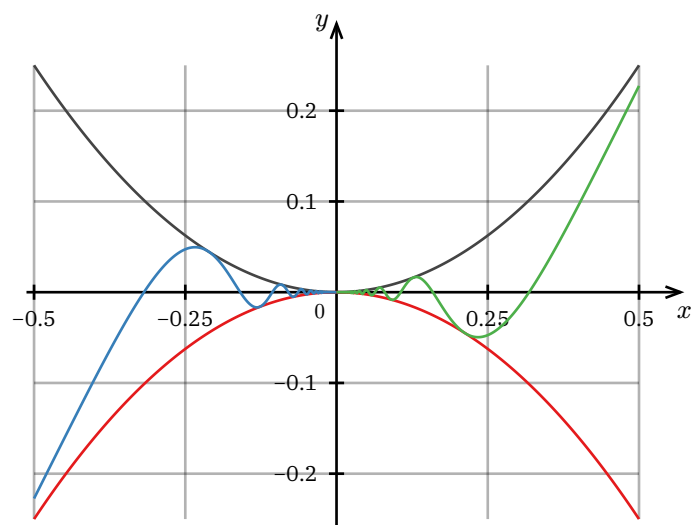
Now, we can find the limits of the two outer functions as x approaches 0:

$$\lim_{x \rightarrow 0} -x^2 = 0$$

$$\lim_{x \rightarrow 0} x^2 = 0$$

Since both limits are equal to 0, by the Squeeze Theorem, we have :

$$\lim_{x \rightarrow 0} x^2 * \sin\left(\frac{1}{x}\right) = 0$$



The Precise Definition of a Limit

1 The “Precise” Definition of a Limit

- Till now, we have been using *intuitive* methods to define the limits of functions. However, in order to be more *precise*, we need a more rigorous definition of limits.

EXAMPLE | A Weird Function

Consider the following function : $\begin{cases} 2x-1 & \text{if } x \neq 3 \\ 6 & \text{if } x = 3 \end{cases}$. What is $f(3)$? What is $\lim_{x \rightarrow 3} f(x)$?

Solution 1 |

$f(3) = 6$, since when $x = 3$, the function outputs 6. However, to find $\lim_{x \rightarrow 3} f(x)$, we need to see what value $f(x)$ approaches as x gets closer and closer to 3. As x approaches 3, $f(x)$ approaches $2(3) - 1 = 5$. Therefore, $\lim_{x \rightarrow 3} f(x) = 5$.

- Consider the example above. How can we more “accurately” define the limit of a function at a point, without relying on intuition or graphs?
- In the problem above, consider the following question.
 - “How close to 3 does x need to be, so that $f(x)$ is within 0.1 of 5?”
 - In other words, we want to find some value δ s.t.

$$\|f(x) - 5\| < 0.1 \quad \text{if} \quad \|x - 3\| < \delta \quad \text{but } x \neq 3$$

If $\|x - 3\| > 0$, then $x \neq 3$, so we can simplify the problem into the following :

- “How close to 3 does x need to be, so that $f(x)$ is within 0.1 of 5, given that $x \neq 3$

$$\|f(x) - 5\| < 0.1 \quad \text{if} \quad 0 < \|x - 3\| < \delta$$

Here, we call 0.1 the “Error Tolerance”. Since we are going to send this to 0, let us replace this with ε for the time being. So the question becomes :

$$\|f(x) - 5\| < \varepsilon \quad \text{if} \quad 0 < \|x - 3\| < \delta$$

Now, we finally have the tools to define the limit of a function more “precisely”.

DEFINITION | Precise Definition of a Limit

Let f be a function defined on an open interval containing a , except possibly at a itself. We say that $\lim_{x \rightarrow a} f(x) = L$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$\|f(x) - L\| < \varepsilon \quad \text{if} \quad 0 < \|x - a\| < \delta$$

To simplify even more, we can write like this(in style):

$$\forall \varepsilon > 0, \exists \delta > 0, \text{ s.t. } 0 < \|x - a\| < \delta \implies \|f(x) - L\| < \varepsilon$$

Since we have to divide them later anyway, we can do it now :

DEFINITION | Precise definition of a left-hand limit

Let f be a function defined on an open interval containing a , except possibly at a itself. We say that $\lim_{x \rightarrow a^-} f(x) = L$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$\|f(x) - L\| < \varepsilon \quad \text{if} \quad a - \delta < x < a$$

To simplify even more, we can write like this:

$$\forall \varepsilon > 0, \exists \delta > 0, \text{ s.t. } a - \delta < x < a \implies \|f(x) - L\| < \varepsilon$$

DEFINITION | Precise definition of a right-hand limit

Let f be a function defined on an open interval containing a , except possibly at a itself. We say that $\lim_{x \rightarrow a^+} f(x) = L$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$\|f(x) - L\| < \varepsilon \quad \text{if} \quad a < x < a + \delta$$

To simplify even more, we can write like this:

$$\forall \varepsilon > 0, \exists \delta > 0, \text{ s.t. } a < x < a + \delta \implies \|f(x) - L\| < \varepsilon$$

One good way to understand the epsilon-delta method is by assuming we chose a “wrong” limit value “L”. In this case, we must be able to find a δ for any arbitrary ε given. Since we chose a wrong limit value, there will always be some x values within the δ -neighborhood of a that make $f(x)$ be outside the ε -neighborhood of “L”. This means that no matter how small we choose δ , there will always be some x values that break the condition. Thus, we can conclude that “L” is not the correct limit value.

NOTE | Usage of the epsilon-delta method

The epsilon-delta method can only be used to “**prove**” that a certain limit value is correct. It cannot be used to “**find**” the limit value itself.

Now lets utilize this to solve some problems.

EXAMPLE | Epsilon-Delta Basics

Prove that $\lim_{x \rightarrow 3} (4x - 5) = 7$ using the epsilon-delta definition of a limit.

Solution 1 |

Let $\varepsilon > 0$ be given. We need to find a $\delta > 0$ such that

$$\|(4x - 5) - 7\| < \varepsilon \quad \text{if} \quad 0 < \|x - 3\| < \delta$$

1. Preliminary Analysis :

Simplifying the left side, we have :

$$\|4x - 12\| = 4 * \|x - 3\|$$

So we want to ensure that

$$4 * \|x - 3\| < \varepsilon \quad \text{if} \quad 0 < \|x - 3\| < \delta$$

To achieve this, we can choose $\delta = \frac{\varepsilon}{4}$. Then, if $0 < \|x - 3\| < \delta$, we have

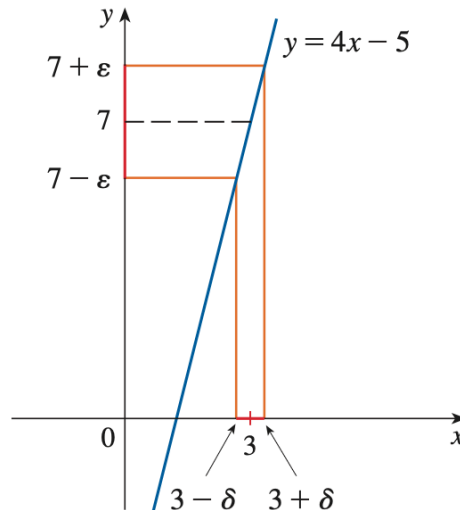
$$\|(4x - 5) - 7\| = 4 * \|x - 3\| < 4 * \delta = \varepsilon$$

2. Formal Proof :

Let $\varepsilon > 0$ be given. Choose $\delta = \frac{\varepsilon}{4}$. Then, if $0 < \|x - 3\| < \delta$, we have

$$\|(4x - 5) - 7\| = 4 * \|x - 3\| < 4 * \delta = \varepsilon$$

Thus, by the epsilon-delta definition of a limit, we conclude that $\lim_{x \rightarrow 3} (4x - 5) = 7$.



EXAMPLE | Epsilon-Delta Basics 2

Describe $\lim_{x \rightarrow a} x^2 = a^2$ using the epsilon-delta definition of a limit.

Solution 1 |

Let $\varepsilon > 0$ be given. We need to find a $\delta > 0$ such that

$$\|x^2 - a^2\| < \varepsilon \quad \text{if} \quad 0 < \|x - a\| < \delta$$

1. Preliminary Analysis :

Simplifying the left side, we have :

$$\|x^2 - a^2\| = \|(x - a)(x + a)\| = \|x - a\| * \|x + a\|$$

To control $\|x + a\|$, we can restrict δ to be less than 1 (an arbitrary number).

This means that if $0 < \|x - a\| < \delta < 1$, then

$$\|x - a\| < 1 \implies \|x\| < \|a\| + 1$$

Thus, we have

$$\|x + a\| \leq \|x\| + \|a\| < (\|a\| + 1) + \|a\| = 2 * \|a\| + 1$$

Therefore, we want to ensure that

$$\|x - a\| * (2 * \|a\| + 1) < \varepsilon \quad \text{if} \quad 0 < \|x - a\| < \delta < 1$$

To achieve this, we can choose $\delta = \min\left(1, \frac{\varepsilon}{2 * \|a\| + 1}\right)$. Then, if $0 < \|x - a\| < \delta$, we have

$$\|x^2 - a^2\| = \|x - a\| * \|x + a\| < \delta * (2 * \|a\| + 1) \leq \left(\frac{\varepsilon}{2 * \|a\| + 1}\right) * (2 * \|a\| + 1) = \varepsilon$$

2. Formal Proof :

Let $\varepsilon > 0$ be given. Choose $\delta = \min\left(1, \frac{\varepsilon}{2 * \|a\| + 1}\right)$. Then, if $0 < \text{norm}(x - a) < \delta$, we have

$$\|x^2 - a^2\| = \|x - a\| * \|x + a\| < \delta * (2 * \|a\| + 1) \leq \left(\frac{\varepsilon}{2 * \|a\| + 1}\right) * (2 * \|a\| + 1) = \varepsilon$$

Thus, by the epsilon-delta definition of a limit, we conclude that $\lim_{x \rightarrow a} x^2 = a^2$.

NOTE | Why can we assign an arbitrary number to delta?

In the preliminary analysis, we restricted δ to be less than 1 to control the value of $\|x + a\|$. This is a common technique in epsilon-delta proofs to ensure that certain expressions remain bounded. By choosing δ to be the minimum of 1 and another expression, we can effectively manage the behavior of the function within the desired neighborhood around the point a . To say intuitively, we are “bounding” the function in the neighborhood of a once more outside of delta.

EXAMPLE | Epsilon-Delta Basics 3

Describe $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2$ using the epsilon-delta definition of a limit.

Solution 1 |

Let $\varepsilon > 0$ be given. We need to find a $\delta > 0$ such that

$$\left\| \frac{x^2 - 1}{x - 1} - 2 \right\| < \varepsilon \quad \text{if} \quad 0 < \|x - 1\| < \delta$$

1. Preliminary Analysis :

Simplifying the left side, we have :

$$\left\| \frac{x^2 - 1}{x - 1} - 2 \right\| = \left\| (x - 1) \frac{x + 1}{x - 1} - 2 \right\| = \|x + 1 - 2\| = \|x - 1\|$$

Therefore, we want to ensure that

$$\|x - 1\| < \varepsilon \quad \text{if} \quad 0 < \|x - 1\| < \delta$$

To achieve this, we can choose $\delta = \varepsilon$. Then, if $0 < \|x - 1\| < \delta$, we have

$$\left\| \frac{x^2 - 1}{x - 1} - 2 \right\| = \|x - 1\| < \delta = \varepsilon$$

2. Formal Proof :

Let $\varepsilon > 0$ be given. Choose $\delta = \varepsilon$. Then, if $0 < \|x - 1\| < \delta$, we have

$$\left\| \frac{x^2 - 1}{x - 1} - 2 \right\| = \|x - 1\| < \delta = \varepsilon$$

Thus, by the epsilon-delta definition of a limit, we conclude that $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = 2$.

EXAMPLE | Epsilon-Delta Basics 4

Describe $\lim_{x \rightarrow 0+} \sqrt{x} = 0$ using the epsilon-delta definition of a limit.

Solution 1 |

Let $\varepsilon > 0$ be given. We need to find a $\delta > 0$ such that

$$\|\sqrt{x} - 0\| < \varepsilon \quad \text{if} \quad 0 < x < \delta$$

1. Preliminary Analysis :

Simplifying the left side, we have :

$$\|\sqrt{x} - 0\| = \sqrt{x}$$

Therefore, we want to ensure that

$$\sqrt{x} < \varepsilon \quad \text{if} \quad 0 < x < \delta$$

To achieve this, we can choose $\delta = \varepsilon^2$. Then, if $0 < x < \delta$, we have

$$\|\sqrt{x} - 0\| = \sqrt{x} < \sqrt{\delta} = \sqrt{\varepsilon^2} = \varepsilon$$

2. Formal Proof :

Let $\varepsilon > 0$ be given. Choose $\delta = \varepsilon^2$. Then, if $0 < x < \delta$, we have

$$\|\sqrt{x} - 0\| = \sqrt{x} < \sqrt{\delta} = \sqrt{\varepsilon^2} = \varepsilon$$

Thus, by the epsilon-delta definition of a limit, we conclude that $\lim_{x \rightarrow 0+} \sqrt{x} = 0$.

Chapter 01.08

Continuity

1 Continuity

Intuitively, continuity can be simply defined as “a function that is connected”...but we don't deal with intuition here. We need a more “precise” definition of continuity.

DEFINITION | Continuity at a Point

A function f is said to be continuous at a point $x = a$ if the following three conditions are satisfied :

1. $f(a)$ is defined.
2. $\lim_{x \rightarrow a} f(x)$ exists.
3. $\lim_{x \rightarrow a} f(x) = f(a)$

If these conditions are not met, we call the function is “discontinuous” at $x = a$, or has a “discontinuity” at $x = a$.

EXAMPLE | Continuity Example

Identify if the functions are discontinuous.

(a) $f(x) = \frac{x^2 - x - 2}{x - 2}$

(b) $f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$

Solution (a) |

The function is discontinuous at $x = 2$. This is because $f(2)$ is not defined, since the denominator becomes 0 at this point. Therefore, the first condition for continuity is not satisfied.

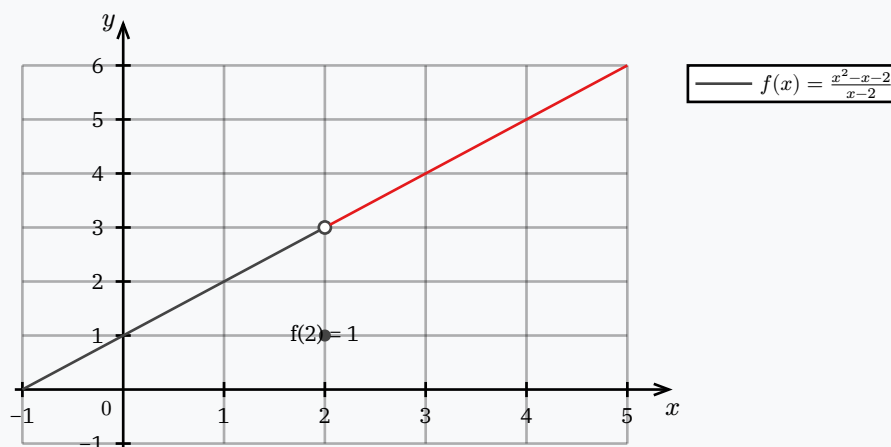
Solution (b) |

The function is continuous at $x = 2$. Let's check the three conditions:

1. $f(2) = 1$, so the first condition is satisfied.
2. To find $\lim_{x \rightarrow 2} f(x)$, we simplify the expression for $x \neq 2$: $f(x) = \frac{x^2 - x - 2}{x - 2} = \frac{(x-2)(x+1)}{x-2} = x + 1$ for $x \neq 2$. Therefore, $\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} (x + 1) = 3$. So the second condition is satisfied.

3. However, $\lim_{x \rightarrow 2} f(x) = 3$ and $f(2) = 1$, so the third condition is not satisfied.

Thus, the function is discontinuous at $x = 2$.



As we did with limits, we can define the left and right continuities for a function too.

DEFINITION | Left Continuity at a Point

A function f is said to be left continuous at a point $x = a$ if the following three conditions are satisfied :

1. $f(a)$ is defined.
2. $\lim_{x \rightarrow a^-} f(x)$ exists.
3. $\lim_{x \rightarrow a^-} f(x) = f(a)$

DEFINITION | Right Continuity at a Point

A function f is said to be right continuous at a point $x = a$ if the following three conditions are satisfied :

1. $f(a)$ is defined.
2. $\lim_{x \rightarrow a^+} f(x)$ exists.
3. $\lim_{x \rightarrow a^+} f(x) = f(a)$

Using this info, we can create the following definition :

DEFINITION | Continuity on an Interval

A function f is said to be continuous on an interval I if it is continuous at every point in I .

- If I is a closed interval $[a, b]$, then we also need to check that f is left continuous at $x = b$ and right continuous at $x = a$.

EXAMPLE | Continuity on an Interval Example

Show that the function $f(x) = 1 - \sqrt{1 - x^2}$ is continuous in the interval $[-1, 1]$.

Solution 1 |

If $-1 < a < 1$, then using the Limit Laws from Section 1.6, we have :

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} (1 - \sqrt{1 - x^2}) = 1 - \lim_{x \rightarrow a} \sqrt{1 - x^2} = 1 - \sqrt{1 - a^2} = f(a)$$

. Thus, f is continuous at every point in the open interval $(-1, 1)$. Now, we need to check the endpoints $x = -1$ and $x = 1$.

- At $x = -1$:

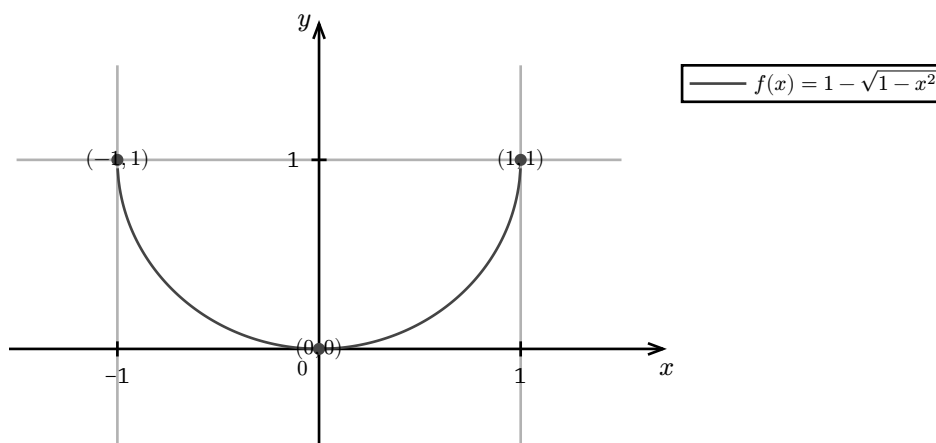
$$\begin{aligned} \lim_{x \rightarrow -1^+} f(x) &= \lim_{x \rightarrow -1^+} (1 - \sqrt{1 - x^2}) \\ &= 1 - \lim_{x \rightarrow -1^+} \sqrt{1 - x^2} \\ &= 1 - \sqrt{1 - (-1)^2} \\ &= 1 - 0 \\ &= 1 = f(-1) \end{aligned}$$

. Thus, f is right continuous at $x = -1$.

- At $x = 1$:

$$\begin{aligned} \lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^-} (1 - \sqrt{1 - x^2}) \\ &= 1 - \lim_{x \rightarrow 1^-} \sqrt{1 - x^2} \\ &= 1 - \sqrt{1 - 1^2} \\ &= 1 - 0 \\ &= 1 = f(1) \end{aligned}$$

. Thus, f is left continuous at $x = 1$. Therefore, f is continuous on the closed interval $[-1, 1]$.



2 Properties of Continuous Functions

Just like limits, continuous functions have some properties that we can use to solve problems more easily.

THEOREM | Properties of Continuous Functions

If f and g are continuous at $x = a$, then the following functions are also continuous at $x = a$:

1. $f + g$
2. $f - g$
3. cf , where c is any constant
4. fg
5. $\frac{f}{g}$, provided that $g(a) \neq 0$
6. $[f(x)]^n$, where n is a positive integer (derived from rule 4)

Proof | Rule 1

$$\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = f(a) + g(a) = (f + g)(a)$$

Thus, since $f(x)$ and $g(x)$ are continuous at $x = a$, $f + g$ is also continuous at $x = a$.

Proof | Rule 2

$$\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x) = f(a) - g(a) = (f - g)(a)$$

Thus, since $f(x)$ and $g(x)$ are continuous at $x = a$, $f - g$ is also continuous at $x = a$.

Proof | Rule 3

$$\lim_{x \rightarrow a} [cf(x)] = c * \lim_{x \rightarrow a} f(x) = c * f(a) = (cf)(a)$$

Thus, since $f(x)$ is continuous at $x = a$, cf is also continuous at $x = a$.

Proof | Rule 4

$$\lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) * \lim_{x \rightarrow a} g(x) = f(a) * g(a) = (fg)(a)$$

Thus, since $f(x)$ and $g(x)$ are continuous at $x = a$, fg is also continuous at $x = a$.

Proof | Rule 5

$$\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{f(a)}{g(a)} = \left(\frac{f}{g} \right)(a)$$

Thus, since $f(x)$ and $g(x)$ are continuous at $x = a$ and $g(a) \neq 0$, $\frac{f}{g}$ is also continuous at $x = a$.

Proof | Rule 6

This rule is derived from Rule 4. Since $f(x)$ is continuous at $x = a$, using Rule 4 repeatedly, we can show that $[f(x)]^n$ is also continuous at $x = a$.

You may have noticed that we didnt even care about continuity during polynomials. This is because of the following theorem :

THEOREM | Continuity of Polynomials and Rational Functions

1. Every polynomial function is continuous for all real numbers; that is, it is continuous on $\mathbb{R} = (-\infty, \infty)$

2. Every rational function is continuous at every point in its domain; that is, it is continuous on its domain.

Proof |

1. Let $P(x)$ be a polynomial function. Since polynomials are formed by adding, subtracting, and multiplying constant functions and the identity function $f(x) = x$, and both of these functions are continuous everywhere, by the Properties of Continuous Functions theorem, $P(x)$ is continuous for all real numbers.
2. Let $R(x) = \frac{P(x)}{Q(x)}$ be a rational function, where $P(x)$ and $Q(x)$ are polynomial functions. Since both $P(x)$ and $Q(x)$ are continuous everywhere (from part 1), and provided that $Q(a) \neq 0$, by the Properties of Continuous Functions theorem, $R(x)$ is continuous at every point in its domain.

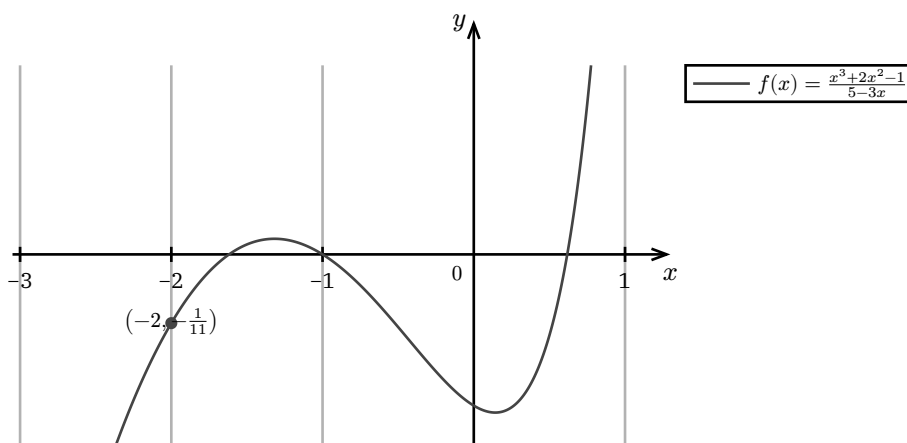
EXAMPLE | Continuity of a Rational Function

Find $\lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$.

Solution 1 |

Since the function is a rational function and 2 is in the domain of the function, we can use the Continuity of Rational Functions theorem.

$$\lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x} = \frac{-2^3 + 2(-2)^2 - 1}{5 - 3(-2)} = \frac{-8 + 8 - 1}{5 + 6} = -\frac{1}{11}$$



3 Intermediate Value Theorem

The Intermediate Value Theorem is a very useful theorem that utilizes the property of continuous functions.

THEOREM | Intermediate Value Theorem

Let f be a function that is continuous on the closed interval $[a, b]$, and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then, there exists at least one number c in the open interval (a, b) such that $f(c) = N$.

Proof |

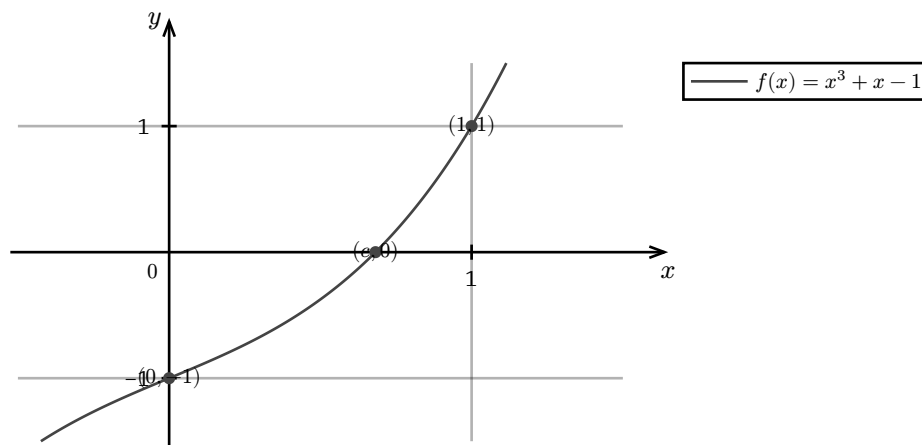
Since f is continuous on the closed interval $[a, b]$, by the definition of continuity, for every x in $[a, b]$, $f(x)$ takes on every value between $f(a)$ and $f(b)$. Therefore, for any number N between $f(a)$ and $f(b)$, there must exist at least one number c in the open interval (a, b) such that $f(c) = N$.

EXAMPLE | Intermediate Value Theorem Example

Show that the equation $x^3 + x - 1 = 0$ has a solution in the interval $[0, 1]$.

Solution 1 |

Let $f(x) = x^3 + x - 1$. We need to show that there exists a number c in the interval $(0, 1)$ such that $f(c) = 0$. First, we check the values of f at the endpoints of the interval: $f(0) = 0^3 + 0 - 1 = -1$ $f(1) = 1^3 + 1 - 1 = 1$ Since $f(0) = -1$ and $f(1) = 1$, and 0 is between -1 and 1 , by the Intermediate Value Theorem, there exists at least one number c in the interval $(0, 1)$ such that $f(c) = 0$. Therefore, the equation $x^3 + x - 1 = 0$ has a solution in the interval $[0, 1]$.



Chapter 02

Derivatives

This chapter turns the limiting process of the previous chapter into a machine: the derivative. Starting from rates of change and the derivative as a function in its own right, it builds the full computational toolkit (the power, product, quotient and chain rules, derivatives of trigonometric functions, and implicit differentiation) before returning to the real world with rates of change in science and society and the idea of linear approximation, where a curve is deliberately replaced by its own tangent line.

Chapter 02.01

Derivatives and Rates of Change

We now should discuss the elephant in the room, **derivatives**.

1 Introduction to Derivatives

- So far, we have discussed limits and how they can be used to find the slope of tangents and instantaneous velocity. However, we have not yet formally defined derivatives.
- Derivatives are a way to represent the rate of change of a function at a specific point. In simpler terms, it tells us how quickly the function's output is changing as we make small changes to its input.

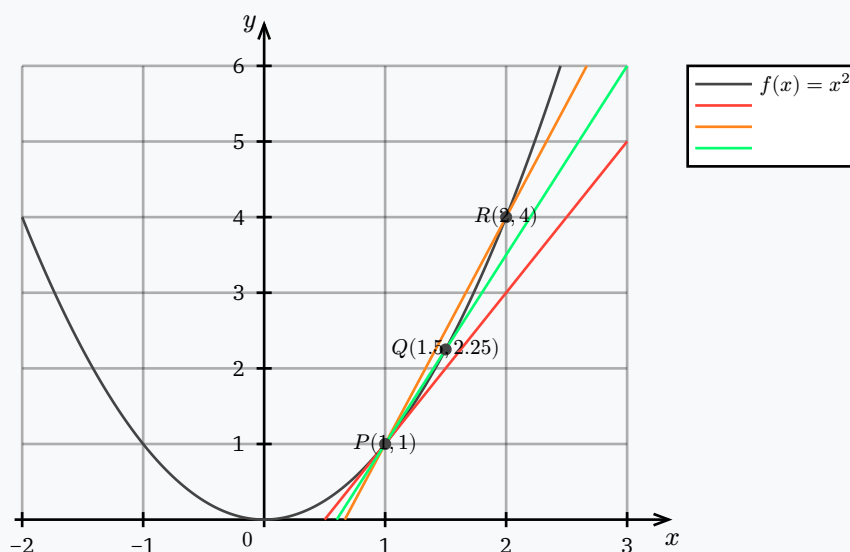
DEFINITION | The Tangent Line

The tangent line to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with the slope

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

. provided that this limit exists.

As we move the limit closer to P , the line more accurately represents the slope of the curve at that point.



EXAMPLE | Derivative Basics

Find the equation of the tangent line to the curve $y = x^2$ at the point $P(2, 4)$.

Solution 1 |

First, we need to find the slope of the tangent line at point P . We can use the definition of the tangent line:

$$m = \lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2}$$

where $f(x) = x^2$ and $f(2) = 4$.

So,

$$m = \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

We can factor the numerator:

$$m = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{x - 2}$$

For $x \neq 2$, we can cancel out $(x - 2)$:

$$m = \lim_{x \rightarrow 2} (x + 2)$$

Now, we can directly substitute $x = 2$:

$$m = 2 + 2 = 4$$

Now that we have the slope of the tangent line, we can use the point-slope form of a line to find its equation:

$$y - y_1 = m(x - x_1)$$

where $(x_1, y_1) = (2, 4)$ and $m = 4$.

Substituting these values in, we get:

$$y - 4 = 4(x - 2)$$

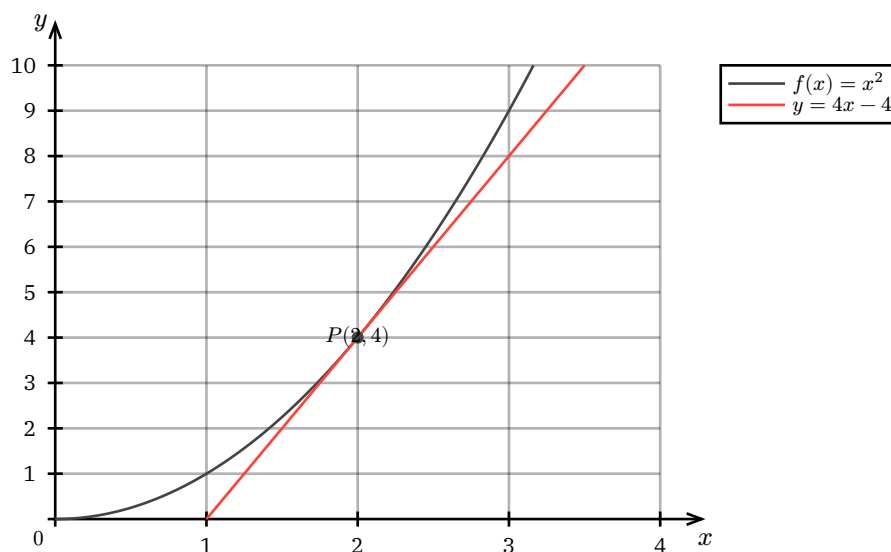
Simplifying this, we get:

$$y - 4 = 4x - 8$$

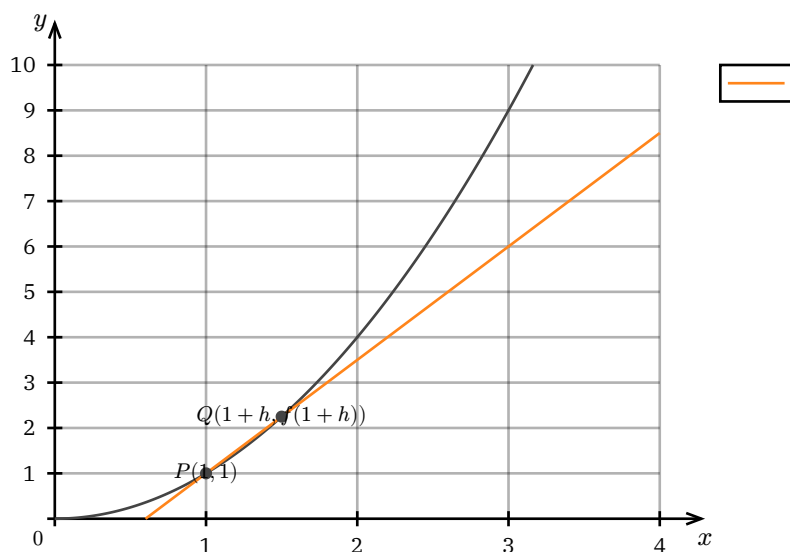
$$y = 4x - 4$$

Therefore, the equation of the tangent line to the curve $y = x^2$ at point $P(2, 4)$ is:

$$y = 4x - 4$$



However, we can have an alternate method for defining derivatives. Imagine we have two points P and Q , separated by a small distance h on the curve of $f(x)$.



DEFINITION | Alternative Definition of the Derivative

The derivative of a function $f(x)$ at a point a is defined as :

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

provided that this limit exists.

Do you remember the velocity problem we discussed in chapter 0? We can use this definition to find the instantaneous velocity of an object at a specific time.

DEFINITION | Instantaneous Velocity

The instantaneous velocity of an object at time $t = a$ is defined as :

$$v(a) = \lim_{h \rightarrow 0} \frac{s(a+h) - s(a)}{h}$$

where $s(t)$ is the position function of the object at time t .

NOTE | Let's go a bit more...

We can connect a few concepts of basic Physics here. For instance, acceleration can be defined as the derivative of velocity with respect to time. Thus, we have :

$$a(t) = v'(t) = s''(t)$$

To write it out in the long method, we have :

$$a(t) = \lim_{h \rightarrow 0} \frac{v(t+h) - v(t)}{h} = \lim_{h \rightarrow 0} \frac{\frac{s(t+h) - s(t)}{h} - v(t)}{h}$$

2 Rate of Change

Derivatives can also be used to find the rate of change of one quantity with respect to another. For example, if we have a function that describes the relationship between distance and time, we can use derivatives to find the rate of change of distance with respect to time, which is velocity.

DEFINITION | Rate of Change

The rate of change of a function $f(x)$ with respect to x at a point a is given by the derivative $f'(a)$. We can represent it as :

$$\text{rate of change} = \frac{\Delta x}{\Delta y} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

EXAMPLE | Rate Of Change Example

A car travels along a straight road, and its position at time t (in seconds) is given by the function $s(t) = 3t^2 + 2t$. Find the car's velocity at time $t = 4$ seconds.

Solution 1 |

To find the car's velocity at time $t = 4$ seconds, we need to find the derivative of the position function $s(t)$, which gives us the velocity function $v(t)$.

The position function is:

$$s(t) = 3t^2 + 2t$$

We can find the derivative $s'(t)$ using the power rule:

$$s'(t) = \frac{d}{dt}(3t^2) + \frac{d}{dt}(2t)$$

$$s'(t) = 6t + 2$$

Therefore, the velocity function is:

$$v(t) = 6t + 2$$

Now, we can find the velocity at time $t = 4$ seconds by substituting $t = 4$ into the velocity function:

$$v(4) = 6(4) + 2$$

$$v(4) = 24 + 2$$

$$v(4) = 26$$

Thus, the car's velocity at time $t = 4$ seconds is 26 units per second.

Chapter 02.02

The Derivative as a Function

1 Derivatives

As we did in the previous section, we can define derivatives using limits.

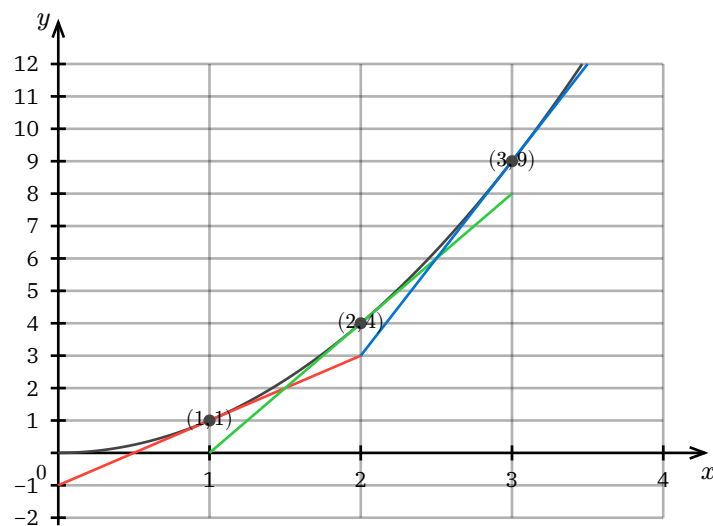
DEFINITION | The Derivative at a Point

The derivative of the function $f(x)$ at the point $x = a$ is defined as:

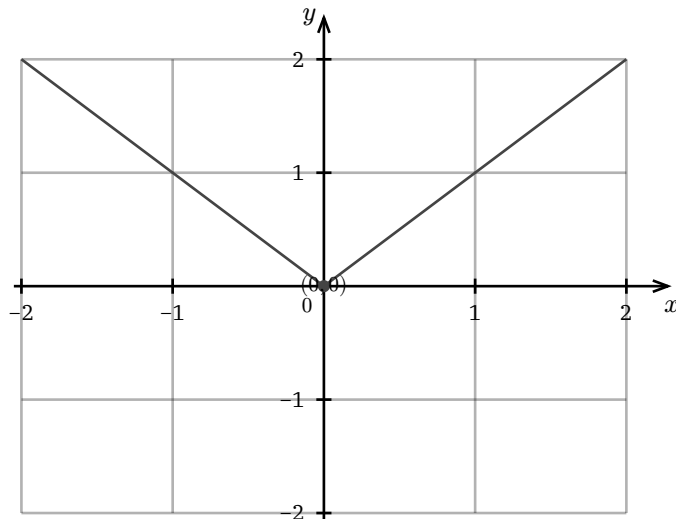
$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

This limit represents the slope of the tangent line to the curve at the point $(a, f(a))$.

We can define a derivative on any point of a function as long as the limit exists.



But, Are there functions that we can't define a derivative on? Yes! For example, consider the function $f(x) = |x|$ at the point $x = 0$.



Since there is a sharp corner at $x = 0$, we cannot define a unique tangent line, and thus the derivative does not exist at that point.

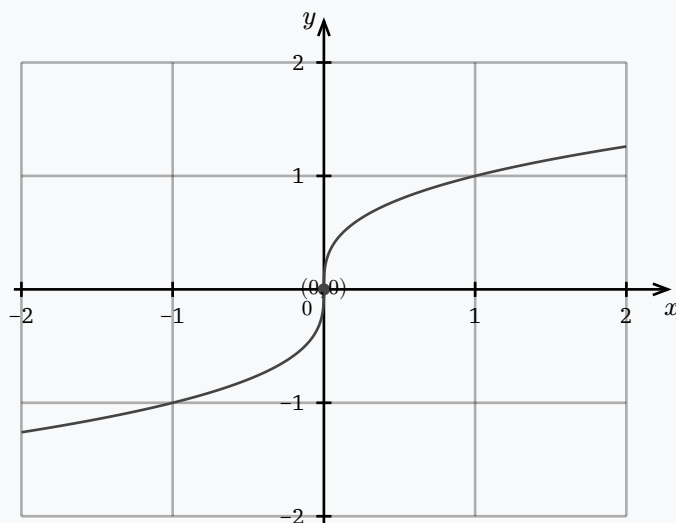
DEFINITION | Non-Differentiable Point

A point $x = a$ is called a non-differentiable point of the function $f(x)$ if the derivative $f'(a)$ does not exist. The opposite is called a differentiable point, where the derivative does exist.

We also call a function differentiable on an interval if it is differentiable at every point in an open interval.

Then, what makes a function non-differentiable at a point? There are three common reasons:

1. The function has a sharp corner or cusp at that point (like $f(x) = |x|$ at $x = 0$).
2. The function has a vertical tangent line at that point (like $f(x) = x^{\frac{1}{3}}$ at $x = 0$).
3. The function is not continuous at that point (like $f(x) = \frac{1}{x}$ at $x = 0$).



2 Other Notations

There are several notations for derivatives. The most common ones are:

- Lagrange's notation: $f'(x)$, $f''(x)$, $f'''(x)$ for the first, second, and third derivatives respectively.
- Leibniz's notation: In Leibniz's notation, the derivative of y with respect to x is denoted as

$$\left. \frac{dy}{dx} \right|_{x=a} \quad \text{or} \quad \left. \frac{dy}{dx} \right]_{x=a}$$

We have noted earlier that if f is differentiable at $x = a$, a limit at that point must exist, and also the function value must be defined at that point. This shows resemblance to the definition of continuity.

THEOREM | Differentiability Implies Continuity

If a function f is differentiable at a point $x = a$, then it is also continuous at that point.

Proof |

Since f is differentiable at $x = a$, the limit

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

exists.

This implies that

$$\lim_{h \rightarrow 0} (f(a+h) - f(a)) = 0$$

because if the limit of the difference quotient exists, the numerator must approach zero as h approaches zero.

Therefore,

$$\lim_{h \rightarrow 0} f(a+h) = f(a)$$

which is the definition of continuity at the point $x = a$.

NOTE | Converse

The converse is not necessarily true; a function can be continuous at a point but not differentiable there. For example, the function $f(x) = |x|$ is continuous at $x = 0$ but not differentiable there.

3 Higher Derivatives.

Till now, we only dealt with the first derivative of a function. However, we can take derivatives of derivatives, leading to higher-order derivatives. The second derivative, denoted as $f''(x)$ or $\frac{d^2y}{(dx)^2}$, represents the rate of change of the first derivative and provides information about the concavity of the function. Similarly, the third derivative, denoted as $f'''(x)$ or $\frac{d^3y}{(dx)^3}$, represents the rate of change of the second derivative.

Chapter 02.03

Differentiation Formulas

1 Differentiation Formulas

- Let's start with the simplest

DEFINITION | Derivative of a Constant Function

If $f(x) = c$, where c is a constant, then $\frac{d}{dx}(c) = 0$.

- Now, let's look at functions with powers of a certain number. Thus, this form : $f(x) = x^n$. What is the derivative of this function?

DEFINITION | Derivate of Power Functions

If $f(x) = x^n$, where n is a real number, then $\frac{d}{dx}(x^n) = n * x^{n-1}$.

Proof |

We can prove this using the definition of the derivative.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Substituting $f(x) = x^n$, we get:

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}$$

Using the Binomial Theorem to expand $(x+h)^n$, we have:

$$(x+h)^n = \sum_{k=0}^n \binom{n}{k} * x^{n-k} * h^k$$

Therefore,

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sum_{k=0}^n \binom{n}{k} * x^{n-k} * h^k - x^n}{h}$$

Simplifying, we get:

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sum_{k=1}^n \binom{n}{k} * x^{n-k} * h^k}{h}$$

$$= \lim_{h \rightarrow 0} \left(\sum_{k=1}^n \binom{n}{k} * x^{n-k} * h^{k-1} \right)$$

As h approaches 0, all terms with $k > 1$ vanish, leaving us with:

$$f'(x) = n * x^{n-1}$$

From this, we can derive the power rule,

THEOREM | Power Rule

If $f(x) = x^n$, where n is a real number, then $\frac{d}{dx}(x^n) = n * x^{n-1}$.

Proof | Method 1

The Formula

$$x^n - a^n = (x - a) * (x^{n-1} + x^{n-2} * a + \dots + x * a^{n-2} + a^{n-1})$$

can be used to find the derivative of $f(x) = x^n$.

$$\begin{aligned} f'(a) &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\ &= \lim_{x \rightarrow a} \left(\frac{x^n - a^n}{x - a} \right) \\ &= \lim_{x \rightarrow a} \frac{(x - a) * (x^{n-1} + x^{n-2} * a + \dots + x * a^{n-2} + a^{n-1})}{x - a} \\ &= \lim_{x \rightarrow a} (x^{n-1} + x^{n-2} * a + \dots + x * a^{n-2} + a^{n-1}) \\ &= a^{n-1} + a^{n-2} * a + \dots + a * a^{n-2} + a^{n-1} = n * a^{n-1} \end{aligned}$$

Proof | Method 2

We can use the binomial theorem to prove this.

$$(x + h)^n = \sum_{k=0}^n \binom{n}{k} * x^{n-k} * h^k$$

Substituting $f(x) = x^n$, we get:

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x + h)^n - x^n}{h}$$

Using the Binomial Theorem to expand $(x + h)^n$, we have:

$$(x + h)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} h^k$$

Therefore,

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sum_{k=0}^n \binom{n}{k} x^{n-k} h^k - x^n}{h}$$

Simplifying, we get:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\sum_{k=1}^n \binom{n}{k} x^{n-k} h^k}{h} \\ &= \lim_{h \rightarrow 0} \left(\sum_{k=1}^n \binom{n}{k} x^{n-k} h^{k-1} \right) \end{aligned}$$

As h approaches 0, all terms with $k > 1$ vanish, leaving us with:

$$f'(x) = n x^{n-1}$$

NOTE | What if $n < 0$?

We can extend the power rule to work for negative powers.

Proof |

$$\begin{aligned} \frac{d}{dx}(x^{-n}) &= \frac{d}{dx} \left(\frac{1}{x^n} \right) \\ &= \frac{x^n \frac{d}{dx}(1) - 1 \frac{d}{dx}(x^n)}{x^{2n}} \\ &= \left(-n \frac{x^{n-1}}{x^{2n}} \right) = -n x^{-n-1} \end{aligned}$$

Then how to we take care of coefficients? For that, we have the constant multiple rule :

THEOREM | Constant Multiple Rule

If $f(x) = c * g(x)$, where c is a constant and $g(x)$ is a differentiable function, then $f'(x) = c * g'(x)$.

Proof |

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Substituting $f(x) = c * g(x)$, we get:

$$f'(x) = \lim_{h \rightarrow 0} \frac{c * g(x+h) - c * g(x)}{h}$$

Factoring out c , we get:

$$f'(x) = c * \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}$$

Since $g(x)$ is differentiable, we can use the definition of the derivative to write:

$$f'(x) = c * g'(x)$$

As we talked about during limit laws and continuity, we need to deal with the arithmetic of function derivatives.

THEOREM | The Sum and Difference Rules

If f and g are both differentiable, then

$$(f(x) \pm g(x))' = f'(x) \pm g'(x)$$

Proof |

Let $l(x) = f(x) \pm g(x)$. Then,

$$\begin{aligned} l'(x) &= \lim_{h \rightarrow 0} \frac{l(x+h) - l(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) \pm g(x+h) - f(x) \mp g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \pm \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\ &= f'(x) \pm g'(x) \end{aligned}$$

THEOREM | The Product Rule

If f and g are both differentiable, then

$$(f(x) * g(x))' = f'(x) * g(x) + f(x) * g'(x)$$

Proof |

Let $h(x) = f(x) * g(x)$. Then,

$$h'(x) = \lim_{h \rightarrow 0} \frac{h(x+h) - h(x)}{h}$$

Substituting $h(x) = f(x) * g(x)$, we get:

$$h'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) * g(x+h) - f(x) * g(x)}{h}$$

We can add and subtract the same value, so lets add and subtract $f(x+h)g(x)$.

$$h'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) * g(x+h) - f(x+h) * g(x) + f(x+h) * g(x) - f(x) * g(x)}{h}$$

Factoring out $g(x)$, we get:

$$h'(x) = \lim_{h \rightarrow 0} \left(g(x) * \frac{f(x+h) - f(x)}{h} + f(x) * \frac{g(x+h) - g(x)}{h} \right)$$

Since $f(x)$ and $g(x)$ have no relation with h , we can pull it out of the limit using the constant multiple rule.

$$h'(x) = g(x) * \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + f(x) * \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}$$

Using the definition of the derivative, we get:

$$h'(x) = g(x) * f'(x) + f(x) * g'(x)$$

THEOREM | The Quotient Rule

If f and g are both differentiable, then

$$\left(\frac{f(x)}{g(x)} \right)' = \frac{f'(x) * g(x) - f(x) * g'(x)}{g(x)^2}$$

Proof |

Let $h(x) = \frac{f(x)}{g(x)}$. Then,

$$h'(x) = \lim_{h \rightarrow 0} \frac{h(x+h) - h(x)}{h}$$

Substituting $h(x) = \frac{f(x)}{g(x)}$, we get:

$$\begin{aligned} h'(x) &= \lim_{h \rightarrow 0} \frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x) - f(x)g(x+h)}{hg(x)g(x+h)} \end{aligned}$$

Now lets add and subtract $f(x)g(x)$ to the numerator.

$$h'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)g(x) - f(x)g(x+h) + f(x)g(x) - f(x)g(x)}{hg(x)g(x+h)}$$

Group into useful groups,

$$\begin{aligned} h'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x) - f(x)g(x+h) + f(x)g(x) - f(x)g(x)}{hg(x)g(x+h)} \\ &= \lim_{h \rightarrow 0} \frac{g(x) \frac{f(x+h)-f(x)}{h} + \frac{f(x)g(x)-f(x)g(x+h)}{h}}{g(x)g(x+h)} \\ &= \frac{g(x)f'(x) - f(x)g'(x)}{g(x)^2} \end{aligned}$$

in the end, the full list of differntial formulas is:

$$\frac{d}{dx}(c) = 0$$

,

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

,

$$(c * f(x))' = c * f'(x)$$

,

$$(f(x) \pm g(x))' = f'(x) \pm g'(x)$$

,

$$(f(x) * g(x))' = f'(x) * g(x) + f(x) * g'(x)$$

,

$$\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x) * g(x) - f(x) * g'(x)}{g(x)^2}$$

Derivative of Trigonometric Functions

1 Trigonometric Functions...

We dealt with normal polynomials in the last section, so now we need to deal with trigonometric functions.

DEFINITION | Derivative of Sine Function

If $f(x) = \sin(x)$, then $f'(x) = \cos(x)$.

Proof |

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sin(x)\cos(h) + \cos(x)\sin(h) - \sin(x)}{h} \\
 &= \lim_{h \rightarrow 0} \left[\frac{\sin(x)\cos(h) - \sin(x)}{h} + \frac{\cos(x)\sin(h)}{h} \right] \\
 &= \lim_{h \rightarrow 0} \left[\sin(x) \left(\frac{\cos(h) - 1}{h} \right) + \cos(x) \left(\frac{\sin(h)}{h} \right) \right] \\
 &= \lim_{h \rightarrow 0} \sin(x) \cdot \lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} + \lim_{h \rightarrow 0} \cos(x) \cdot \lim_{h \rightarrow 0} \frac{\sin(h)}{h}
 \end{aligned}$$

Since we treat x as a constant during limits with h , we can take $\sin(x)$ and $\cos(x)$ out of the limits.

$$= \sin(x) \cdot \lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} + \cos(x) \cdot \lim_{h \rightarrow 0} \frac{\sin(h)}{h}$$

We'll later prove that $\lim_{h \rightarrow 0} \frac{\sin(h)}{h} = 1$, and $\lim_{h \rightarrow 0} \frac{\cos(h)-1}{h} = 0$.

Thus, we get $\cos(x)$.

Using the exact same method, we can prove the derivative of cosine.

DEFINITION | Derivative of Cosine Function

If $f(x) = \cos(x)$, then $f'(x) = -\sin(x)$.

Proof |

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cos(x)\cos(h) - \sin(x)\sin(h) - \cos(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[\frac{\cos(x)\cos(h) - \cos(x)}{h} - \frac{\sin(x)\sin(h)}{h} \right] \\ &= \lim_{h \rightarrow 0} \left[\cos(x) \left(\frac{\cos(h) - 1}{h} \right) - \sin(x) \left(\frac{\sin(h)}{h} \right) \right] \\ &= \lim_{h \rightarrow 0} \cos(x) \cdot \lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} - \lim_{h \rightarrow 0} \sin(x) \cdot \lim_{h \rightarrow 0} \frac{\sin(h)}{h} \end{aligned}$$

Since we treat x as a constant during limits with h , we can take $\sin(x)$ and $\cos(x)$ out of the limits.

$$= \cos(x) \cdot \lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} - \sin(x) \cdot \lim_{h \rightarrow 0} \frac{\sin(h)}{h}$$

We'll later prove that $\lim_{h \rightarrow 0} \frac{\sin(h)}{h} = 1$, and $\lim_{h \rightarrow 0} \frac{\cos(h)-1}{h} = 0$.

Thus, we get $-\sin(x)$.

Since we know the derivatives of sine and cosine, we can easily think of the derivative of the tangent function, since $\tan(x) = \frac{\sin(x)}{\cos(x)}$.

DEFINITION | Derivative of Tangent Function

If $f(x) = \tan(x)$, then $f'(x) = \sec^2(x)$.

Proof |

Using the Quotient Rule :

$$\begin{aligned}
f'(x) &= \frac{\cos(x) \cdot \frac{d}{dx}[\sin(x)] - \sin(x) \cdot \frac{d}{dx}[\cos(x)]}{(\cos(x))^2} \\
&= \frac{\cos(x) \cdot \cos(x) - \sin(x) \cdot (-\sin(x))}{(\cos(x))^2} \\
&= \frac{\cos^2(x) + \sin^2(x)}{(\cos(x))^2} \\
&= \frac{1}{(\cos(x))^2} \\
&= \sec^2(x)
\end{aligned}$$

The derivatives of $\csc(x)$, $\sec(x)$, $\cot(x)$ can be derived using a similar method. In the end, we have the following results :

$$\frac{d}{dx}(\sin(x)) = \cos(x)$$

$$\frac{d}{dx}(\cos(x)) = -\sin(x)$$

$$\frac{d}{dx}(\tan(x)) = \sec^2(x)$$

$$\frac{d}{dx}(\csc(x)) = -\csc(x) \cot(x)$$

$$\frac{d}{dx}(\sec(x)) = \sec(x) \tan(x)$$

$$\frac{d}{dx}(\cot(x)) = -\csc^2(x)$$

EXAMPLE | Trigonometric Derivative Example

An object fastened to the end of a vertical string stretched 4cm from rest is released at time $t = 0$. Its position at time t is

$$s = f(t) = 4 \cos(t)$$

Find the velocity and acceleration of the object at time t .

Solution 1 |

The velocity is the derivative of the position function:

$$v(t) = f'(t) = \frac{d}{dt}[4 \cos(t)] = -4 \sin(t)$$

The acceleration is the derivative of the velocity function:

$$a(t) = v'(t) = \frac{d}{dt}[-4 \sin(t)] = -4 \cos(t)$$

2 Special Trigonometric Limits

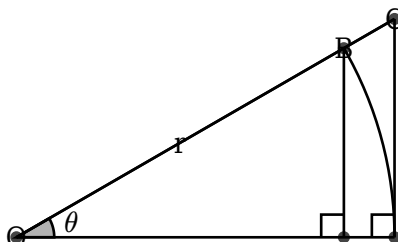
- There are two special trigonometric limits that are very useful in calculus.

THEOREM | Special Trigonometric Limits

1.
$$\lim_{x \rightarrow 0} \left(\frac{\sin(x)}{x} \right) = 1$$
2.
$$\lim_{x \rightarrow 0} \left(\frac{1 - \cos(x)}{x} \right) = 0$$

Proof | 1

We will utilize the Squeeze Theorem to prove this.



Here, we know that the inequality

$$\overline{BD} \leq \widehat{AB} \leq \overline{AC}$$

If we take the limit $\theta \rightarrow 0$, we get

$$\lim_{\theta \rightarrow 0} \overline{BD} \leq \lim_{\theta \rightarrow 0} \widehat{AB} \leq \lim_{\theta \rightarrow 0} \overline{AC}$$

$$\lim_{\theta \rightarrow 0} \overline{OB} \sin(\theta) \leq \lim_{\theta \rightarrow 0} r\theta \leq \lim_{\theta \rightarrow 0} \overline{OA} \tan(\theta)$$

Divide both sides by $\sin(\theta)$.

$$= \lim_{\theta \rightarrow 0} \cancel{x} \leq \lim_{\theta \rightarrow 0} \frac{\cancel{x}\theta}{\sin(\theta)} \leq \lim_{\theta \rightarrow 0} \frac{\cancel{x}}{\cos(\theta)}$$

Take the inverse of each fraction.

$$\lim_{\theta \rightarrow 0} \cos(\theta) \leq \lim_{\theta \rightarrow 0} \frac{\sin(\theta)}{\theta} \leq \lim_{\theta \rightarrow 0} 1$$

Since $\lim_{\theta \rightarrow 0} \cos(\theta) = 1$, we get

$$1 \leq \lim_{\theta \rightarrow 0} \frac{\sin(\theta)}{\theta} \leq 1$$

By the Squeeze Theorem, we get

$$\lim_{\theta \rightarrow 0} \frac{\sin(\theta)}{\theta} = 1$$

Proof | 2

Multiply the numerator and denominator by $\cos(\theta) + 1$ to turn the numerator into a nice form.

$$\begin{aligned} \lim_{\theta \rightarrow 0} \frac{\cos(\theta) - 1}{\theta} &= \lim_{\theta \rightarrow 0} \frac{\cancel{\cos(\theta)^2} - 1 - \sin(\theta)^2}{\theta(\cos(\theta) + 1)} \\ &= - \lim_{\theta \rightarrow 0} \frac{\sin(\theta)}{\theta} \cdot \lim_{\theta \rightarrow 0} \frac{\sin(\theta)}{\cos(\theta) + 1} \\ &= -1 * 0 = 0 \end{aligned}$$

Chapter 02.05

The Chain Rule

1 The Chain Rule

Say we want to differentiate $f(x) = \sqrt{\sin(x)}$. How do we do this? The chain rule is a powerful tool to solve the derivative of composite functions.

DEFINITION | Chain Rule

If $f(x) = a(b(x))$, then $f'(x) = a'(b(x))b'(x)$.

Proof |

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{a(b(x+h)) - a(b(x))}{h} \\ &= \lim_{h \rightarrow 0} \frac{a(b(x+h)) - a(b(x))}{b(x+h) - b(x)} \cdot \frac{b(x+h) - b(x)}{h} \\ &= a'(b(x)) \cdot b'(x) \end{aligned}$$

However, if $b(x)$ is a constant function, or for sake has a place where the denominator is 0, it directly leads to a divide by zero error. Thus, people found a better method for that :

Proof | Foolproof

We must know a small theorem before we continue :

THEOREM |

If a function is differentiable, the actual change is equal to the linear approximation at the point plus a small error.

$$\Delta y = \underbrace{f'(a)\Delta x}_{\text{Linear Projection}} + \underbrace{\varepsilon\Delta x}_{\text{Error Term}}$$

Let's utilize this. Let $b = g(a)$. If we have $u = g(x)$ and $y = f(u)$, we can derive the two equations :

$$\Delta u = g'(a)\Delta x + \varepsilon_1\Delta x = [g'(a) + \varepsilon_1]\Delta x$$

$$\Delta y = f'(b)\Delta u + \varepsilon_2\Delta u = [f'(b) + \varepsilon_2]\Delta u$$

Plug in Δu to the lower equation, we get

$$\frac{\Delta y}{\Delta x} = [f'(b) + \varepsilon_2][g'(a) + \varepsilon_1]\Delta x$$

Now if we take the limit of both sides, we get :

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} [f'(b) + \varepsilon_2][g'(a) + \varepsilon_1]$$

Since the error will near zero once $\lim_{\Delta x \rightarrow 0}$, we get $f'(b)g'(a) = f'(g(a))g'(a)$.

NOTE | Leibniz Notation

If you prefer the leibniz notation, you can write

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

.

EXAMPLE | Chain Rule Basics

Find $F'(x)$ if $F = \sqrt{x^2 + 1}$.

Solution 1 |

Using the chain rule, we get $F'(x) = \left(\frac{1}{2}\right)(x^2 + 1)^{-\frac{1}{2}} \cdot 2x = \frac{x}{\sqrt{x^2 + 1}}$

EXAMPLE | Chain Rule with Trigonometric Functions

Find $F'(x)$ if $F = \sin(\sqrt{x})$.

Solution 1 |

Using the chain rule, we get $F'(x) = \cos(\sqrt{x}) \cdot \left(\frac{1}{2}\right)x^{-\frac{1}{2}} = \frac{\cos(\sqrt{x})}{2\sqrt{x}}$.

One useful form to note is the Power Rule combined with the Chain Rule :

DEFINITION | Power Rule with Chain Rule

If $f(x) = (g(x))^n$, then $f'(x) = n(g(x))^{n-1}g'(x)$.

EXAMPLE | Power Rule with Chain Rule

Differentiate $y = (x^3 - 1)^{100}$

Solution 1 |

Using the power rule with chain rule, we get

$$y' = 100(x^3 - 1)^{99} \cdot 3x^2 = 300x^2(x^3 - 1)^{99}$$

Chapter 02.06

Implicit Differentiation

1 Implicit Function Differentiation

Implicit functions are functions that don't take the form of $y = f(x)$. Some examples include $x^2 + y^2 = 1$ and $\frac{x^2}{4} - \frac{y^2}{9} = 1$.

How do we find the derivative of these functions? We can use a very cool method called **Implicit Differentiation**.

DEFINITION | Implicit Differentiation

Implicit Differentiation means to take $\frac{d}{dx}$ of both sides in order to relieve the final result in the form of $\frac{dy}{dx}$.

EXAMPLE | Implicit Differentiation

Find $\frac{dy}{dx}$ if $x^2 + y^2 = 25$. Then find the equation of the tangent to the circle $x^2 + y^2 = 25$ at point $(3, 4)$.

Solution 1 |

Take $\frac{d}{dx}$ on both sides.

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}25$$

$$\frac{d}{dx}x^2 + \frac{d}{dx}y^2 = 0$$

Since y is a function of x , we can use the chain rule.

$$2x + 2\frac{y(dy)}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

Therefore, at the point $(3, 4)$, we get $\frac{dy}{dx} = -\frac{3}{4}$.

2 Second derivatives

EXAMPLE | Second Derivative of Implicit Functions

Find y'' if $x^4 + y^4 = 16$.

Solution 1 |

Differentiate once to get $4x^3 + 4y^3y' = 0$. Solving for y' gives $y' = -\frac{x^3}{y^3}$.
Differentiate again to get

$$\begin{aligned}y'' &= \frac{d}{dx} \left(-\frac{x^3}{y^3} \right) = -\frac{y^3 \frac{d}{dx} x^3 - x^3 \frac{d}{dx} y^3}{y^6} \\&= -\frac{y^3 \cdot 3x^2 - x^3 \cdot 3y^2 y'}{y^6}\end{aligned}$$

this can be simplified into

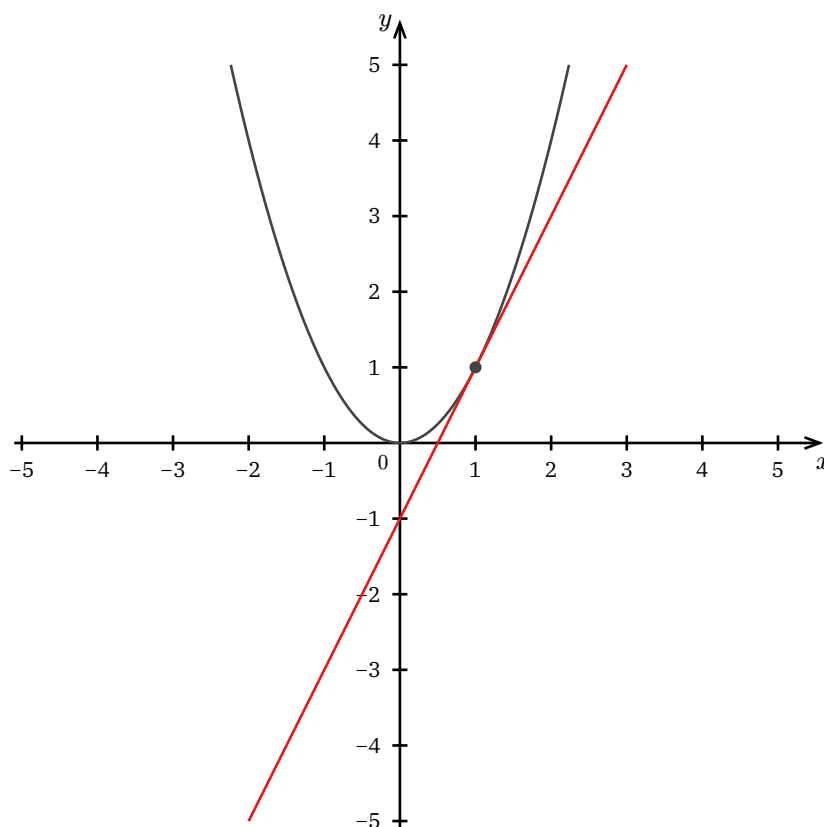
$$y'' = -48 \frac{x^2}{y^7}$$

Rates of Change in Science and Society

1 Linear approximation

Remember our proof of the **chain rule** a few sections back? That was our preview of the topic of **linear approximation**. We can define a tangent line at a point $(a, f(a))$ as

$$L(x) = f(a) + f'(a)(x - a)$$



We can use this linearized function to predict a value near a . For example, if we want to find $f(1.1)$, we can use the linearized function to predict $f(1.1) = 2.1$

DEFINITION | Linear Approximation

If $f(x)$ is differentiable at a , then the linear approximation of $f(x)$ at a is $L(x) = f(a) + f'(a)(x - a)$. We can use this to predict $f(x)$ near a .

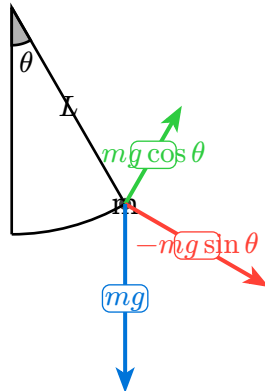
$$f(x) \approx L(x) = f(a) + f'(a)(x - a)$$

So where is this used? It is a famous thing that $\sin(\theta) \approx \theta$ near 0. This is used numerous for physics problems, such as finding the period of a pendulum.

EXAMPLE | Period of a Pendulum

Find the period of a pendulum with length L .

Solution 1 |



The equation of motion for a pendulum is derived from Newton's second law:

$$mL \frac{d^2\theta}{dt^2} = -mg \sin(\theta)$$

This simplifies to:

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L} \sin(\theta)$$

For small angles, we use the linear approximation $\sin(\theta) \approx \theta$ (since $\sin'(0) = 1$):

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L} \theta$$

This matches the standard form of simple harmonic motion:

$$\frac{d^2\theta}{dt^2} = -\omega^2 \theta$$

Comparing the two equations, we identify:

$$\omega^2 = \frac{g}{L} \Rightarrow \omega = \sqrt{\frac{g}{L}}$$

The period is therefore:

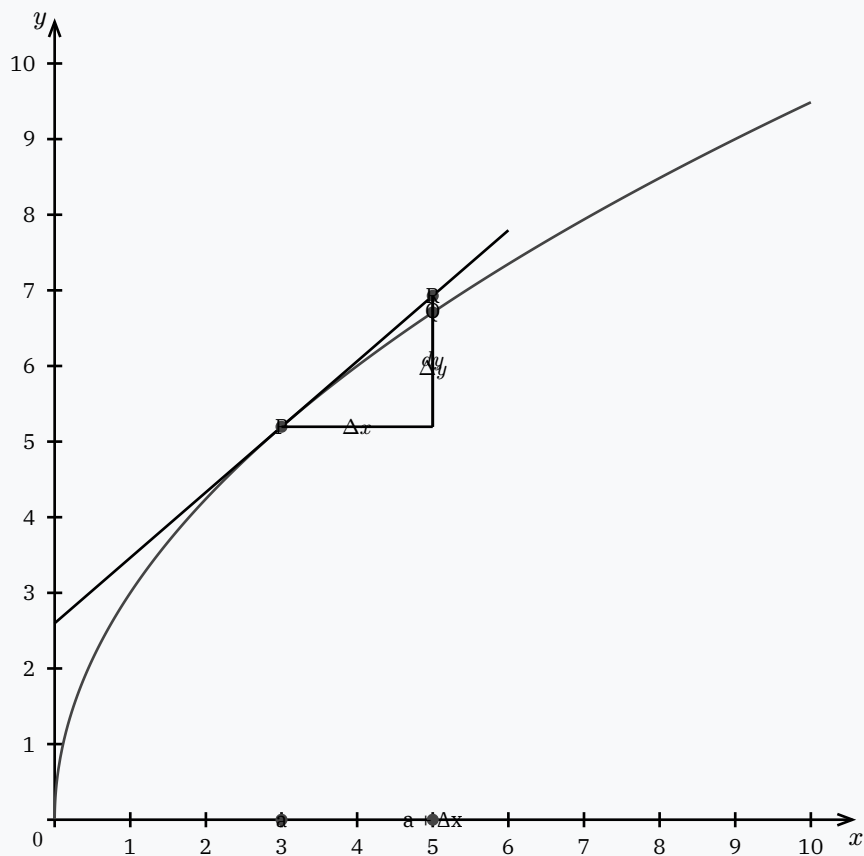
$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{L}{g}}$$

2 Differentials

- We dealt with $\frac{dy}{dx}$ a few times before. However, we can actually use this to obtain dy and dx .

DEFINITION | Differential

If $f(x)$ is differentiable at x , then $dy = f'(x)dx$.

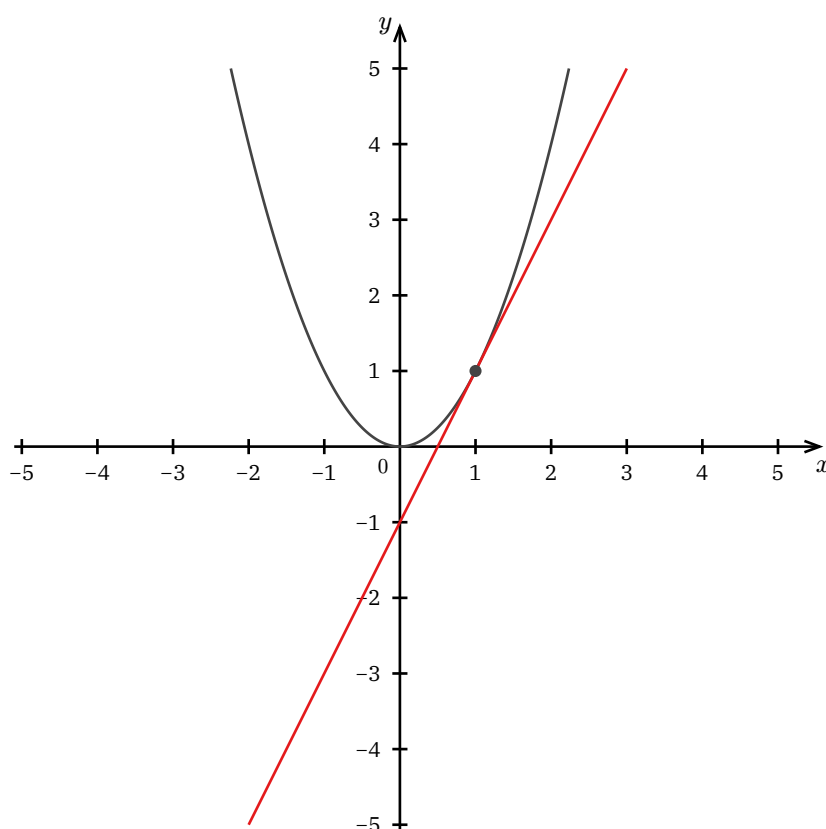


Linear Approximation & Differentials

1 Linear approximation

Remember our proof of the **chain rule** a few sections back? That was our preview of the topic of **linear approximation**. We can define a tangent line at a point $(a, f(a))$ as

$$L(x) = f(a) + f'(a)(x - a)$$



We can use this linearized function to predict a value near a . For example, if we want to find $f(1.1)$, we can use the linearized function to predict $f(1.1) = 2.1$

DEFINITION | Linear Approximation

If $f(x)$ is differentiable at a , then the linear approximation of $f(x)$ at a is $L(x) = f(a) + f'(a)(x - a)$. We can use this to predict $f(x)$ near a .

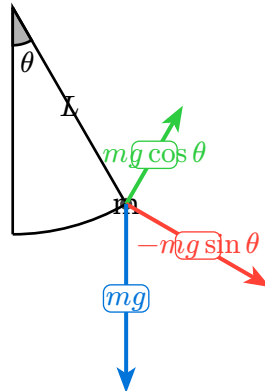
$$f(x) \approx L(x) = f(a) + f'(a)(x - a)$$

So where is this used? It is a famous thing that $\sin(\theta) \approx \theta$ near 0. This is used numerous for physics problems, such as finding the period of a pendulum.

EXAMPLE | Period of a Pendulum

Find the period of a pendulum with length L .

Solution 1 |



The equation of motion for a pendulum is derived from Newton's second law:

$$mL \frac{d^2\theta}{dt^2} = -mg \sin(\theta)$$

This simplifies to:

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L} \sin(\theta)$$

For small angles, we use the linear approximation $\sin(\theta) \approx \theta$ (since $\sin'(0) = 1$):

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L} \theta$$

This matches the standard form of simple harmonic motion:

$$\frac{d^2\theta}{dt^2} = -\omega^2 \theta$$

Comparing the two equations, we identify:

$$\omega^2 = \frac{g}{L} \Rightarrow \omega = \sqrt{\frac{g}{L}}$$

The period is therefore:

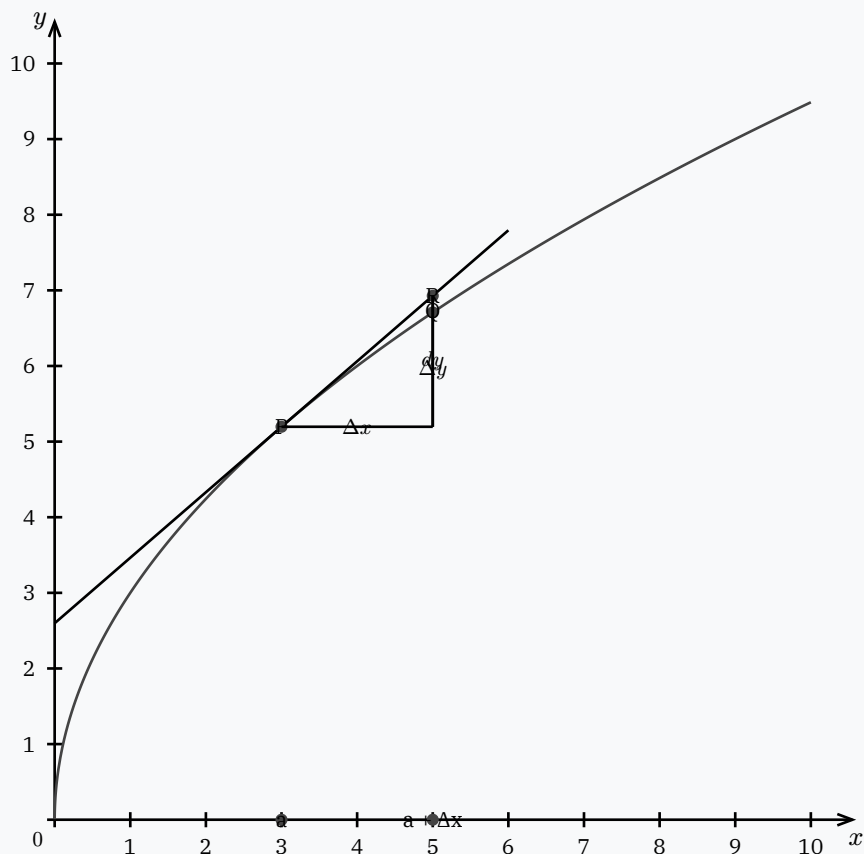
$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{L}{g}}$$

2 Differentials

- We dealt with $\frac{dy}{dx}$ a few times before. However, we can actually use this to obtain dy and dx .

DEFINITION | Differential

If $f(x)$ is differentiable at x , then $dy = f'(x)dx$.



Chapter 03

Applications of Differentiation

Here the derivative stops being a formula and starts being an instrument. The Extreme Value Theorem and the Mean Value Theorem provide the theory; increasing-decreasing behavior, concavity, and asymptotes provide the vocabulary for reading the shape of any graph. The chapter culminates in systematic curve sketching, optimization problems that turn geometry and physics into calculus, Newton's method for hunting roots numerically, and a first look at antiderivatives, the doorway into integration.

Maximum and Minimum Values

1 Optimizaition problems

- Say we have velocity graph of a car moving. In a logical scenario, we might want to find **the maximum/minimum acceleration of the car**. How can we do this?

DEFINITION | Local&Absolute Maxima&minima

Let c be a number in the domain D of a function f . Then, $f(c)$ is the :

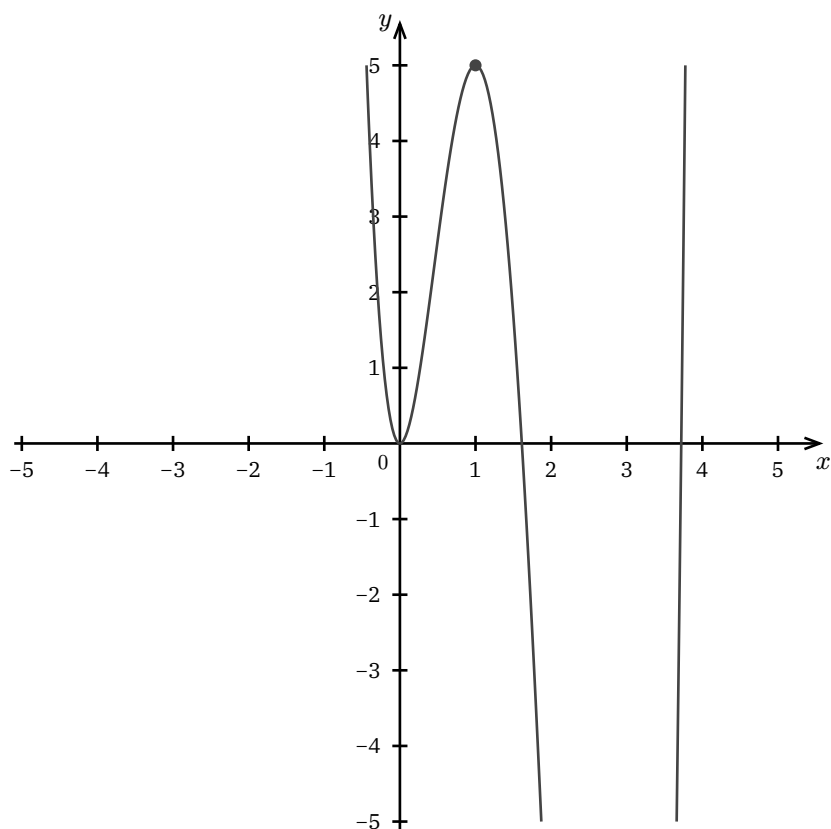
- **Absolute maximum** if $f(c) \geq f(x)$ for all x in D .
- **Absolute minimum** if $f(c) \leq f(x)$ for all x in D .
- **Local maximum** if $f(c) \geq f(x)$ for x near c .
- **Local minimum** if $f(c) \leq f(x)$ for x near c .

EXAMPLE | Local Maximum of a Function

Take the function $f(x) = 3x^4 - 16x^3 + 18x^2$ Near the point at $x = 1$, what is the local maxima?

Solution 1 |

We get the following graph when drawn :



Thus we can know that the local maxima near point 1 is 5.

You may notice that if a function is continuous and limited to a certain closed region, an absolute minima and maxima exists in the region.

THEOREM | Extreme Value Theorem

Inside a closed region $[a, b]$, if the function is continuous within the region, the function has an absolute maxima and minima values c and d .

Now, let's consider the meaning of a "local maxima". At the local maxima, any point near the local maxima (for simplicity let's assume it is at a) is always smaller than $f(a)$. Thus, we can tell that the differentiated function changes signs at $x \approx a$. From this, we can deduce a very cool and useful theorem;

THEOREM | Fermat's Theorem

If f has a local maximum or minimum at c , and if $f'(c)$ exists, we can find that $f'(c) = 0$.

Proof | Intuition

This can be proven easily using the logic above plus the intermediate value theorem.

Proof | Definition of Derivatives

Let's suppose that $f(x)$ has a local maxima at $x = c$. This implies that if h is sufficiently near 0, we can find the following.

$$f(c) \geq f(c + h)$$

If we simplify the inequality, we get

$$\frac{f(c + h) - f(c)}{h} \leq 0$$

Take the right hand limit of both sides, we get

$$\lim_{h \rightarrow 0^+} \frac{f(c + h) - f(c)}{h} \leq \lim_{h \rightarrow 0^+} 0 = 0$$

Since $f'(c)$ exists, we can define LHS as $f'(c)$.

$$f'(c) \leq 0$$

If we do the same for the left hand side, we get

$$f'(c) \geq 0$$

Therefore, $f'(c) = 0$.

From this, we can get a number of points that is of our interest in a certain area.

DEFINITION | Critical Numbers

A critical number of a function f is a number c in the domain of f that is either $f'(c) = 0$ or $f'(c)$ does not exist.

EXAMPLE | Critical Numbers Example

For the function

$$f(x) = x - 2 \sin(x) \quad x \leq x \leq 2\pi$$

Use calculus to find the exact minimum and maximum values.

Solution 1 |

since $f(x)$ is continuous within the given interval, we can deduce the fact that critical values exist on the points where $f'(x) = 0$ or $f'(x)$ doesn't exist. Since $f'(x) = 1 - 2\cos(x)$, we have $f'(x) = 0$ when $\cos(x) = \frac{1}{2}$. This is at $x = \frac{\pi}{3}$ or $x = \frac{5\pi}{3}$. Since the endpoints are discontinuous, $f(0)$ and $f(2\pi)$ are both critical values.

In the end, we can find that $x_c = 0, \frac{\pi}{3}, \frac{5\pi}{3}, 2\pi$

Chapter 03.02

The Mean Value Theorem

1 Rolle's Theorem

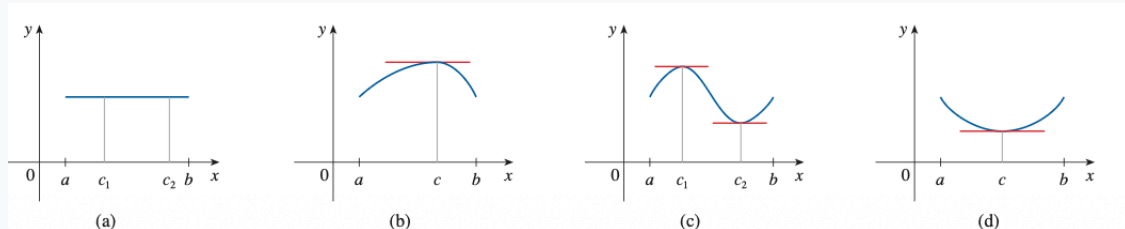
Let us define a simple theorem before we go into the mean value theorem.

THEOREM | Rolle's Theorem

Let f be a function that suffices the following :

1. $f(x)$ is continuous on the closed interval $[a, b]$.
2. $f(x)$ is differentiable on the open interval (a, b) .
3. $f(a) = f(b)$

Then there is a number c in (a, b) such that $f'(c) = 0$.



Proof | Case I

$f'(x) = 0$ for all $x \in (a, b)$.

Proof | Case II

$f(x) > f(a)$ for some point $x \in (a, b)$.

By the Extreme Value Theorem, $f(x)$ has a maximum somewhere inside (a, b) . Thus $f(x)$ has a local maximum at c , and therefore $f'(c) = 0$ by Fermat's Theorem.

Proof | Case III

$f(x) < f(a)$ for some point $x \in (a, b)$.

By the Extreme Value Theorem, $f(x)$ has a minimum somewhere inside (a, b) . Thus $f(x)$ has a local minimum at c , and therefore $f'(c) = 0$ by Fermat's Theorem.

From this, we can derive the **Mean Value Theorem**.

THEOREM | Mean Value Theorem

If f is a function that satisfies the following hypotheses :

1. f is continuous on $[a, b]$
2. f is differentiable on (a, b)

Then, there is a number c that suffices

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

or

$$f(b) - f(a) = f'(c)(b - a)$$

Of course, there could be multiple c s.

By intuition, we can know that this works for all cases, where the function is concave, convex, or a mixed form. However, what if the function is a perfect line? We can't talk concavity on that.

THEOREM | Lines and Constant functions

If $f'(x) = 0$ for all x in (a, b) , f is constant on (a, b) .

Also, if $f'(x) = g'(x)$ for all x in interval (a, b) , then $f(x) = g(x) + c$.

Derivatives & The Shape of a Graph

1 Some Tests we can do

Recall that a derivative($f'(x)$) displays the literal “slope” at a certain part of the graph. Using this, we can infer a few basic tests :

NOTE | Increasing/Decreasing Test (abb. I/D Test)

1. If $f'(x) > 0$ on an interval, then f increases.
2. If $f'(x) < 0$ on an interval, then f decreases.
3. Since the points of interest are when $f'(x) = 0$, we can easily divide and estimate the shape of the function via splitting the function on it's critical points.

From (3) at the note above, we can estimate the “type” of critical point we are dealing with.

THEOREM | The First Derivative Test

Suppose c is a critical number for a continuous function f .

1. If f' goes from $+$ to $-$ at c , c is a local maxima of f .
2. If f' goes from $-$ to $+$ at c , c is a local minimum of f .
3. If f' 's sign does not change, c is not a local maxima/minima.

NOTE | Second Derivatives

We can define this using second derivatives too.

1. If $f''(c) < 0$, c is a local maxima.
2. If $f''(c) > 0$, c is a local minima.
3. If $f''(c) = 0$, c is neither. We call it *inconclusive*.

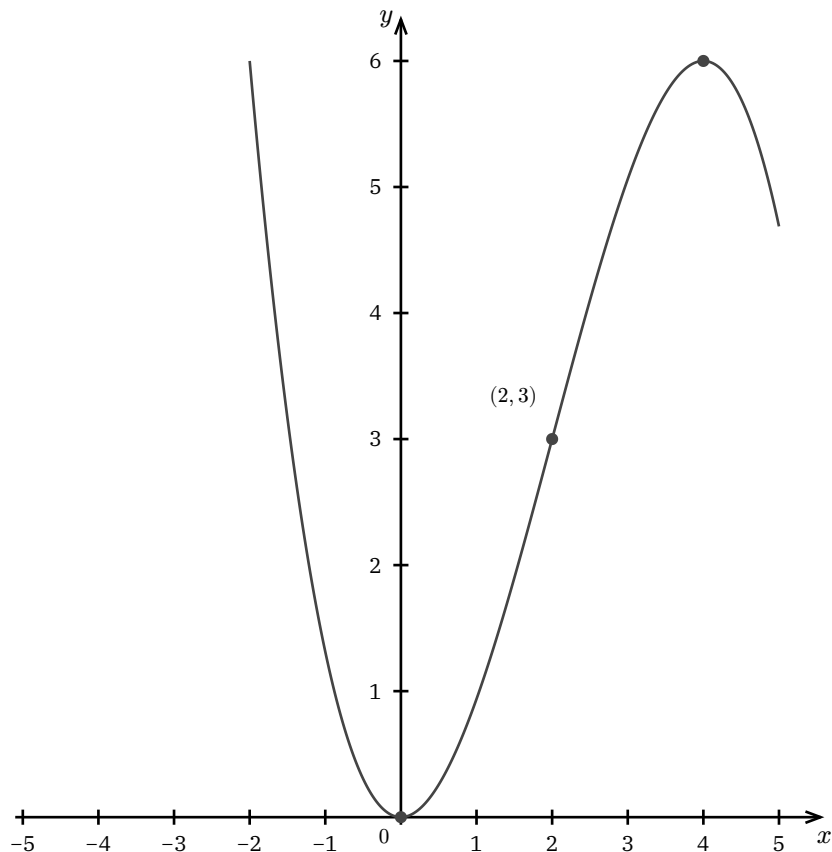
If not at a critical point, the second derivative can also describe the concavity of a function.

DEFINITION | Concave

A function is **concave upward** on an interval if all tangents of f in that region lie below f . A function is **concave downward** on an interval if all tangents of f in that region lie above f .

We can test this easily by looking at the sign of the second derivative of $f(f'')$. If $f''(x) > 0$ in the interval $\mathbb{I}$, then the graph of f is concave upward and vice versa.

But consider functions like the following.



Note that the concavity inverts at the point $(2, 3)$. We call such a point an **inflection point**.

DEFINITION | Inflection Point

An **inflection point** is a point where the curve changes from concave upward to concave downward.

2 Curve Sketching

Using the info we gathered in the tests above, we can sketch the basic shape of an arbitrary function using basic calculus.

NOTE | The Steps to Sketching a Function

1. Find Intervals
2. Inside those intervals, find increase/decrease
3. Inside those intervals, find concavity
4. Draw

Limits at Infinity:

Horizontal Asymptotes

1 Limits to Infinity

Lets consider a limit to infinity. we would probably note it like this :

$$\lim_{x \rightarrow \infty}$$

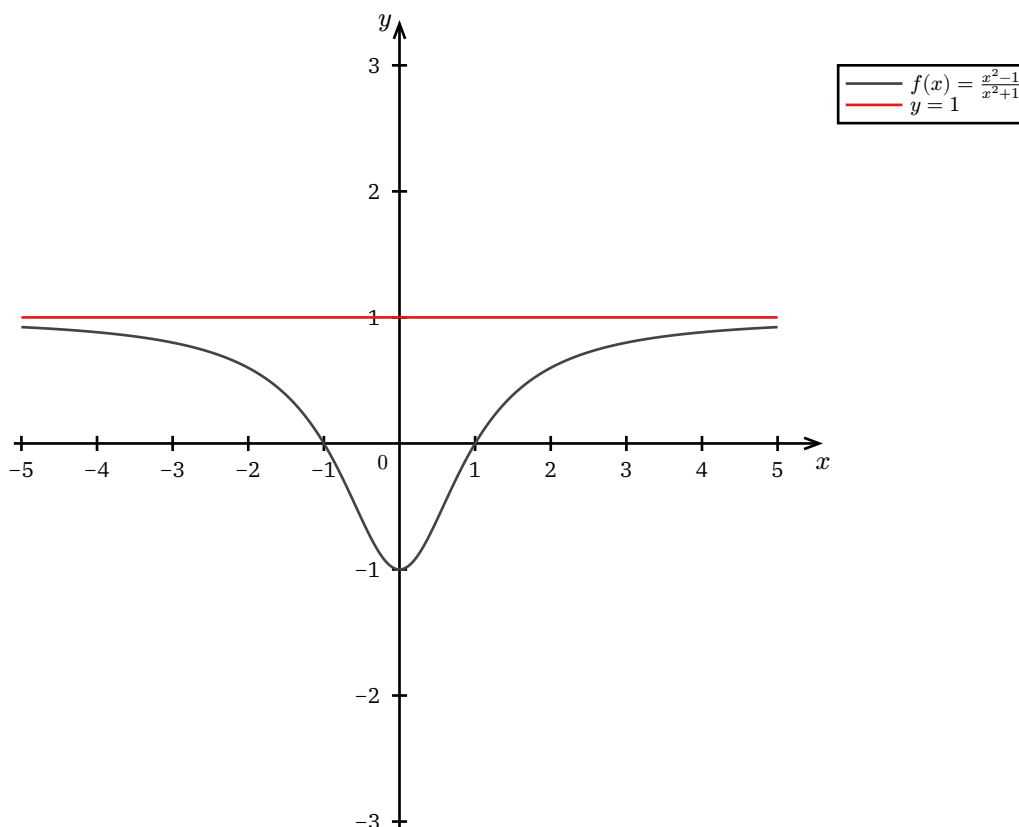
Which is quite intuitive on it's own. But what happens if we add this to a function?

EXAMPLE | Limits at Infinity

Consider the behavior of the function $f(x) = \frac{x^2-1}{x^2+1}$ when $\lim_{x \rightarrow \infty}$.

Solution 1 |

If we graph the function, we get the following :



from the graph, we can intuitively tell that the function will approach $y = 1$ indefinitely when $x \rightarrow \infty$.

As we did with normal limits, let us look at the intuitive definition of a limit first.

DEFINITION | Intuitive Definition of a Limit at Infinity

If a function f is defined on an arbitrary interval $(-\infty, \alpha)$, then

$$\lim_{x \rightarrow -\infty} f(x) = L$$

means that the value of $f(x)$ can be made arbitrarily close to L by making x a sufficiently large negative.

From this, we can define what we call a **horizontal asymptote**.

DEFINITION | Horizontal Asymptotes

We can define a horizontal asymptote if

$$\lim_{x \rightarrow -\infty} f(x) = L$$

or

$$\lim_{x \rightarrow \infty} f(x) = L$$

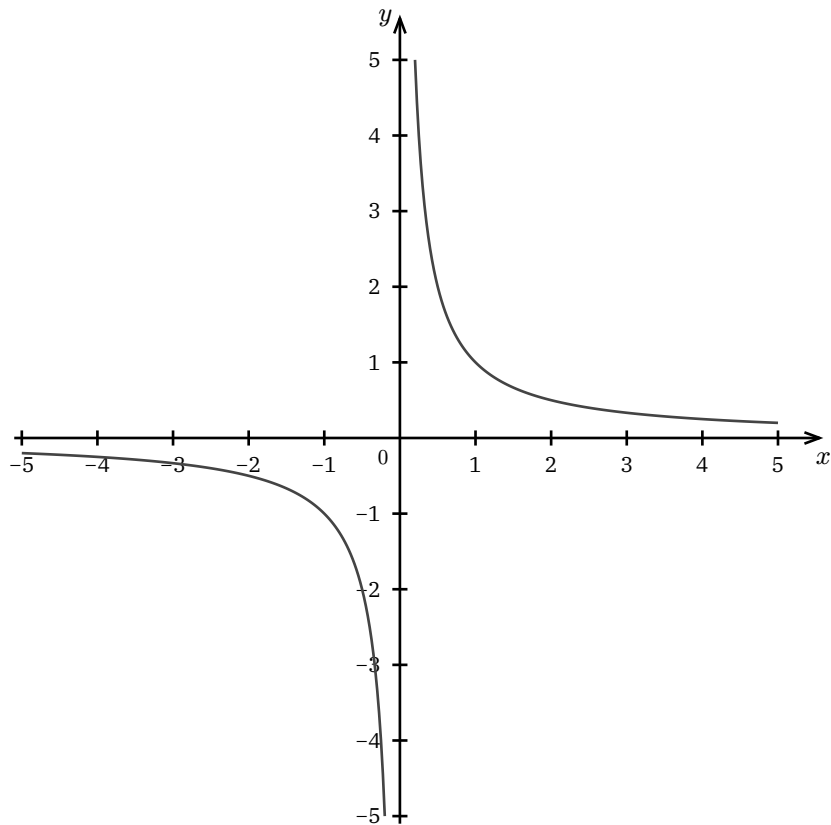
.

EXAMPLE | Infinite limits

Find

$$\lim_{x \rightarrow \infty} \frac{1}{x}$$

Solution 1 |



From the graph, we can tell that the function will approach 0 when going near $x \rightarrow \infty$.

In general, we can derive the following :

THEOREM | Limits to Infinity

If $r > 0$, functions in the form $f(x) = \frac{1}{x^r}$ limit to 0 when pushed to $\pm\infty$.

2 Infinite Limits at infinity

Consider a very basic function(eg. $y = x$). What happens when $\lim_{x \rightarrow \infty}$? y also goes to infinity. we call this **Infinite Limits at Infinity**.

DEFINITION | Infinite Limits at Infinity

we can write a limit as

$$\lim_{x \rightarrow \infty} f(x) = \infty$$

Since infinity itself is not a number, we cannot apply limit laws to this limit. Most questions ask for the sign of the infinite limit at infinity.

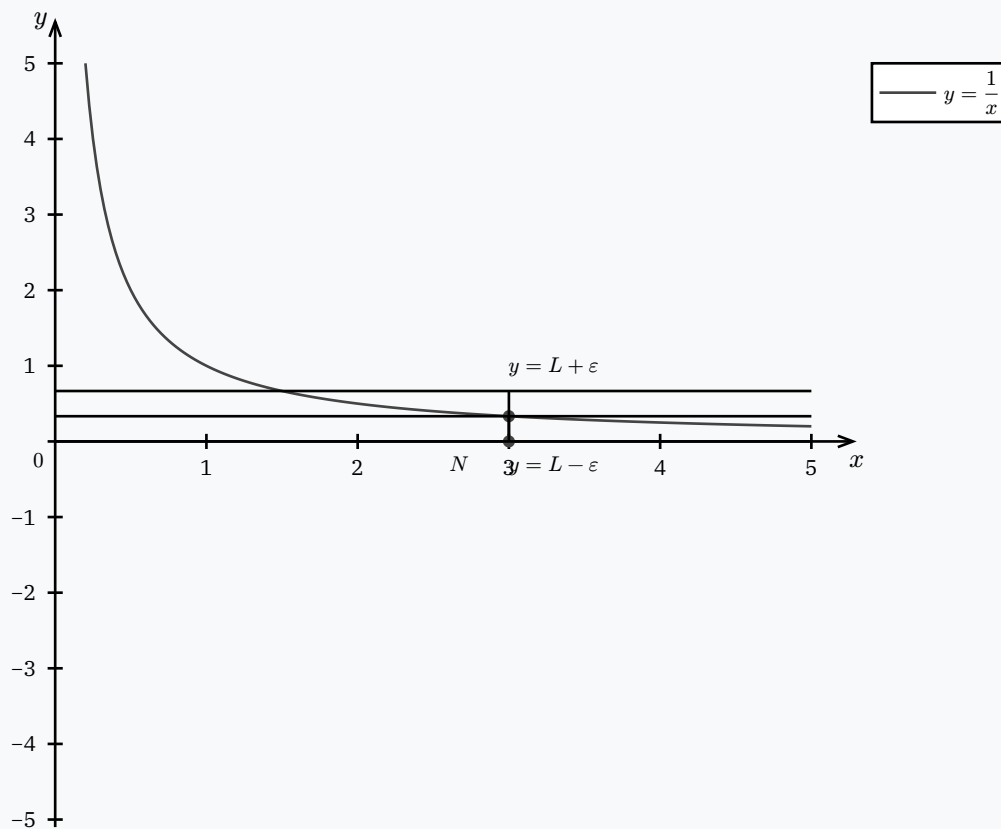
How do we write this in the $\varepsilon - \delta$ method? Since there is no “number” larger than ∞ , we only have to look at one side of the equation.

DEFINITION | Precise Definition of a Limit at Infinity

Let f be a function defined on some interval $(\sim, \infty)$. For any $\varepsilon > 0$, there is a corresponding number N s.t.

$$\text{if } x > N \quad \text{then} \quad |f(x) - L| < \varepsilon$$

As a graph, we can draw it like this :



DEFINITION | Precise Definition of an Infinite Limit at Infinity

Let f be a function defined on some interval $(\sim, \infty)$. For any M , there is a corresponding number N s.t.

$$\text{if } x > N \quad \text{then} \quad f(x) > M$$

Chapter 03.05

Summary of Curve Sketching

We consider a bunch of elements when sketching a function.

- critical points
- domain
- intersects
- the derivative
- the second derivative
- ...

We can define sort of a “always working” formula for sketching curves.

NOTE | Curve Sketching

1. Domain
2. Intercepts
3. Symmetry
 1. Periodic functions
4. Asymptotes
5. Intervals of increase/decrease
6. Local maximum & minimum (Critical points)
7. Concavity

With these things considered, we can much easily draw a curve.

EXAMPLE | Curve Sketching Basics (1)

Sketch the curve

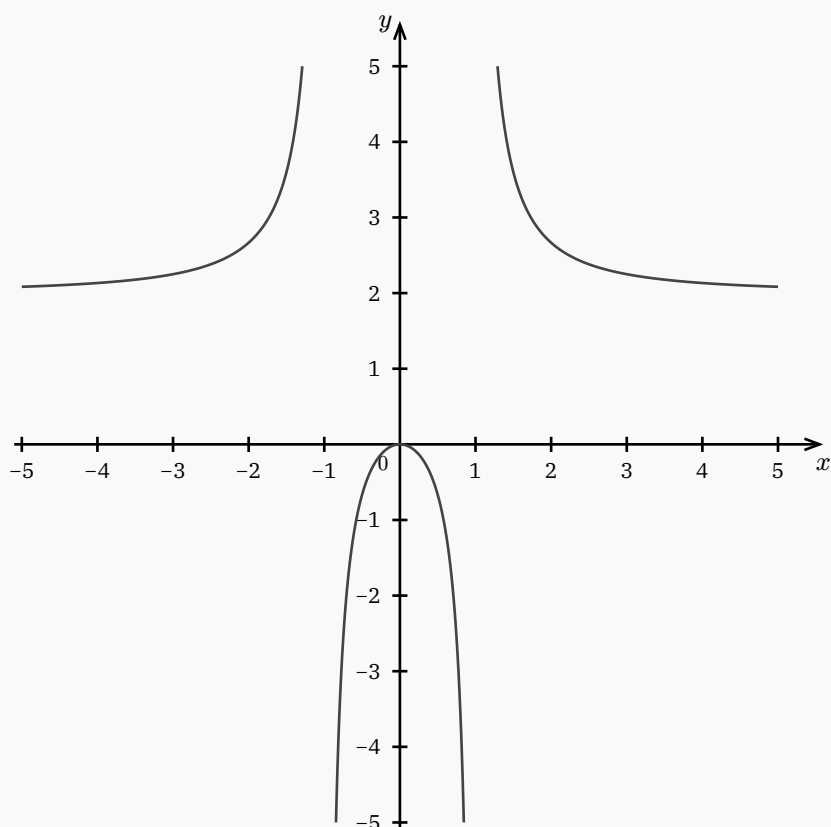
$$y = \frac{2x^2}{x^2 - 1}$$

Solution 1 |

1. The domain is $x \neq \pm 1$. In different form, $(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$.
2. Plug in $x = 0$ and $y = 0$. The function passes $(0, 0)$.

3. Consider its limits to its ends of the domains $(-\infty, -1, 1, \infty)$. We can see that it approaches 2 at $x = \pm\infty$ and $\pm\infty$ at $-1^+, -1^-, 1^+, 1^-$. These are the asymptotes.
4. Use the I/D test.
5. We can know that the critical numbers for this function lie at $-\infty, -1, 0, 1, \infty$.
6. Concavity can be determined by examining the second derivative.

We end up with a graph like this :



EXAMPLE | A Complete Portrait of a Cubic

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = x^3 - 3x^2 + 4$$

1. Find the intervals of increase or decrease of $f(x)$.
2. Find the local maximum and minimum values of $f(x)$.
3. Find the intervals of concavity and the points of inflection.
4. Sketch the graph of $f(x)$ using the information from (1) to (3).

Solution 1 |

(1) Increase and decrease. Differentiate and factor :

$$f'(x) = 3x^2 - 6x = 3x(x - 2)$$

The critical numbers are $x = 0$ and $x = 2$. Testing the sign of f' :

- on $(-\infty, 0) : f' > 0$, increasing
- on $(0, 2) : f' < 0$, decreasing
- on $(2, \infty) : f' > 0$, increasing

(2) Local extrema. By the First Derivative Test, f has a local maximum at $x = 0$ with

$$f(0) = 4$$

and a local minimum at $x = 2$ with

$$f(2) = 8 - 12 + 4 = 0$$

(3) Concavity. The second derivative is

$$f''(x) = 6x - 6$$

so f is concave downward on $(-\infty, 1)$ and concave upward on $(1, \infty)$, with an inflection point at

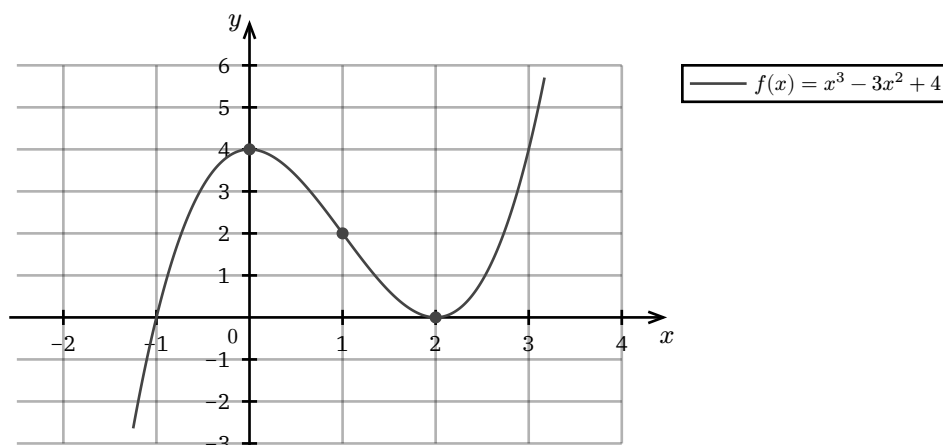
$$(1, f(1)) = (1, 2)$$

This point is exactly halfway between the two extrema, as always happens for a cubic.

(4) The sketch. One more free gift before drawing : the polynomial factors as

$$f(x) = (x + 1)(x - 2)^2$$

so the graph crosses the x -axis at $x = -1$ and **touches** it at $x = 2$, which is consistent with the local minimum value 0 found above.

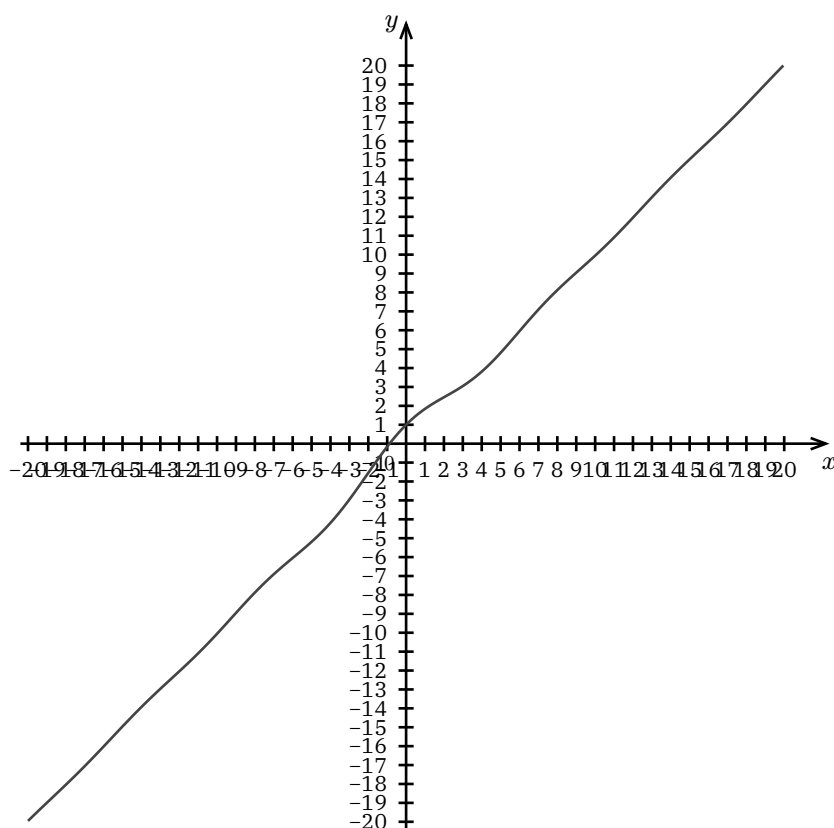


The three marked points (local maximum, inflection, local minimum) are the entire skeleton of the curve. Everything else is just connecting them with the right bends.

1 Slant Asymptotes

Till now we have only considered asymptotes in the form $x = \alpha$ and $y = \alpha$. But why can't it be in the form of $y = \alpha x + \beta$? Slant asymptotes is exactly that.

Consider the function $y = x + \frac{\sin(x)}{x}$. If we draw the function, we get the following :



This follows a “slanted asymptote”.

DEFINITION | Slanted Asymptotes

The line $y = mx + b$ is a slant asymptote of f if

$$\lim_{x \rightarrow \infty} [f(x) - (mx + b)] = 0$$

(or the same with $x \rightarrow -\infty$).

Graphing with Calculus & Tech

1 Graphing with calculus and technology

A graphing device draws thousands of points faster than we ever could, and then happily shows us the **wrong picture**. The screen only displays the window we ask for. Calculus is how we know what to ask for.

NOTE | The viewing window problem

Every plot lives inside a rectangle $[x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}]$. Choose it too small and the interesting features fall outside the frame. Choose it too large and they flatten into invisibility. The derivative tells us **where** the features are, so the window can be built around them.

EXAMPLE | A Cubic that Hides its Personality

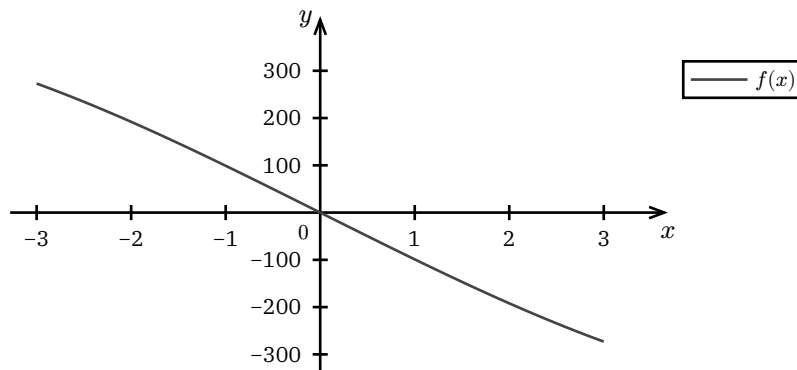
Graph the function

$$f(x) = x^3 - 100x$$

and choose a viewing window that shows all of its important features.

Solution 1 |

A naive first attempt : plot it on $-3 \leq x \leq 3$, the sort of default window every calculator loves.



The graph looks like a boring straight-ish line sloping downhill. No hint of a cubic at all, and nothing on the screen warns us that we are being deceived.

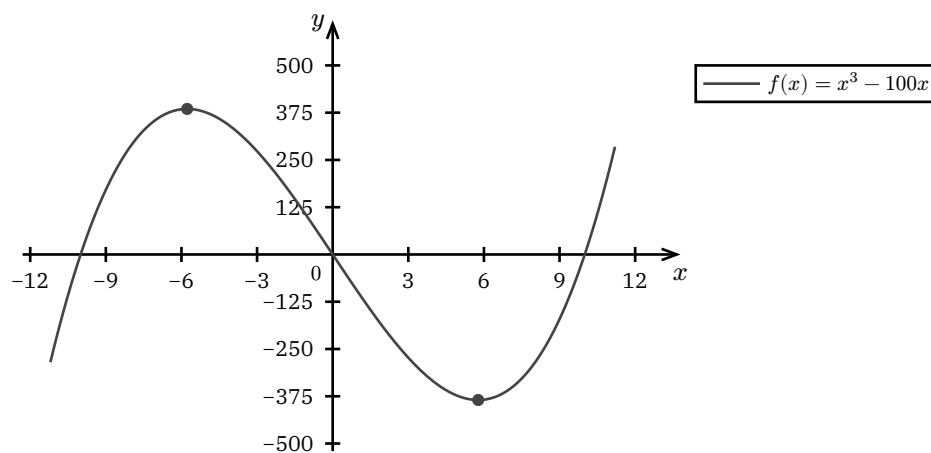
Now consult calculus instead of the screen :

$$f'(x) = 3x^2 - 100 = 0 \implies x = \pm \frac{10}{\sqrt{3}} \approx \pm 5.77$$

Both critical points lie **outside** the window we chose. Their values are

$$f\left(\pm \frac{10}{\sqrt{3}}\right) = \mp \frac{2000}{3\sqrt{3}} \approx \mp 385$$

So the honest window must reach past $x = \pm 5.77$ horizontally and past $y = \pm 385$ vertically :



Same function, same machine, but this frame was designed by the derivative, and now the two extrema, the inflection at the origin, and the three roots ($x = 0, \pm 10$) are all on display.

NOTE | The division of labor

Technology is superb at **drawing** and terrible at **knowing**. Calculus is the reverse. The workflow that actually functions :

1. Differentiate to locate critical points, inflection points, and asymptotes.
2. Build a window that contains them.
3. Let the machine draw.
4. If the picture contradicts the calculus, distrust the picture first : resolution, rounding, and sampling all lie.

EXAMPLE | When the Screen Cannot Win

What goes wrong if we ask a machine to graph

$$f(x) = \sin(100x)$$

on the interval $[0, 2\pi]$?

Solution 1 |

The function completes 100 full oscillations across the window. A screen a few hundred pixels wide gets roughly one or two pixels per period, so what it draws is **not the function** but an interference pattern between the true frequency and the pixel grid : the same aliasing that makes wheels spin backward on film.

Calculus knows better without drawing anything : the graph is an ordinary sine wave with amplitude exactly 1 and period

$$\frac{2\pi}{100} \approx 0.063$$

A trustworthy view needs a window only a few periods wide, such as $[0, 0.2]$, a window nobody would guess without first reading the formula.

NOTE | What this section is really about

Neither tool replaces the other. The machine supplies speed and precision ; calculus supplies the guarantee that nothing important is hiding off-screen. Use each one to audit the other.

Chapter 03.07

Optimization Problems

1 Overview

We have a lot of optimization problems in our daily lives.

- Minimize cost · time · labor etc...
- Maximize gain · area · distance etc...

These problems have been discussed in calculus, but we can examine them more in detail in this chapter.

NOTE | The basic steps to solving optimization problems

1. Understand the constraints
2. Draw a diagram
3. Introduce notation (Let Q be the objective function and $a_1 \dots a_n$ be notations for unknown quantities)
4. Express Q in $a_1 \dots a_n$
5. Find the absolute maximum/minimum of the graph $Q = f(a_1, \dots, a_n)$.

EXAMPLE | Optimization problem example

A farmer is trying to build a ranch near a river. One side of the river is taken care by the river. The farmer has a total of 1200m worth of fence. What is the maximum area the farmer can claim?

Solution 1 |

Let the rectangular side covered by the river be length x ($0 < x < 1200$). Then, our objective function Q becomes

$$\begin{aligned} Q &= x * \frac{1200 - x}{2} \\ &= x * \left(600 - \frac{x}{2}\right) \\ &= -\frac{1}{2}x^2 + 600x \end{aligned}$$

If we take the derivative of this function,

$$Q' = -x + 600$$

Investigating the point of $Q' = 0$ (Since the original function is a second-order quadratic function, we only have one critical point), we get $x = 600(m)$.

$$\therefore x = 600m, y = 300m \text{ with total area of } 180000m^2$$

Is there any simple way to find an absolute maximum/minimum, even if it is not a simple function like above? Of course there is.

THEOREM | First Derivative Test for Absolute Extreme Values

$$f'(x) > 0 \quad \forall x < c \quad \wedge \quad f'(x) < 0 \quad \forall x > c \implies f(c) = \text{absolute maximum}$$

$$f'(x) < 0 \quad \forall x < c \quad \wedge \quad f'(x) > 0 \quad \forall x > c \implies f(c) = \text{absolute minimum}$$

Chapter 03.08

Newton's Method

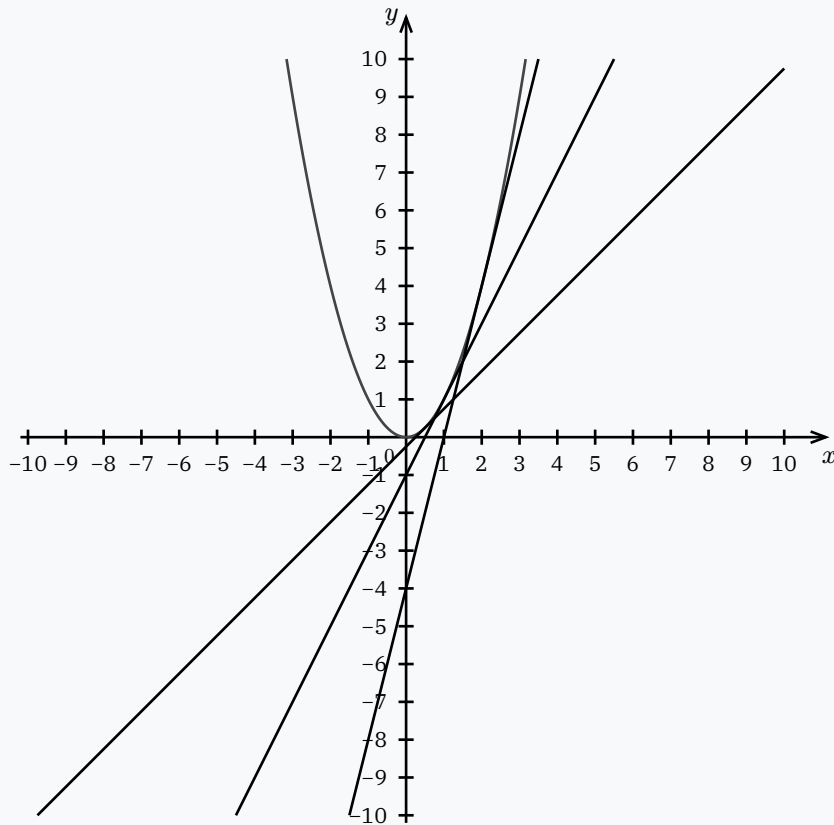
Say we have an equation, and we have to find the equation. For normal functions, this is relatively easy. However, what about functions like

$$48x(1+x)^{60} - (1+x)^{60} + 1 = 0$$

?

Take a random point. Find it's intersection with the x-axis. Find the point on the function with that x coordinate. Repeat this process. If we continue this indefinitely, we can reach a solution of the function.

THEOREM | The Newton-Raphson Method



Let the x-intercept of the function f by the i th line from the starting point P be Q_i . We can define one “action” as the following :

$$Q_0 = x(P)$$

$$g_i(x) = f'(Q_{i-1})(x - Q_{i-1}) + f(Q_{i-1}) \quad \text{where } i \geq 1$$

$$\text{define } Q_i \text{ s.t. } g_i(Q_i) = 0$$

Or, to write it in a one-liner fashion,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'}(x_n)$$

This is an iterative process. Computers use this method to find fairly accurate solutions of a equation.

EXAMPLE | Newton-Rhapson Method Basics

Starting with $x_1 = 2$, find the third approximation x_3 to the solution of the equation $x^3 - 2x - 5 = 0$.

Solution 1 |

We apply the Newton-Rhapson Method.

$$x_{n+1} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2}$$

$$x_2 = 2.1$$

$$x_3 = 2.0946$$

Chapter 03.09

Antiderivatives

Consider this.

$$f \xrightarrow[\text{???}]{\text{derivative}} f'$$

What do we call that action of finding f from f' ?

DEFINITION | Antiderivative

A function F is called an **antiderivative** of f on an interval I if $F'(x) = f(x) \forall x \in I$. The most general form of an antiderivative in this case is $F(x) + C$ where C is the antiderivative constant.

1 Antidifferentiation Formulae

Basically the opposite of differentiation formulae.

EXAMPLE | Antidifferentiation Basics

Find all functions g s.t. $g'(x) = 4 \sin(x) + \frac{2x^5 - \sqrt{x}}{x}$.

Solution 1 |

$$g'(x) = 4 \sin(x) + 2x^4 - \frac{1}{\sqrt{x}}$$

$$\therefore g = -4 \cos(x) + \frac{2}{5}x^5 - 2\sqrt{x} + C$$

Chapter 04

Integrals

This chapter solves the area and distance problems posed at the very start of the course. Riemann sums chop a region into thin rectangles, and their limit defines the definite integral: a single number recording accumulated change. The Fundamental Theorem of Calculus then reveals the astonishing fact that accumulation and differentiation undo one another, collapsing area computations into antiderivative lookups. The chapter closes with the substitution rule, the chain rule run backward, and the symmetry shortcuts it explains.

Chapter 04.01

The Area and Distance Problems

Remember the preview chapter? We kept talking about area and distance by chopping them into little pieces. Now let us really **visualize** those pieces.

1 The area problem

- Take a nonnegative function on an interval $[a, b]$. If we draw many thin rectangles under the curve, the sum of their areas starts to resemble the true region.

DEFINITION | Riemann Sum

Suppose we divide $[a, b]$ into n equal parts. Let

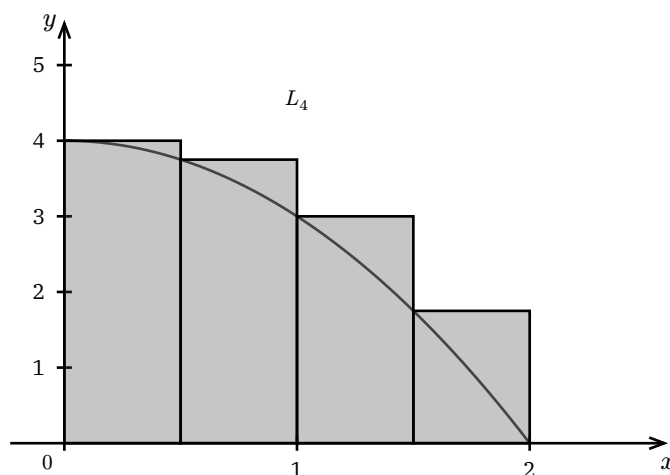
$$\Delta x = \frac{b - a}{n}$$

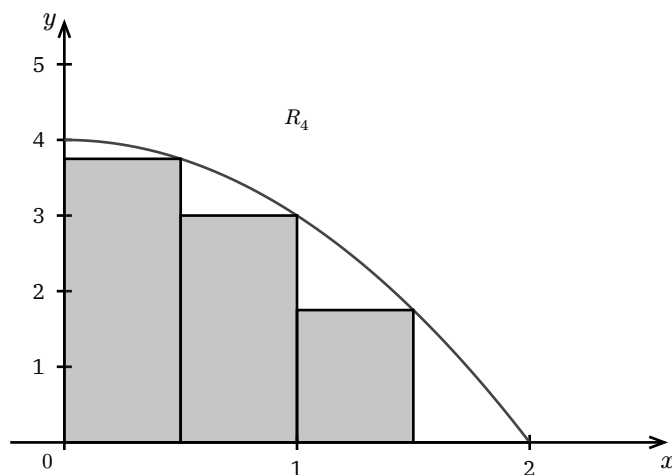
and choose a sample point x_i^* in the i th subinterval. Then

$$A_n = \sum_{i=1}^n f(x_i^*) \Delta x$$

is called a **Riemann sum** for f on $[a, b]$.

- The most common choices of sample points are the **left endpoints**, the **right endpoints**, and the **midpoints**. Watch how differently they hug the same curve :





NOTE | Overestimates and underestimates

Because $y = 4 - x^2$ is **decreasing** on $[0, 2]$, every left rectangle pokes above the curve and every right rectangle hides below it. So

$$R_n \leq A \leq L_n$$

and the true area A is squeezed between the two. As n grows, the squeeze tightens.

EXAMPLE | Approximating the Area under $y = 4 - x^2$

Approximate the area under $y = 4 - x^2$ from $x = 0$ to $x = 2$ using 4 left-endpoint rectangles, then using 4 right-endpoint rectangles.

Solution 1 |

Here

$$\Delta x = \frac{2 - 0}{4} = \frac{1}{2}$$

and the left endpoints are

$$x = 0, \frac{1}{2}, 1, \frac{3}{2}$$

So

$$L_4 = \frac{1}{2} \left[4 + \left(4 - \frac{1}{4} \right) + (4 - 1) + \left(4 - \frac{9}{4} \right) \right]$$

$$= \frac{1}{2} \left[4 + \frac{15}{4} + 3 + \frac{7}{4} \right]$$

$$= \frac{1}{2} \cdot \frac{25}{2} = \frac{25}{4}$$

The right endpoints are

$$x = \frac{1}{2}, 1, \frac{3}{2}, 2$$

so

$$R_4 = \frac{1}{2} \left[\frac{15}{4} + 3 + \frac{7}{4} + 0 \right] = \frac{1}{2} \cdot \frac{17}{2} = \frac{17}{4}$$

Therefore the true area A satisfies

$$\frac{17}{4} \leq A \leq \frac{25}{4}$$

Each rectangle family errs in one consistent direction because the parabola is decreasing, which is exactly why we get a two-sided bound.

- Approximation is nice, but we can already do better. If we know how to add the numbers $1^2, 2^2, \dots, n^2$, we can let $n \rightarrow \infty$ and get the **exact** area.

NOTE | Summation formulas worth memorizing

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2$$

EXAMPLE | The Exact Area as a Limit

Find the exact area under $y = 4 - x^2$ from $x = 0$ to $x = 2$ by taking the limit of right-endpoint Riemann sums.

Solution 1 |

With n rectangles,

$$\Delta x = \frac{2}{n}$$

and the right endpoints are

$$x_i = \frac{2i}{n}$$

So

$$\begin{aligned} R_n &= \sum_{i=1}^n \left[4 - \left(\frac{2i}{n} \right)^2 \right] \frac{2}{n} \\ &= \frac{8}{n} \sum_{i=1}^n 1 - \frac{8}{n^3} \sum_{i=1}^n i^2 \\ &= 8 - \frac{8}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} \end{aligned}$$

As $n \rightarrow \infty$, the fraction

$$\frac{n(n+1)(2n+1)}{n^3} \rightarrow 2$$

so

$$\lim_{n \rightarrow \infty} R_n = 8 - 8 \cdot \frac{2}{6} = 8 - \frac{8}{3} = \frac{16}{3}$$

The area is exactly $\frac{16}{3}$, safely inside the bracket $[\frac{17}{4}, \frac{25}{4}]$ we found before.

2 The distance problem

Now replace “height” by “velocity”. Each little rectangle becomes velocity times a tiny time interval, so the picture becomes accumulated distance.

DEFINITION | Distance from a Velocity Function

If an object moves with velocity $v(t)$ on $[a, b]$, then an approximation of the traveled distance is

$$D_n = \sum_{i=1}^n v(t_i^*) \Delta t$$

where

$$\Delta t = \frac{b - a}{n}$$

NOTE | What should we picture here?

Imagine the velocity graph. Each rectangle under it tells us how far the object would travel if the velocity stayed almost constant on that tiny interval.

EXAMPLE | Distance Approximation with a Curved Velocity Graph

A particle moves with velocity

$$v(t) = t^3 - 2t + 4$$

on $[0, 2]$. Use 4 midpoint rectangles to approximate the distance traveled.

Solution 1 |

We have

$$\Delta t = \frac{2 - 0}{4} = \frac{1}{2}$$

and the midpoints are

$$\frac{1}{4}, \frac{3}{4}, \frac{5}{4}, \frac{7}{4}$$

So

$$\begin{aligned} D_4 &= \frac{1}{2} \left[v\left(\frac{1}{4}\right) + v\left(\frac{3}{4}\right) + v\left(\frac{5}{4}\right) + v\left(\frac{7}{4}\right) \right] \\ &= \frac{1}{2} \left[\frac{225}{64} + \frac{187}{64} + \frac{221}{64} + \frac{375}{64} \right] \\ &= \frac{1}{2} \cdot \frac{1008}{64} = \frac{63}{8} \end{aligned}$$

Therefore the approximate distance is $\frac{63}{8}$ units.

Chapter 04.02

The Definite Integral

1 The definite integral

The limit of the Riemann sums is what we really care about. That final number records the full accumulation, not just one rough sketch.

DEFINITION | Definite Integral

If the limit of the Riemann sums for f on $[a, b]$ exists, then we write

$$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*)\Delta x$$

and call it the **definite integral** of f from a to b .

Here :

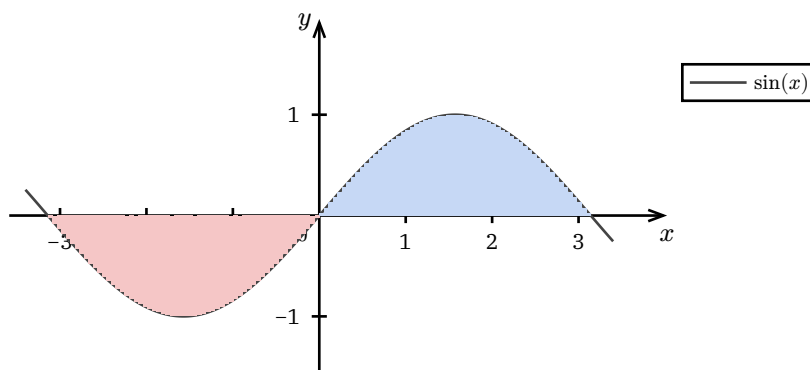
- $f(x)$ is the **integrand**.
- a is the **lower limit**.
- b is the **upper limit**.
- x is only the **variable of integration**.

THEOREM | Which Functions are Integrable?

If f is continuous on $[a, b]$, or has only finitely many jump discontinuities there, then f is integrable on $[a, b]$: the limit of Riemann sums exists and gives the same value for every choice of sample points.

NOTE | Signed area

The integral remembers orientation. Regions above the axis contribute positively, and regions below the axis contribute negatively.



EXAMPLE | A Geometric Integral with a Curved Boundary

Evaluate

$$\int_{-2}^2 \sqrt{4 - x^2} dx$$

using geometry.

Solution 1 |

The graph $y = \sqrt{4 - x^2}$ is the upper semicircle of radius 2 centered at the origin. So the integral is just the area of that semicircle :

$$\int_{-2}^2 \sqrt{4 - x^2} dx = \frac{1}{2} \pi (2)^2 = 2\pi$$

EXAMPLE | A Net Area Example

Evaluate

$$\int_{-\pi}^{\pi} \sin(x) dx$$

Solution 1 |

The graph of $\sin(x)$ is odd, so the part on $[-\pi, 0]$ is the mirror image of the part on $[0, \pi]$ with opposite sign. The positive and negative regions cancel, so

$$\int_{-\pi}^{\pi} \sin(x) dx = 0$$

2 Reading a limit of sums backwards

The definition is a two-way street. Whenever a complicated limit of sums has the shape $\sum f(x_i)\Delta x$, we can **recognize** it as a definite integral and evaluate it with geometry or (soon) antiderivatives.

NOTE | The recognition recipe

Look for the block $\frac{1}{n}$: that is Δx after dividing $[0, 1]$ into n pieces (or $\frac{c}{n}$ for the interval $[0, c]$). Then look for the block $\frac{i}{n}$: that is the sample point x_i . Everything else is the integrand f .

EXAMPLE | A Limit of Sums in Disguise

Evaluate

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2}{n} \left[\frac{2i}{n} + \sqrt{4 - \left(\frac{2i}{n} \right)^2} \right]$$

Solution 1 |

The repeated block is

$$\frac{2i}{n}$$

so set

$$x_i = \frac{2i}{n}$$

These are right endpoints of $[0, 2]$ with

$$\Delta x = \frac{2}{n}$$

The sum is therefore a Riemann sum for

$$f(x) = x + \sqrt{4 - x^2}$$

on $[0, 2]$, and the limit is

$$\int_0^2 [x + \sqrt{4 - x^2}] dx$$

The first piece is a triangle of base 2 and height 2, so it contributes 2. The second piece is a quarter circle of radius 2, so it contributes $\frac{1}{4}\pi(2)^2 = \pi$.

Hence

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2}{n} \left[\frac{2i}{n} + \sqrt{4 - \left(\frac{2i}{n}\right)^2} \right] = 2 + \pi$$

EXAMPLE | A Sum Hiding a Square Root

Evaluate

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{k\sqrt{n^2 - k^2}}{n^3}$$

Solution 1 |

Pull the powers of n inside :

$$\frac{k\sqrt{n^2 - k^2}}{n^3} = \frac{k}{n} \cdot \sqrt{1 - \left(\frac{k}{n}\right)^2} \cdot \frac{1}{n}$$

With $x_k = \frac{k}{n}$ and $\Delta x = \frac{1}{n}$, this is a Riemann sum for

$$f(x) = x\sqrt{1 - x^2}$$

on $[0, 1]$. So the limit equals

$$\int_0^1 x\sqrt{1 - x^2} dx$$

We will soon compute such integrals by substitution ; for now, notice that

$$\frac{d}{dx} \left[-\frac{1}{3}(1 - x^2)^{\frac{3}{2}} \right] = x\sqrt{1 - x^2}$$

so the value is

$$\left[-\frac{1}{3}(1-x^2)^{\frac{3}{2}}\right]_0^1 = 0 - \left(-\frac{1}{3}\right) = \frac{1}{3}$$

THEOREM | Basic Properties of the Definite Integral

If the integrals exist, then :

$$1. \quad \int_a^a f(x)dx = 0$$

$$2. \quad \int_a^b f(x)dx = -\int_b^a f(x)dx$$

$$3. \quad \int_a^b cf(x)dx = c \int_a^b f(x)dx$$

$$4. \quad \int_a^b [f(x) + g(x)]dx = \int_a^b f(x)dx + \int_a^b g(x)dx$$

$$5. \quad \int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

Proof | From the Riemann sums

The picture is still useful here : every property comes from what happens to the rectangles. But we can also write it directly in symbols.

For (1), if the interval is $[a, a]$, then

$$\Delta x = \frac{a - a}{n} = 0$$

so every Riemann sum is

$$\sum_{i=1}^n f(x_i^*)\Delta x = 0$$

Hence

$$\int_a^a f(x)dx = 0$$

For (2), reversing the interval changes the width from Δx to $-\Delta x$, so

$$\int_b^a f(x)dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*)(-\Delta x) = -\int_a^b f(x)dx$$

For (3),

$$\begin{aligned}\int_a^b cf(x)dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n cf(x_i^*)\Delta x \\ &= c \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*)\Delta x = c \int_a^b f(x)dx\end{aligned}$$

For (4),

$$\begin{aligned}\int_a^b [f(x) + g(x)]dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n [f(x_i^*) + g(x_i^*)]\Delta x \\ &= \lim_{n \rightarrow \infty} \left[\sum_{i=1}^n f(x_i^*)\Delta x + \sum_{i=1}^n g(x_i^*)\Delta x \right] \\ &= \int_a^b f(x)dx + \int_a^b g(x)dx\end{aligned}$$

For (5), if we split $[a, b]$ at c , then the full Riemann sum splits into the two partial sums on $[a, c]$ and $[c, b]$, so in the limit

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$$

THEOREM | Comparison Properties

Suppose $a < b$ and the integrals exist.

1. If $f(x) \geq 0$ on $[a, b]$, then

$$\int_a^b f(x)dx \geq 0$$

2. If $f(x) \geq g(x)$ on $[a, b]$, then

$$\int_a^b f(x)dx \geq \int_a^b g(x)dx$$

3. If $m \leq f(x) \leq M$ on $[a, b]$, then

$$m(b-a) \leq \int_a^b f(x)dx \leq M(b-a)$$

Proof | Rectangles inherit inequalities

Every Riemann sum of a nonnegative function is a sum of nonnegative terms, so its limit is nonnegative. That is (1).

For (2), apply (1) to the function $f(x) - g(x) \geq 0$ and use additivity.

For (3), apply (2) to the constant functions m and M : the integral of a constant c over $[a, b]$ is $c(b - a)$, which gives both bounds.

EXAMPLE | Estimating an Integral Without Computing It

Show that

$$\frac{1}{5} \leq \int_1^2 \frac{1}{1+x^2} dx \leq \frac{1}{2}$$

without evaluating the integral.

Solution 1 |

On $[1, 2]$, the function $\frac{1}{1+x^2}$ is decreasing, so its maximum is at $x = 1$ and its minimum is at $x = 2$:

$$\frac{1}{5} \leq \frac{1}{1+x^2} \leq \frac{1}{2}$$

By the comparison property with $b - a = 1$,

$$\frac{1}{5} \leq \int_1^2 \frac{1}{1+x^2} dx \leq \frac{1}{2}$$

No antiderivative was harmed in the making of this estimate.

The Fundamental Theorem of Calculus

1 The Fundamental Theorem of Calculus

This is the moment where the picture clears up. The process of **accumulating** area and the process of **measuring** instantaneous change undo one another.

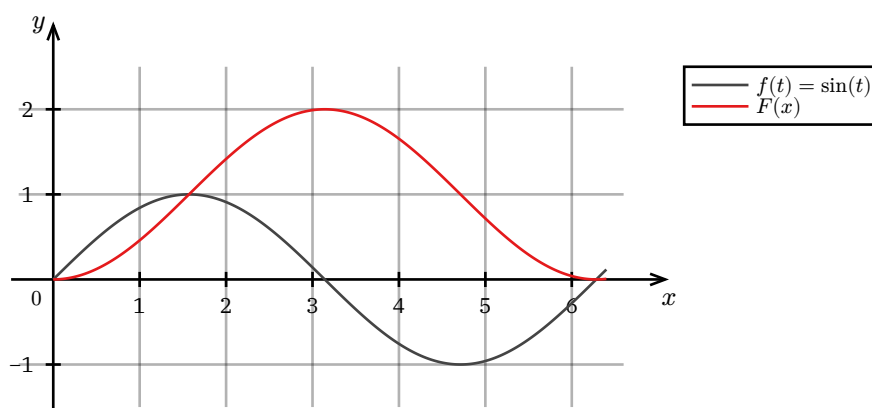
DEFINITION | Accumulation Function

If f is continuous on $[a, b]$, the function

$$F(x) = \int_a^x f(t) dt$$

is called an **accumulation function** : it records the signed area collected from a up to the moving position x .

- Before any theorem, look at a concrete pair. Below, the accumulation of $f(t) = \sin(t)$ starting at 0 is $F(x) = 1 - \cos(x)$. Where $\sin$ is positive, F climbs ; where $\sin$ is negative, F drains back down.



THEOREM | Fundamental Theorem of Calculus Part I

If f is continuous on $[a, b]$, and if

$$F(x) = \int_a^x f(t) dt$$

then

$$F'(x) = f(x)$$

Proof | Think of a thin strip

Compare $F(x + h)$ and $F(x)$:

$$F(x + h) - F(x) = \int_x^{x+h} f(t) dt$$

Geometrically, this is the area of a very thin strip of width h . Because f is continuous, the height across that strip is almost constant and looks like $f(x)$.

More precisely,

$$\frac{F(x + h) - F(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t) dt$$

and as $h \rightarrow 0$, the average height on that tiny interval approaches $f(x)$.

Therefore

$$F'(x) = \lim_{h \rightarrow 0} \frac{F(x + h) - F(x)}{h} = f(x)$$

EXAMPLE | Differentiating an Accumulation Function

Let

$$F(x) = \int_1^{x^3} \ln(1 + t^2) dt$$

Find $F'(x)$.

Solution 1 |

First picture the inside quantity x^3 as the moving upper bound. By Part I, the derivative of the area function is the integrand evaluated at that moving bound, and then we multiply by the derivative of the bound.

So

$$\begin{aligned}
 F'(x) &= \ln(1 + (x^3)^2) \cdot \frac{d}{dx}(x^3) \\
 &= 3x^2 \ln(1 + x^6)
 \end{aligned}$$

EXAMPLE | Differentiate First, Integrate Second

Evaluate

$$\int_0^1 \frac{d}{dx} \left[\int_3^{x^2} \frac{1}{t^2 + 1} dt \right] dx$$

Solution 1 |

Work from the inside out. By Part I with the moving bound x^2 ,

$$\frac{d}{dx} \int_3^{x^2} \frac{1}{t^2 + 1} dt = \frac{1}{(x^2)^2 + 1} \cdot 2x = \frac{2x}{x^4 + 1}$$

Notice that the lower bound 3 has completely vanished : a constant starting line never affects the **rate** of accumulation.

Now integrate this from 0 to 1. Substituting $u = x^2$ (so $du = 2x dx$),

$$\int_0^1 \frac{2x}{x^4 + 1} dx = \int_0^1 \frac{1}{u^2 + 1} du = [\tan^{-1}(u)]_0^1 = \frac{\pi}{4}$$

EXAMPLE | A Nested Accumulation Function

Let

$$f(x) = \int_0^{g(x)} \frac{1}{\sqrt{1+t^3}} dt$$

where

$$g(x) = \int_0^{\cos(x)} [1 + \sin(t^2)] dt$$

Find $f'(\frac{\pi}{2})$.

Solution 1 |

Differentiate the outer layer with Part I and the chain rule :

$$f'(x) = \frac{1}{\sqrt{1+g(x)^3}} \cdot g'(x)$$

The inner layer has moving bound $\cos(x)$, so

$$g'(x) = [1 + \sin(\cos^2(x))] \cdot (-\sin(x))$$

Now evaluate at $x = \frac{\pi}{2}$. Since $\cos(\frac{\pi}{2}) = 0$,

$$g\left(\frac{\pi}{2}\right) = \int_0^0 [1 + \sin(t^2)] dt = 0$$

and

$$g'\left(\frac{\pi}{2}\right) = [1 + \sin(0)] \cdot (-1) = -1$$

Therefore

$$f'\left(\frac{\pi}{2}\right) = \frac{1}{\sqrt{1+0}} \cdot (-1) = -1$$

EXAMPLE | A Function Born from Riemann Sums

For $x > 0$, define

$$f(x) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{x}{n} \sqrt{1 + \left(\frac{kx}{n}\right)^2}$$

Find $f'(2)$.

Solution 1 |

Read the sum backwards, exactly as in the previous section. With

$$\Delta t = \frac{x}{n}$$

and sample points

$$t_k = \frac{kx}{n}$$

on $[0, x]$, the sum is a Riemann sum for $\sqrt{1+t^2}$. Hence

$$f(x) = \int_0^x \sqrt{1+t^2} dt$$

We have no idea (yet) how to compute this integral in closed form, and we do not need to : Part I gives

$$f'(x) = \sqrt{1+x^2}$$

so

$$f'(2) = \sqrt{5}$$

THEOREM | Fundamental Theorem of Calculus Part II

If f is continuous on $[a, b]$ and F is any antiderivative of f , then

$$\int_a^b f(x) dx = F(b) - F(a)$$

Proof | Area and antiderivatives differ only by a constant

Define

$$G(x) = \int_a^x f(t) dt$$

Then Part I tells us

$$G'(x) = f(x)$$

But $F'(x) = f(x)$ as well, so

$$G'(x) = F'(x)$$

This means $G(x) - F(x)$ has derivative 0, so it must be a constant.

Evaluate at $x = a$:

$$G(a) = \int_a^a f(t) dt = 0$$

Hence

$$G(x) = F(x) - F(a)$$

Putting $x = b$ gives

$$\int_a^b f(x)dx = G(b) = F(b) - F(a)$$

EXAMPLE | Evaluating a Definite Integral with a Logarithmic Antiderivative

Evaluate

$$\int_1^e \frac{2 \ln(x)}{x} dx$$

Solution 1 |

We look for an antiderivative. Since

$$\frac{d}{dx}(\ln(x))^2 = 2 \frac{\ln(x)}{x}$$

we can take

$$F(x) = (\ln(x))^2$$

Therefore

$$\begin{aligned} \int_1^e \frac{2 \ln(x)}{x} dx &= [(\ln(x))^2]_1^e \\ &= (\ln(e))^2 - (\ln(1))^2 = 1 - 0 = 1 \end{aligned}$$

NOTE | The two parts in one sentence

Part I says differentiation undoes accumulation. Part II says accumulation can be computed by reversing differentiation. Together : the tangent problem and the area problem from chapter 0 were secretly the same problem.

Chapter 04.04

Indefinite Integrals and the Net Change Theorem

We already saw antiderivatives in chapter 3. Here they stop being isolated tricks and become part of a bigger picture.

1 Indefinite integrals

DEFINITION | Indefinite Integral

The collection of all antiderivatives of $f(x)$ is called the **indefinite integral** of f and is written as

$$\int f(x)dx = F(x) + C$$

where $F'(x) = f(x)$ and C is an arbitrary constant.

THEOREM | Basic Indefinite Integral Formulae

1. $\int x^n dx = \frac{x^{n+1}}{n+1} + C$
if $n \neq -1$
2. $\int \cos(x)dx = \sin(x) + C$
3. $\int \sin(x)dx = -\cos(x) + C$
4. $\int e^x dx = e^x + C$
5. $\int \frac{1}{x} dx = \ln|x| + C$
6. $\int \sec^2(x)dx = \tan(x) + C$
7. $\int \sec(x) \tan(x)dx = \sec(x) + C$
8. $\int \frac{1}{1+x^2} dx = \tan^{-1}(x) + C$

Proof | Differentiate the answers

The fastest way to see these is to go backward from derivative rules we already know. For instance,

$$\frac{d}{dx} \frac{x^{n+1}}{n+1} = x^n$$

and

$$\frac{d}{dx} \sin(x) = \cos(x)$$

The same check works for the other formulae, so each right-hand side really is an antiderivative of the left-hand side. Adding a constant changes nothing because constants differentiate to 0.

EXAMPLE | The Most General Antiderivative

Find the most general antiderivative of

$$f(t) = t^{-\frac{2}{5}} + \sin(t)$$

Solution 1 |

Integrate term by term. The power rule with $n = -\frac{2}{5}$ gives

$$\int t^{-\frac{2}{5}} dt = \frac{t^{\frac{3}{5}}}{\frac{3}{5}} = \frac{5}{3} t^{\frac{3}{5}}$$

and

$$\int \sin(t) dt = -\cos(t)$$

One subtlety : $t^{-\frac{2}{5}}$ is undefined at $t = 0$, so the domain splits into two separate intervals, and each interval carries its **own** arbitrary constant. The most general antiderivative is therefore

$$F(t) = \begin{cases} \frac{5}{3} t^{\frac{3}{5}} - \cos(t) + C_1 & \text{if } t < 0 \\ \frac{5}{3} t^{\frac{3}{5}} - \cos(t) + C_2 & \text{if } t > 0 \end{cases}$$

A single C would silently claim that the two branches move up and down together, but nothing connects them across the gap at $t = 0$.

EXAMPLE | Finding an Indefinite Integral with Fractional Powers

Find

$$\int \frac{x^4 - 3x^2 + 2}{\sqrt{x}} dx$$

Solution 1 |

First rewrite the integrand :

$$\frac{x^4 - 3x^2 + 2}{\sqrt{x}} = x^{\frac{7}{2}} - 3x^{\frac{3}{2}} + 2x^{-\frac{1}{2}}$$

Now integrate term by term :

$$\int x^{\frac{7}{2}} dx = \frac{2}{9} x^{\frac{9}{2}}$$

$$\int -3x^{\frac{3}{2}} dx = -\frac{6}{5} x^{\frac{5}{2}}$$

$$\int 2x^{-\frac{1}{2}} dx = 4\sqrt{x}$$

Therefore

$$\int \frac{x^4 - 3x^2 + 2}{\sqrt{x}} dx = \frac{2}{9} x^{\frac{9}{2}} - \frac{6}{5} x^{\frac{5}{2}} + 4\sqrt{x} + C$$

2 The net change theorem

THEOREM | Net Change Theorem

If a quantity changes at the rate $F'(x)$, then the total change from a to b is

$$\int_a^b F'(x) dx = F(b) - F(a)$$

Proof | Visualize rate piling up into total change

The function $F'(x)$ tells us the tiny change per tiny input change. Adding all those tiny changes from a to b should recover the full difference in the original quantity.

Formally, F is an antiderivative of F' , so the Fundamental Theorem of Calculus Part II gives

$$\int_a^b F'(x)dx = F(b) - F(a)$$

EXAMPLE | Net Change of a Quantity

Suppose a quantity C changes according to

$$C'(t) = 3t^2 - \frac{2}{t}$$

for $1 \leq t \leq 3$. Find the total change in C on this interval.

Solution 1 |

By the net change theorem,

$$C(3) - C(1) = \int_1^3 \left(3t^2 - \frac{2}{t} \right) dt$$

An antiderivative is

$$t^3 - 2\ln(t)$$

so

$$\begin{aligned} \int_1^3 \left(3t^2 - \frac{2}{t} \right) dt &= [t^3 - 2\ln(t)]_1^3 \\ &= (27 - 2\ln(3)) - (1 - 0) \\ &= 26 - 2\ln(3) \end{aligned}$$

Therefore the total change is $26 - 2\ln(3)$.

3 Displacement versus distance

For a moving particle, the velocity graph carries **two** different accumulation stories at once.

DEFINITION | Displacement and Total Distance

If $v(t)$ is the velocity of a particle on $[a, b]$, then

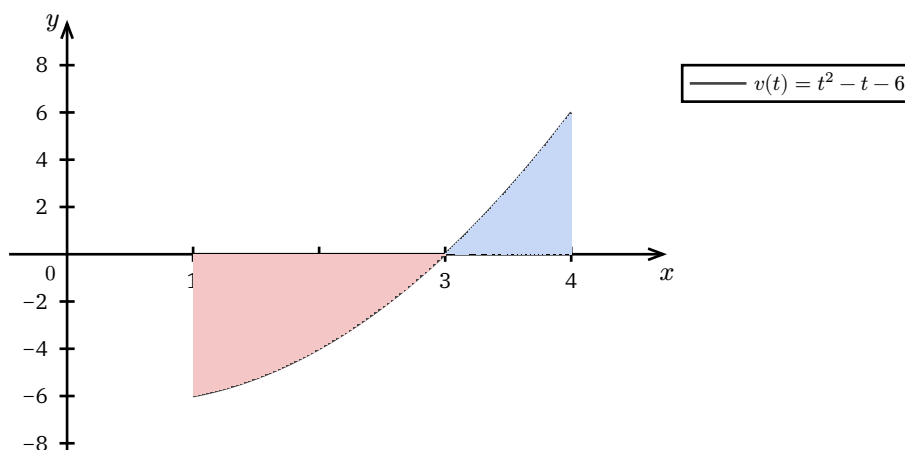
$$\text{displacement} = \int_a^b v(t) dt$$

and

$$\text{total distance} = \int_a^b |v(t)| dt$$

NOTE | Read it off the picture

Displacement lets the region below the axis cancel the region above it. Total distance flips the below-axis region up before adding, so nothing cancels.



EXAMPLE | Displacement and Distance from the Same Velocity

A particle moves along a line with velocity

$$v(t) = t^2 - t - 6$$

for $1 \leq t \leq 4$. Find the displacement and the total distance traveled.

Solution 1 |

The displacement is

$$\int_1^4 (t^2 - t - 6) dt = \left[\frac{t^3}{3} - \frac{t^2}{2} - 6t \right]_1^4$$

$$= \left(\frac{64}{3} - 8 - 24 \right) - \left(\frac{1}{3} - \frac{1}{2} - 6 \right) = -\frac{9}{2}$$

So the particle ends up $\frac{9}{2}$ units to the **left** of where it started.

For the distance, factor

$$v(t) = (t - 3)(t + 2)$$

so on $[1, 3]$ the velocity is negative and on $[3, 4]$ it is positive. Then

$$\int_1^4 |v(t)| dt = -\int_1^3 v(t) dt + \int_3^4 v(t) dt$$

Compute each piece :

$$\int_1^3 v(t) dt = \left[\frac{t^3}{3} - \frac{t^2}{2} - 6t \right]_1^3 = -\frac{22}{3}$$

$$\int_3^4 v(t) dt = \left[\frac{t^3}{3} - \frac{t^2}{2} - 6t \right]_3^4 = \frac{17}{6}$$

Therefore the total distance is

$$\frac{22}{3} + \frac{17}{6} = \frac{61}{6}$$

which is much larger than $|\frac{9}{2}|$, exactly because the backward trip was wasted motion.

Chapter 04.05

The Substitution Rule

1 The substitution rule

The chain rule told us what happens when functions are nested. Substitution is what happens when we reverse that nesting during integration.

DEFINITION | Substitution Rule

If $u = g(x)$ and $du = g'(x)dx$, then

$$\int f(g(x))g'(x)dx = \int f(u)du$$

For definite integrals, we also change the bounds :

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du$$

NOTE | What should we try to see?

Look for an **outside function** and an **inside function**. If some derivative of the inside piece is sitting nearby, substitution is usually the right picture.

EXAMPLE | A Power hiding inside another power

Evaluate

$$\int x^3 \sqrt{x^4 + 1} dx$$

Solution 1 |

The inner expression is

$$x^4 + 1$$

and its derivative is

$$4x^3$$

which is almost sitting there already.

Let

$$u = x^4 + 1$$

Then

$$du = 4x^3 dx$$

so

$$x^3 dx = \frac{1}{4} du$$

Therefore

$$\begin{aligned}\int x^3 \sqrt{x^4 + 1} dx &= \frac{1}{4} \int u^{\frac{1}{2}} du \\ &= \frac{1}{4} \cdot \frac{2}{3} u^{\frac{3}{2}} + C \\ &= \frac{1}{6} (x^4 + 1)^{\frac{3}{2}} + C\end{aligned}$$

EXAMPLE | Substitution with Trigonometric Bounds

Evaluate

$$\int_0^{\frac{\pi}{4}} \frac{\sin(2x)}{1 + \cos^2(2x)} dx$$

Solution 1 |

Here the inside function is

$$\cos(2x)$$

because its derivative gives a multiple of $\sin(2x)$.

Let

$$u = \cos(2x)$$

Then

$$du = -2 \sin(2x) dx$$

so

$$\sin(2x)dx = -\frac{1}{2}du$$

Change the bounds : If $x = 0$, then $u = 1$. If $x = \frac{\pi}{4}$, then $u = 0$.

Therefore

$$\begin{aligned}\int_0^{\frac{\pi}{4}} \frac{\sin(2x)}{1 + \cos^2(2x)} dx &= -\frac{1}{2} \int_1^0 \frac{1}{1 + u^2} du \\ &= \frac{1}{2} \int_0^1 \frac{1}{1 + u^2} du \\ &= \frac{1}{2} [\tan^{-1}(u)]_0^1 = \frac{1}{2} \cdot \frac{\pi}{4} = \frac{\pi}{8}\end{aligned}$$

EXAMPLE | Substituting to Kill a Square Root

Evaluate

$$\int_0^1 \frac{1}{(1 + \sqrt{x})^3} dx$$

Solution 1 |

The annoying part is $\sqrt{x}$, so make the whole parenthesis the new variable.
Let

$$u = 1 + \sqrt{x}$$

Then

$$\sqrt{x} = u - 1$$

so

$$x = (u - 1)^2$$

and

$$dx = 2(u - 1)du$$

Change the bounds : If $x = 0$, then $u = 1$. If $x = 1$, then $u = 2$.

Therefore

$$\begin{aligned}
\int_0^1 \frac{1}{(1+\sqrt{x})^3} dx &= \int_1^2 \frac{2(u-1)}{u^3} du \\
&= 2 \int_1^2 (u^{-2} - u^{-3}) du \\
&= 2 \left[-\frac{1}{u} + \frac{1}{2u^2} \right]_1^2 \\
&= 2 \left[\left(-\frac{1}{2} + \frac{1}{8} \right) - \left(-1 + \frac{1}{2} \right) \right] = 2 \cdot \frac{1}{8} = \frac{1}{4}
\end{aligned}$$

2 Integrals of symmetric functions

Substitution also explains why symmetry is such a powerful shortcut on symmetric intervals.

THEOREM | Symmetry Theorem

Suppose f is continuous on $[-a, a]$.

1. If f is even, so $f(-x) = f(x)$, then

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$

2. If f is odd, so $f(-x) = -f(x)$, then

$$\int_{-a}^a f(x) dx = 0$$

Proof | Substitute in the left half

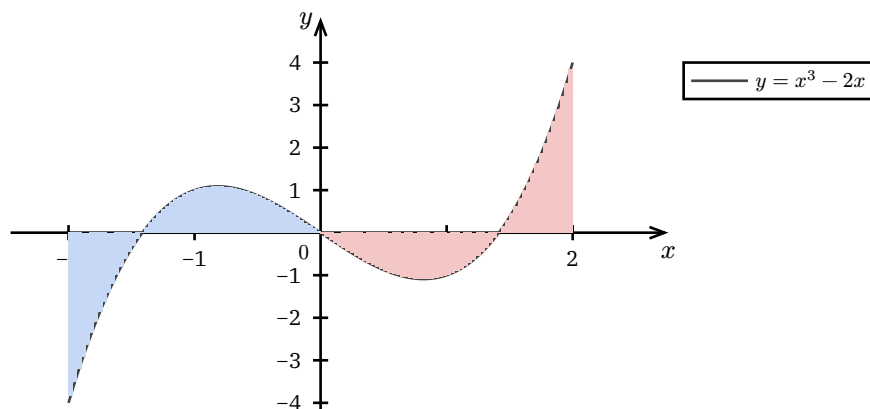
Split the integral :

$$\int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$$

In the first piece, substitute $u = -x$, so $du = -dx$:

$$\int_{-a}^0 f(x) dx = \int_0^a f(-u) du$$

If f is even, this equals $\int_0^a f(u)du$ and the two halves add. If f is odd, this equals $-\int_0^a f(u)du$ and the two halves cancel.



EXAMPLE | Symmetry Doing All the Work

Evaluate

$$\int_{-3}^3 \frac{x^5 + x^3 \sin^2(x)}{x^4 + 2} dx$$

Solution 1 |

Check the symmetry of the integrand. Replacing x by $-x$:

- x^5 and x^3 flip sign,
- $\sin^2(x)$ and $x^4 + 2$ stay the same.

So the whole integrand is **odd**, and the interval $[-3, 3]$ is symmetric.

Therefore

$$\int_{-3}^3 \frac{x^5 + x^3 \sin^2(x)}{x^4 + 2} dx = 0$$

with no computation at all. The shaded picture above shows exactly why: the two lobes are congruent with opposite signs.

Chapter 05

Applications of Integration

One idea powers this whole chapter: slice the object, understand one slice, and let the integral add the slices back up. Vertical and horizontal slices measure areas between curves; slices revolved around an axis become disks, washers, and cylindrical shells that measure volume; slices of a rope or a tank of water measure the work done against gravity; and dividing accumulated area by the width of the interval defines the average value of a function.

Chapter 05.01

Areas Between Curves

1 Areas between curves

When two curves trap a region, the easiest way to understand it is to imagine thin slices filling the gap from one boundary to the other.

DEFINITION | Area between Two Curves

If $f(x) \geq g(x)$ on $[a, b]$, then the area between the curves $y = f(x)$ and $y = g(x)$ is

$$A = \int_a^b [f(x) - g(x)] dx$$

In short :

$$\text{Area} = \int (\text{top} - \text{bottom}) dx$$

NOTE | Visualize the slice

A vertical slice has height

$$f(x) - g(x)$$

and width

$$dx$$

so its area is approximately

$$[f(x) - g(x)] dx$$

EXAMPLE | A Quick First Region

Find the area of the region

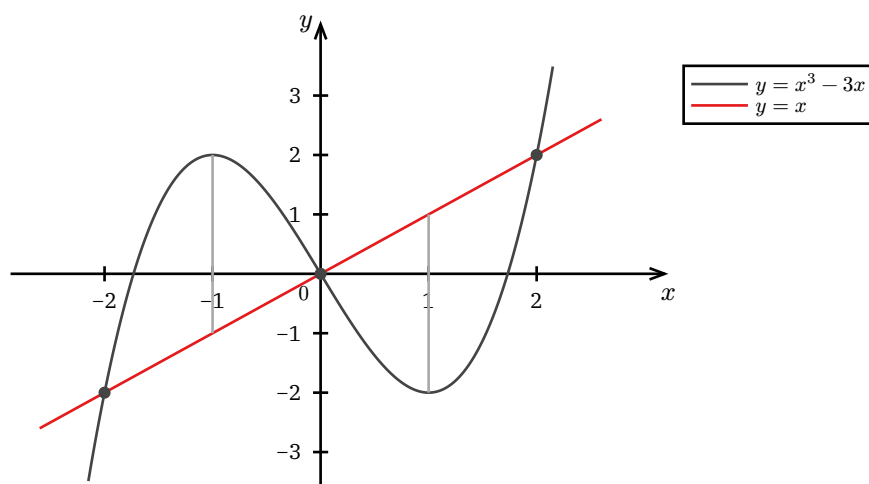
$$S = \{(x, y) : y \leq 9 - x^2, x \geq 0, y \geq 0\}$$

Solution 1 |

The region sits in the first quadrant, under the parabola $y = 9 - x^2$, which meets the axes at $(0, 9)$ and $(3, 0)$. Here the “bottom curve” is just $y = 0$, so

$$A = \int_0^3 (9 - x^2) dx = \left[9x - \frac{x^3}{3} \right]_0^3 = 27 - 9 = 18$$

- When the curves cross, “top” and “bottom” can swap roles. Always find the intersection points first, then handle each stretch separately.



EXAMPLE | A Region where the Top Curve Changes

Find the area enclosed by

$$y = x^3 - 3x$$

and

$$y = x$$

Solution 1 |

First find the intersection points :

$$x^3 - 3x = x$$

$$x^3 - 4x = 0$$

$$x(x^2 - 4) = 0$$

so

$$x = -2, 0, 2$$

On $[-2, 0]$, the curve $y = x^3 - 3x$ lies above $y = x$. On $[0, 2]$, the line $y = x$ lies above $y = x^3 - 3x$. The two gray slices in the picture show the top and bottom swapping.

Therefore

$$\begin{aligned} A &= \int_{-2}^0 [(x^3 - 3x) - x] dx + \int_0^2 [x - (x^3 - 3x)] dx \\ &= \int_{-2}^0 (x^3 - 4x) dx + \int_0^2 (4x - x^3) dx \end{aligned}$$

By symmetry, these two areas are equal, so

$$\begin{aligned} A &= 2 \int_0^2 (4x - x^3) dx \\ &= 2 \left[2x^2 - \frac{x^4}{4} \right]_0^2 \\ &= 2(8 - 4) = 8 \end{aligned}$$

2 Slicing horizontally

Some regions hate vertical slices : a single vertical line may enter and exit the region through the **same** curve. In that case, flip your head sideways and slice horizontally.

DEFINITION | Area with Respect to y

If $x = f(y)$ lies to the right of $x = g(y)$ for $c \leq y \leq d$, then the area between them is

$$A = \int_c^d [f(y) - g(y)] dy$$

In short :

$$\text{Area} = \int (\text{right} - \text{left}) dy$$

EXAMPLE | A Region that Prefers Horizontal Slices

Find the area enclosed by

$$x = y^2 - 4$$

and

$$x = 2y - 1$$

Solution 1 |

Find the intersections by setting the two right-hand sides equal :

$$y^2 - 4 = 2y - 1$$

$$y^2 - 2y - 3 = 0$$

$$(y - 3)(y + 1) = 0$$

so $y = -1$ and $y = 3$.

For $-1 \leq y \leq 3$, the line $x = 2y - 1$ lies to the **right** of the parabola $x = y^2 - 4$.
So

$$\begin{aligned} A &= \int_{-1}^3 [(2y - 1) - (y^2 - 4)] dy \\ &= \int_{-1}^3 (3 + 2y - y^2) dy \\ &= \left[3y + y^2 - \frac{y^3}{3} \right]_{-1}^3 \\ &= (9 + 9 - 9) - \left(-3 + 1 + \frac{1}{3} \right) = 9 + \frac{5}{3} = \frac{32}{3} \end{aligned}$$

A vertical-slice version of this computation would need two separate integrals and a square root. Horizontal slices see the region the way it wants to be seen.

Chapter 05.02

Volumes

1 Volumes

The one idea of this section : a volume is an integral of cross-sectional areas. Everything else (disks, washers, weird solids) is just a different formula for one slice.

DEFINITION | Volume by Slicing

Suppose a solid lies between $x = a$ and $x = b$, and the cross-section at position x (perpendicular to the x -axis) has area $A(x)$. Then

$$V = \int_a^b A(x) dx$$

NOTE | Visualize one cross-section

Freeze the solid, cut it with a plane, and stare at the flat face you produced. If you can write the area of that face as a function of the cut position, the volume is already yours.

EXAMPLE | A Solid Built on an Elliptical Base

The base of a solid S is the elliptical region bounded by

$$4x^2 + 9y^2 = 36$$

Cross-sections perpendicular to the x -axis are isosceles right triangles with hypotenuse in the base. Find the volume of S .

Solution 1 |

Rewrite the ellipse as

$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

so x runs over $[-3, 3]$ and, at each x , the base segment reaches from

$$y = -\frac{2}{3}\sqrt{9-x^2}$$

to

$$y = \frac{2}{3}\sqrt{9-x^2}$$

The hypotenuse of the triangle is that full segment, of length

$$h = \frac{4}{3}\sqrt{9-x^2}$$

An isosceles right triangle with hypotenuse h has legs $h/\sqrt{2}$ and area

$$A = \frac{1}{2} \left(\frac{h}{\sqrt{2}} \right)^2 = \frac{h^2}{4}$$

So

$$A(x) = \frac{1}{4} \cdot \frac{16}{9}(9-x^2) = \frac{4}{9}(9-x^2)$$

and

$$V = \int_{-3}^3 \frac{4}{9}(9-x^2)dx = \frac{4}{9} \left[9x - \frac{x^3}{3} \right]_{-3}^3 = \frac{4}{9} \cdot 36 = 16$$

2 Solids of revolution

When a plane region rotates, a thin slice turns into a solid piece. The main question is : what does one slice become?

DEFINITION | Disk and Washer Methods

If a slice perpendicular to the axis of rotation has outer radius $R(x)$ and inner radius $r(x)$, then

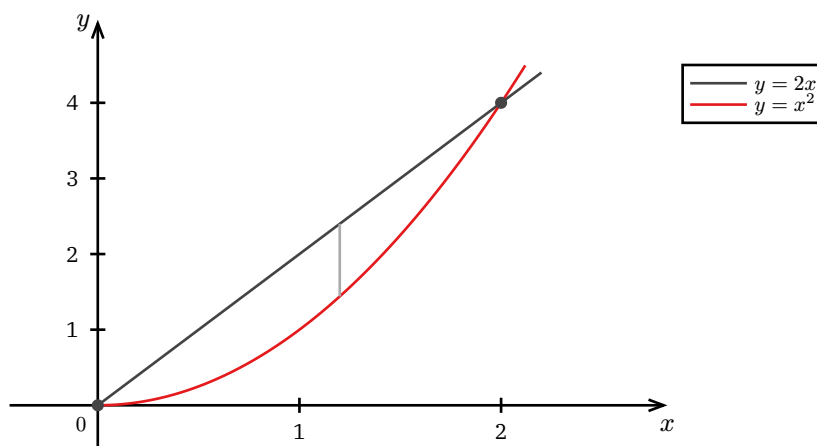
$$V = \pi \int_a^b [R(x)^2 - r(x)^2] dx$$

If there is no hole, then $r(x) = 0$ and we get the **disk method** :

$$V = \pi \int_a^b R(x)^2 dx$$

NOTE | Disk or washer?

A slice becomes a disk if it fills in the middle. It becomes a washer if there is an empty hole in the middle, which happens exactly when the region does not touch the axis of rotation.



EXAMPLE | A Washer Method Example

Find the volume obtained by rotating the region between

$$y = 2x$$

and

$$y = x^2$$

on $[0, 2]$ about the x -axis.

Solution 1 |

On $[0, 2]$, the outer curve is

$$y = 2x$$

and the inner curve is

$$y = x^2$$

The gray slice in the picture, once revolved, becomes a washer whose outer rim is traced by the line and whose hole is traced by the parabola.

So the outer radius is

$$R(x) = 2x$$

and the inner radius is

$$r(x) = x^2$$

Therefore

$$\begin{aligned} V &= \pi \int_0^2 \left[(2x)^2 - (x^2)^2 \right] dx \\ &= \pi \int_0^2 (4x^2 - x^4) dx \\ &= \pi \left[4\frac{x^3}{3} - \frac{x^5}{5} \right]_0^2 \\ &= \pi \left[\frac{32}{3} - \frac{32}{5} \right] = 64\frac{\pi}{15} \end{aligned}$$

3 Rotating about other lines

Nothing sacred about the coordinate axes. If the axis of rotation is the line $x = k$ or $y = k$, the only change is that every radius is now a **distance to that line**.

EXAMPLE | Rotation about a Vertical Line

Let R be the region enclosed by

$$y = \ln(x)$$

the x -axis, and the line $x = 2$. Find the volume of the solid obtained by rotating R about the line $x = 3$, using y as the variable of integration.

Solution 1 |

The region lives over $1 \leq x \leq 2$, and its height runs from $y = 0$ up to $y = \ln(2)$.

Slice horizontally at height y . Inside the region, x runs from the curve $x = e^y$ to the wall $x = 2$. Rotating about $x = 3$:

- the **far** edge of the slice is $x = e^y$, at distance $R(y) = 3 - e^y$ from the axis,
- the **near** edge is $x = 2$, at distance $r(y) = 3 - 2 = 1$.

So each slice becomes a washer, and

$$\begin{aligned}
 V &= \pi \int_0^{\ln(2)} [(3 - e^y)^2 - 1^2] dy \\
 &= \pi \int_0^{\ln(2)} [8 - 6e^y + e^{2y}] dy \\
 &= \pi \left[8y - 6e^y + \frac{e^{2y}}{2} \right]_0^{\ln(2)}
 \end{aligned}$$

At $y = \ln(2)$: $8 \ln(2) - 12 + 2$. At $y = 0$: $-6 + \frac{1}{2}$.

Therefore

$$V = \pi \left[(8 \ln(2) - 10) - \left(-\frac{11}{2} \right) \right] = \pi \left[8 \ln(2) - \frac{9}{2} \right]$$

Volumes by Cylindrical Shells

1 Volumes by cylindrical shells

Sometimes washers force us to rewrite everything in a clumsy way. Shells let us keep the original variable and picture thin hollow cylinders instead.

DEFINITION | Shell Method

When a region is rotated about an axis, a vertical strip can create a cylindrical shell. Its volume is approximately

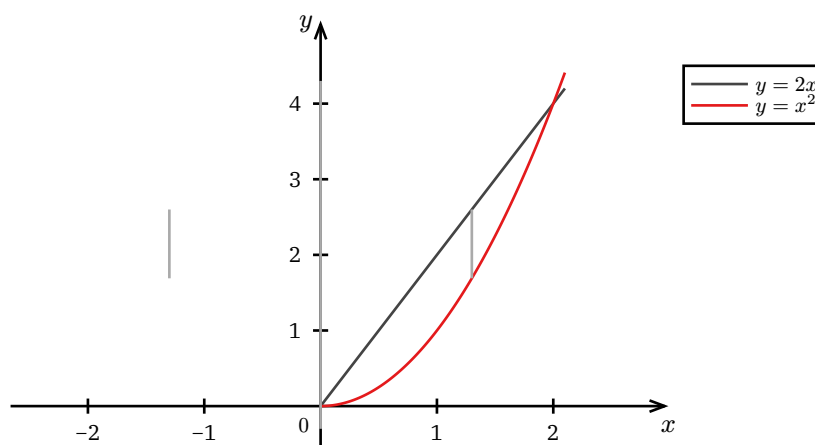
$$\Delta V \approx 2\pi(\text{radius})(\text{height})(\text{thickness})$$

So in the limit,

$$V = 2\pi \int_a^b (\text{radius})(\text{height})dx$$

NOTE | Why shells are visual

You should imagine wrapping a thin rectangle around the axis. The width becomes thickness, the height stays height, and the distance from the axis becomes radius. Unrolling the shell gives a flat slab of dimensions $2\pi r \times h \times dx$. That is where the formula comes from.



EXAMPLE | Using Shells on the Same Region

Rotate the region between

$$y = 2x$$

and

$$y = x^2$$

on $[0, 2]$ about the y -axis. Find the volume.

Solution 1 |

A shell at position x has

- radius x
- height $2x - x^2$

The two gray strips in the picture are the same strip seen twice : once at x , once swung around to $-x$ mid-rotation.

Therefore

$$\begin{aligned} V &= 2\pi \int_0^2 x(2x - x^2)dx \\ &= 2\pi \int_0^2 (2x^2 - x^3)dx \\ &= 2\pi \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_0^2 \\ &= 2\pi \left[\frac{16}{3} - 4 \right] = 8\frac{\pi}{3} \end{aligned}$$

EXAMPLE | A Shell Integral that Substitution Loves

Find the volume of the solid generated by rotating the region bounded by

$$y = \sqrt{4 + x^2}, \quad y = 0, \quad x = 0, \quad x = 2$$

about the y -axis.

Solution 1 |

Washers would be painful here : solving $y = \sqrt{4 + x^2}$ for x splits the solid into two different pieces. Shells keep everything in x .

A shell at position x has radius x and height $\sqrt{4 + x^2}$, so

$$V = 2\pi \int_0^2 x\sqrt{4 + x^2} dx$$

Let

$$u = 4 + x^2$$

Then

$$du = 2x dx$$

and the bounds become $u = 4$ to $u = 8$:

$$\begin{aligned} V &= \pi \int_4^8 \sqrt{u} du = \pi \left[\frac{2}{3} u^{\frac{3}{2}} \right]_4^8 \\ &= \frac{2\pi}{3} \left[8^{\frac{3}{2}} - 4^{\frac{3}{2}} \right] = \frac{2\pi}{3} (16\sqrt{2} - 8) \end{aligned}$$

So

$$V = \frac{16\pi}{3} (2\sqrt{2} - 1)$$

NOTE | Washers or shells? How to decide

Ask one question : is your slicing variable **parallel** or **perpendicular** to the axis of rotation?

- Strip parallel to the axis $\rightarrow$ it wraps around $\rightarrow$ **shell**.
- Strip perpendicular to the axis $\rightarrow$ it spins in place $\rightarrow$ **disk or washer**.

Then pick whichever choice lets you describe the curves without solving for the inverse function.

Chapter 05.04

Work

1 Work

Work is not hard to picture : a force acts while something moves. If the force changes, then we add many tiny pieces of force times distance.

DEFINITION | Work done by a Variable Force

If a force $F(x)$ acts along a line from $x = a$ to $x = b$, then the work done is

$$W = \int_a^b F(x)dx$$

DEFINITION | Hooke's Law

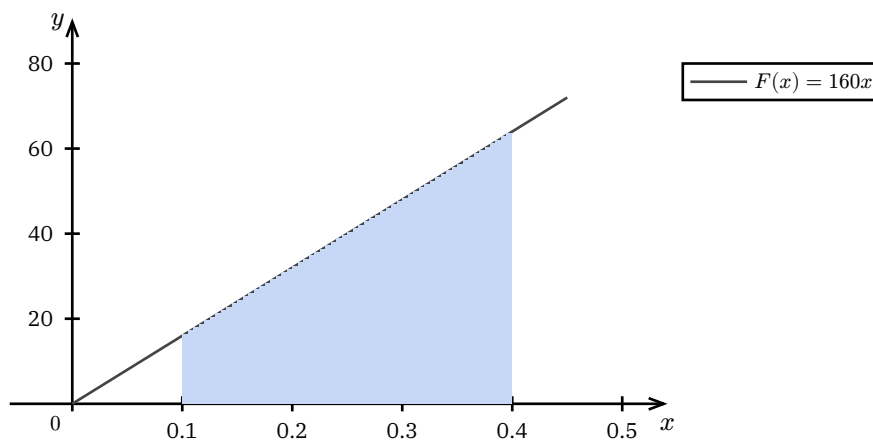
For a spring, the force needed to hold it stretched by x units is

$$F(x) = kx$$

where k is the spring constant.

NOTE | What should we picture?

A spring gets harder to stretch the farther we pull it. So the force graph rises, and the work becomes the area under that rising line.



EXAMPLE | Stretching a Spring over an Interval

A force of 40 newtons is required to hold a spring stretched 0.25 meters beyond its natural length. Find the work required to stretch the spring from 0.10 meters to 0.40 meters beyond its natural length.

Solution 1 |

By Hooke's Law,

$$F(x) = kx$$

and since

$$40 = k(0.25)$$

we get

$$k = 160$$

Therefore

$$F(x) = 160x$$

The required work is the shaded area above :

$$\begin{aligned} W &= \int_{0.10}^{0.40} 160x dx \\ &= 160 \left[\frac{x^2}{2} \right]_{0.10}^{0.40} \\ &= 80(0.16 - 0.01) = 12 \end{aligned}$$

So the work required is 12 joules.

2 When the force is gravity

For lifting problems, the “variable force” usually comes from a changing **amount of stuff** still being lifted, or a changing **distance** each piece must travel. Slice the object, and ask each slice two questions : how heavy are you, and how far do you go?

EXAMPLE | Hauling up a Hanging Cable

A uniform cable 20 meters long with mass density 3 kg/m hangs from the top of a tall building. How much work is required to pull the entire cable up to the top? (Use $g = 9.8 \text{ m/s}^2$.)

Solution 1 |

Slice the cable. The piece at depth x below the roof has :

- mass $3dx$ kilograms, hence weight $3 \cdot 9.8dx$ newtons,
- a trip of length x to reach the top.

So the work for that one piece is

$$29.4x dx$$

and the total is

$$W = \int_0^{20} 29.4x dx = 29.4 \cdot \frac{400}{2} = 5880$$

The answer is 5880 joules. Notice the structure : the **slice** was easy, and the integral did the bookkeeping.

EXAMPLE | Pumping Water out of a Tank

A cylindrical tank of radius 2 meters and height 5 meters is full of water (density 1000 kg/m^3). Find the work required to pump all of the water over the top rim. (Use $g = 9.8 \text{ m/s}^2$.)

Solution 1 |

Slice the water horizontally. The layer at height y (measured from the bottom) is a thin disk :

- volume $\pi(2)^2 dy = 4\pi dy$ cubic meters,
- weight $1000 \cdot 9.8 \cdot 4\pi dy$ newtons,
- lifting distance $5 - y$ meters up to the rim.

Therefore

$$\begin{aligned} W &= \int_0^5 9800 \cdot 4\pi(5 - y) dy \\ &= 39200\pi \int_0^5 (5 - y) dy \\ &= 39200\pi \cdot \frac{25}{2} = 490000\pi \end{aligned}$$

So the work required is $490000\pi \approx 1.54 \times 10^6$ joules. Deep layers cost more, shallow layers cost less, and the integral averages the whole story.

Average Value of a Function

1 Average value of a function

The average value of a function is the height of a single horizontal line that would give the same total area over the interval.

DEFINITION | Average Value of a Function

The average value of $f(x)$ on $[a, b]$ is

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx$$

NOTE | The water-leveling picture

Pour the region under the graph into a rectangular aquarium of the same width $b - a$ and let the water settle flat. The final water level is exactly f_{avg} .

EXAMPLE | A Quick Average

Find the average value of $\sin(x)$ on $[0, \pi]$.

Solution 1 |

$$f_{\text{avg}} = \frac{1}{\pi} \int_0^{\pi} \sin(x) dx = \frac{1}{\pi} [-\cos(x)]_0^{\pi} = \frac{2}{\pi}$$

So the famous sine arch, flattened out, is only about 0.64 tall.

THEOREM | Mean Value Theorem for Integrals

If f is continuous on $[a, b]$, then there exists some number c in $[a, b]$ such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx$$

Proof | Picture a horizontal balancing line

Because f is continuous on a closed interval, it has a minimum value m and a maximum value M . So for every x in $[a, b]$,

$$m \leq f(x) \leq M$$

Integrating across the whole interval gives

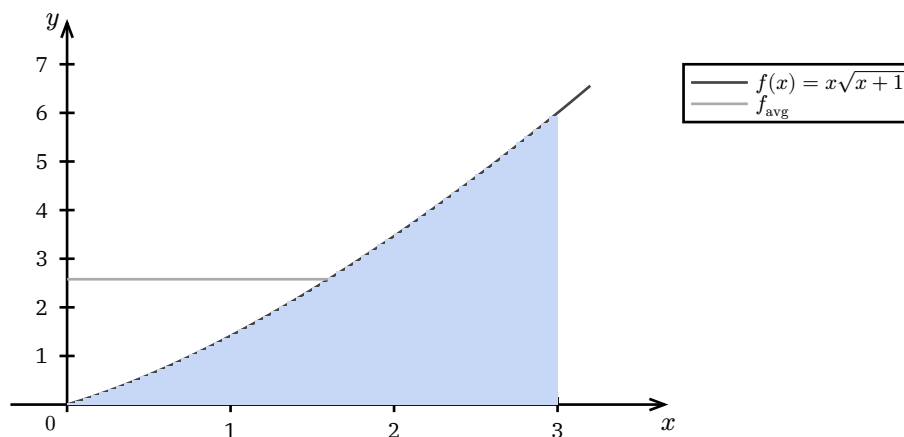
$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$

Divide by $b-a$:

$$m \leq \frac{1}{b-a} \int_a^b f(x) dx \leq M$$

So the average value lies between the minimum and maximum heights of the graph. Since f is continuous, it takes every intermediate height, so there must be some c with

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx$$



EXAMPLE | Average Value with a Substitution-Friendly Function

Find the average value of

$$f(x) = x\sqrt{x+1}$$

on $[0, 3]$.

Solution 1 |

By definition,

$$f_{\text{avg}} = \frac{1}{3} \int_0^3 x\sqrt{x+1} dx$$

Let

$$u = x + 1$$

Then

$$x = u - 1$$

and

$$du = dx$$

The bounds become : If $x = 0$, then $u = 1$. If $x = 3$, then $u = 4$.

Therefore

$$\begin{aligned} \int_0^3 x\sqrt{x+1} dx &= \int_1^4 (u-1)u^{\frac{1}{2}} du \\ &= \int_1^4 \left(u^{\frac{3}{2}} - u^{\frac{1}{2}}\right) du \\ &= \left[\frac{2}{5}u^{\frac{5}{2}} - \frac{2}{3}u^{\frac{3}{2}}\right]_1^4 \\ &= \frac{116}{15} \end{aligned}$$

Hence

$$f_{\text{avg}} = \frac{1}{3} \cdot \frac{116}{15} = \frac{116}{45}$$

In the picture, the shaded area under the curve equals the area of the rectangle capped by the gray line $y = \frac{116}{45}$. The curve crosses that line at the point c promised by the Mean Value Theorem for Integrals.

Chapter 06

Inverse Functions

Reflection across the line $y = x$ generates an entire zoo of new functions. The natural logarithm is born from the area under $1/t$, its inverse is the exponential function, and together they explain every other base, growth and decay processes, and logarithmic differentiation. Restricting the trigonometric functions yields their inverses, exponentials in disguise yield the hyperbolic functions, and the chapter ends with l'Hospital's Rule, the definitive tool for the indeterminate forms these functions constantly produce.

Inverse Functions and Their Derivatives

1 Inverse functions and their derivatives

An inverse function undoes what the original function does. The key picture is reflection across the line $y = x$, but we still need the derivative formula in actual symbols.

DEFINITION | Inverse Function

A function f has an inverse if it is one-to-one. Its inverse is written f^{-1} and satisfies

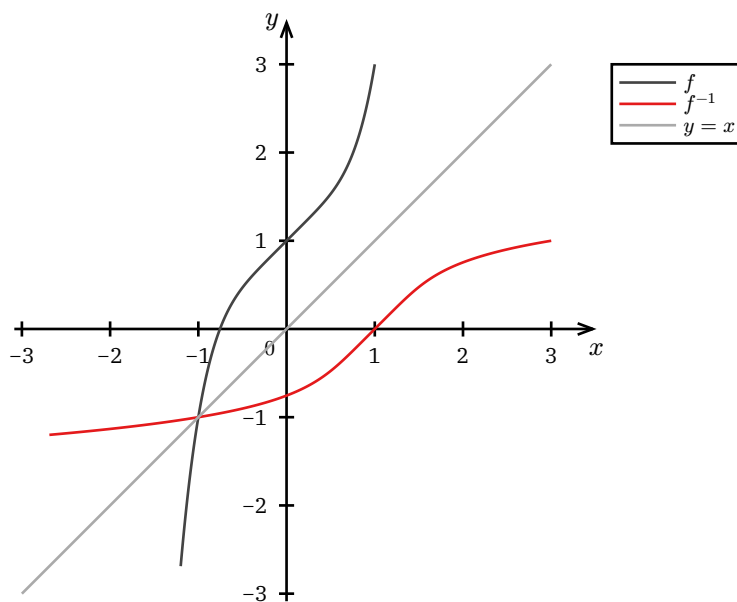
$$f(f^{-1}(x)) = x$$

and

$$f^{-1}(f(x)) = x$$

NOTE | What should we visualize?

The graph of f^{-1} is the reflection of the graph of f across the line $y = x$. So a steep graph turns into a shallow one, which is exactly why the derivative gets inverted.



THEOREM | Derivative of an Inverse Function

If f is differentiable and one-to-one, and if $f'(f^{-1}(a)) \neq 0$, then

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}$$

Proof | Differentiate the identity

Let

$$y = f^{-1}(x)$$

Then

$$f(y) = x$$

Differentiate both sides with respect to x :

$$f'(y) \frac{dy}{dx} = 1$$

Therefore

$$\frac{dy}{dx} = \frac{1}{f'(y)}$$

Since $y = f^{-1}(x)$, we get

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

Replacing x by a gives the formula.

EXAMPLE | Derivative of an Inverse at a Hidden Point

Let

$$f(x) = x^5 + x + 1$$

Find $(f^{-1})'(3)$.

Solution 1 |

We first need the point on the original graph that maps to 3. Since

$$f(1) = 1^5 + 1 + 1 = 3$$

we know

$$f^{-1}(3) = 1$$

Also,

$$f'(x) = 5x^4 + 1$$

so

$$f'(1) = 6$$

Therefore

$$(f^{-1})'(3) = \frac{1}{6}$$

EXAMPLE | An Inverse of an Accumulation Function

Let

$$f(x) = \int_3^x \sqrt{1+t^4} dt$$

for $x \in \mathbb{R}$. Find $(f^{-1})'(0)$.

Solution 1 |

The integrand $\sqrt{1+t^4}$ is always positive, so f is strictly increasing and really does have an inverse.

Where does f take the value 0? An accumulation function vanishes at its own starting line :

$$f(3) = \int_3^3 \sqrt{1+t^4} dt = 0$$

so

$$f^{-1}(0) = 3$$

By the Fundamental Theorem of Calculus Part I,

$$f'(x) = \sqrt{1+x^4}$$

hence

$$f'(3) = \sqrt{1+81} = \sqrt{82}$$

Therefore

$$(f^{-1})'(0) = \frac{1}{\sqrt{82}}$$

We never needed a formula for f itself, only its derivative and one special point.

EXAMPLE | The Second Derivative of an Inverse

Let f be one-to-one with a continuous second derivative, and suppose $f'(x) \neq 0$ on its domain. Writing $g = f^{-1}$, show that

$$g''(x) = -\frac{f''(g(x))}{(f'(g(x)))^3}$$

Solution 1 |

Start from the first-derivative formula :

$$g'(x) = \frac{1}{f'(g(x))}$$

Differentiate both sides with the chain rule and the reciprocal rule :

$$g''(x) = -\frac{f''(g(x)) \cdot g'(x)}{(f'(g(x)))^2}$$

Now substitute $g'(x) = 1/f'(g(x))$ once more :

$$g''(x) = -\frac{f''(g(x))}{(f'(g(x)))^3}$$

Read the sign : if f is increasing ($f' > 0$) and concave upward ($f'' > 0$), the formula forces $g'' < 0$, so the inverse is concave **downward**. Reflection across $y = x$ flips concavity for increasing functions, and now we have proved it, not just stared at it.

The Natural Logarithmic Function

1 The natural logarithmic function

For the starred school track, logarithms are born from area. That is a very visual definition, but we will still push it all the way to algebra and derivatives.

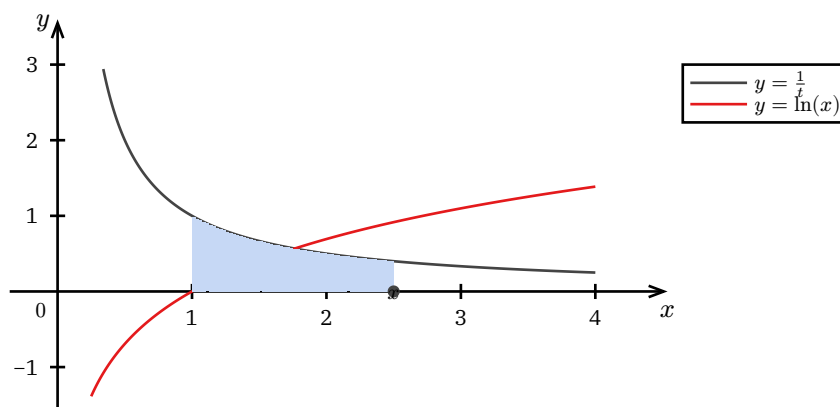
DEFINITION | Natural Logarithm

For $x > 0$, the natural logarithmic function is defined by

$$\ln(x) = \int_1^x \frac{1}{t} dt$$

NOTE | The geometry behind it

If $x > 1$, then $\ln(x)$ is the area under $y = \frac{1}{t}$ from 1 to x . If $0 < x < 1$, the interval is reversed, so the value becomes negative.



THEOREM | Properties of $\ln(x)$

For positive numbers a and b :

1. $\ln(1) = 0$
2. $\ln(ab) = \ln(a) + \ln(b)$

3. $\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$
4. $\ln(a^r) = r \ln(a)$

Proof | Split the area, then write the algebra

The first property is immediate from the definition :

$$\ln(1) = \int_1^1 \frac{1}{t} dt = 0$$

For the product law,

$$\ln(ab) = \int_1^{ab} \frac{1}{t} dt = \int_1^a \frac{1}{t} dt + \int_a^{ab} \frac{1}{t} dt$$

In the second integral let

$$t = au$$

so

$$dt = a du$$

Then

$$\int_a^{ab} \frac{1}{t} dt = \int_1^b \frac{1}{au} \cdot a du = \int_1^b \frac{1}{u} du = \ln(b)$$

Hence

$$\ln(ab) = \ln(a) + \ln(b)$$

Now

$$\ln\left(\frac{a}{b}\right) + \ln(b) = \ln\left(\left(\frac{a}{b}\right)b\right) = \ln(a)$$

so

$$\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$$

Finally, for rational r we repeatedly use the product law to get

$$\ln(a^r) = r \ln(a)$$

and then extend this to real r by continuity.

THEOREM | Derivative of the Natural Logarithm

$$\frac{d}{dx} \ln(x) = \frac{1}{x}$$

Proof | Use the Fundamental Theorem directly

Since

$$\ln(x) = \int_1^x \frac{1}{t} dt$$

and the integrand $\frac{1}{t}$ is continuous for $x > 0$, the Fundamental Theorem of Calculus Part I gives

$$\frac{d}{dx} \ln(x) = \frac{1}{x}$$

EXAMPLE | Differentiating a Logarithm with Several Layers

Differentiate

$$y = \ln\left(\frac{(x^2 + 1)^3}{\sqrt{x^5 + 1}}\right)$$

Solution 1 |

First use logarithm laws :

$$y = 3 \ln(x^2 + 1) - \frac{1}{2} \ln(x^5 + 1)$$

Now differentiate term by term :

$$\begin{aligned} y' &= 3 \cdot 2 \frac{x}{x^2 + 1} - \frac{1}{2} \cdot 5 \frac{x^4}{x^5 + 1} \\ &= 6 \frac{x}{x^2 + 1} - 5 \frac{x^4}{2(x^5 + 1)} \end{aligned}$$

2 Logarithmic differentiation

When a function is a pile of products, quotients, and powers, do not differentiate it directly. Take the logarithm first : the pile collapses into a sum, and sums are easy.

NOTE | The three-step routine

1. Take $\ln$ of both sides and simplify with logarithm laws.
2. Differentiate implicitly, remembering that $\frac{d}{dx} \ln(y) = \frac{y'}{y}$.
3. Multiply back by y at the end.

EXAMPLE | Taming a Monster Quotient

Differentiate

$$y = \frac{x^{\frac{3}{4}} \sqrt{x^2 + 1}}{(3x + 2)^5}$$

Solution 1 |

Take logarithms :

$$\ln(y) = \frac{3}{4} \ln(x) + \frac{1}{2} \ln(x^2 + 1) - 5 \ln(3x + 2)$$

Differentiate :

$$\frac{y'}{y} = \frac{3}{4x} + \frac{x}{x^2 + 1} - \frac{15}{3x + 2}$$

Therefore

$$y' = \frac{x^{\frac{3}{4}} \sqrt{x^2 + 1}}{(3x + 2)^5} \left[\frac{3}{4x} + \frac{x}{x^2 + 1} - \frac{15}{3x + 2} \right]$$

EXAMPLE | An Implicit Equation Written in Logarithms

Suppose

$$\ln(xy) = \ln(x) \ln(y)$$

defines y as a function of x near the point where $x = e^2$. Find $\frac{dy}{dx}$ at $x = e^2$.

Solution 1 |

First expand the left side :

$$\ln(x) + \ln(y) = \ln(x) \ln(y)$$

Find the matching y at $x = e^2$, where $\ln(x) = 2$:

$$2 + \ln(y) = 2 \ln(y)$$

so

$$\ln(y) = 2$$

which gives $y = e^2$ as well.

Now differentiate the equation implicitly :

$$\frac{1}{x} + \frac{y'}{y} = \frac{\ln(y)}{x} + \ln(x) \cdot \frac{y'}{y}$$

Substitute $x = y = e^2$ and $\ln(x) = \ln(y) = 2$:

$$\frac{1}{e^2} + \frac{y'}{e^2} = \frac{2}{e^2} + \frac{2y'}{e^2}$$

Multiplying through by e^2 :

$$1 + y' = 2 + 2y'$$

so

$$y' = -1$$

The Natural Exponential Function

1 The natural exponential function

Since $\ln(x)$ is increasing, it has an inverse. That inverse is the function e^x , and almost everything about it can be read off from the fact that it undoes $\ln(x)$.

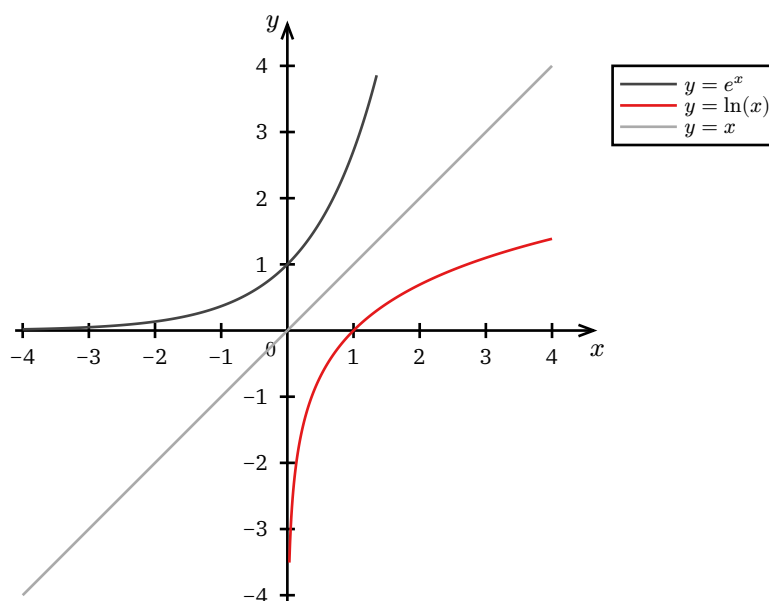
DEFINITION | Natural Exponential Function

The inverse of $\ln(x)$ is denoted by e^x . So

$$e^{\ln(x)} = x \quad \text{for } x > 0$$

and

$$\ln(e^x) = x$$



THEOREM | Properties of e^x

1. $e^0 = 1$
2. $e^{x+y} = e^x e^y$
3. $e^{x-y} = \frac{e^x}{e^y}$

$$4. \quad \frac{d}{dx} e^x = e^x$$

$$5. \quad \int e^x dx = e^x + C$$

Proof | Use the inverse relationship with $\ln(x)$

Since $\ln(1) = 0$, applying the inverse function gives

$$e^0 = 1$$

For the product law, compare the logarithms of both sides :

$$\ln(e^{x+y}) = x + y$$

and

$$\ln(e^x e^y) = \ln(e^x) + \ln(e^y) = x + y$$

Since $\ln(x)$ is one-to-one,

$$e^{x+y} = e^x e^y$$

Then

$$e^{x-y} e^y = e^x$$

so

$$e^{x-y} = \frac{e^x}{e^y}$$

For the derivative, let

$$y = e^x$$

Then

$$x = \ln(y)$$

Differentiate :

$$1 = \frac{1}{y} d \frac{y}{dx}$$

Hence

$$d \frac{y}{dx} = y = e^x$$

The integral formula follows immediately because a function whose derivative is e^x is itself e^x .

EXAMPLE | Solving an Exponential Equation with a Hidden Quadratic

Solve

$$e^{2x} - 5e^x + 4 = 0$$

Solution 1 |

Let

$$u = e^x$$

Then the equation becomes

$$u^2 - 5u + 4 = 0$$

so

$$(u - 1)(u - 4) = 0$$

Therefore

$$u = 1 \quad \text{or} \quad u = 4$$

which means

$$e^x = 1 \quad \text{or} \quad e^x = 4$$

Hence

$$x = 0 \quad \text{or} \quad x = \ln(4)$$

EXAMPLE | Differentiating a Product with an Exponential Core

Differentiate

$$y = (x^2 + 1)e^{x^3}$$

Solution 1 |

Use the product rule and the chain rule :

$$\begin{aligned} y' &= 2xe^{x^3} + (x^2 + 1)e^{x^3} \cdot 3x^2 \\ &= e^{x^3}[2x + 3x^2(x^2 + 1)] \end{aligned}$$

2 A full portrait : the logistic curve

Here is a function built entirely out of e^x that shows up everywhere, from probability to machine learning. Sketching it uses every tool from chapter 3 at once.

EXAMPLE | Sketching $y = \frac{e^x}{1+e^x}$

In a game without ties, let x be the natural logarithm of the **odds**, that is, the probability of winning divided by the probability of losing. Solving for the probability p in terms of x gives

$$p(x) = \frac{e^x}{1 + e^x}$$

Sketch the graph of $y = p(x)$: asymptotes, intercepts, monotonicity, and concavity.

Solution 1 |

Domain and intercepts. The denominator is never zero, so the domain is all of $\mathbb{R}$, and

$$p(0) = \frac{1}{2}$$

The function is always strictly between 0 and 1, which is a sanity check, since it is supposed to be a probability.

Asymptotes. As $x \rightarrow \infty$, divide top and bottom by e^x :

$$p(x) = \frac{1}{e^{-x} + 1} \rightarrow 1$$

As $x \rightarrow -\infty$, $e^x \rightarrow 0$, so $p(x) \rightarrow 0$. Horizontal asymptotes : $y = 0$ and $y = 1$.

Monotonicity. By the quotient rule,

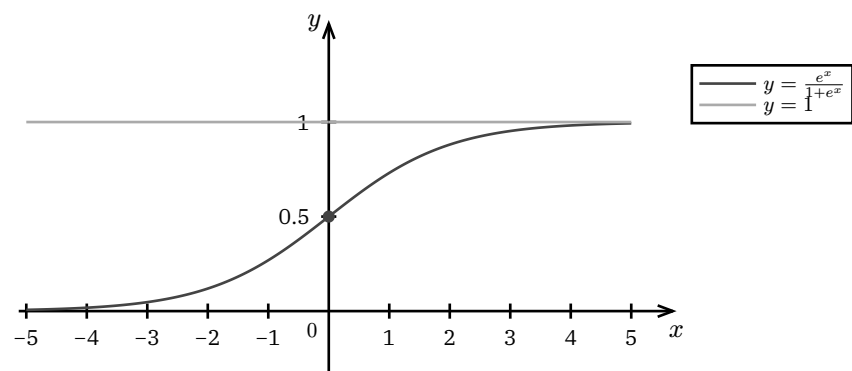
$$p'(x) = \frac{e^x(1 + e^x) - e^x \cdot e^x}{(1 + e^x)^2} = \frac{e^x}{(1 + e^x)^2} > 0$$

so p is increasing everywhere : better log-odds always means better probability.

Concavity. Write $p'(x) = e^x(1 + e^x)^{-2}$ and differentiate :

$$\begin{aligned} p''(x) &= e^x(1 + e^x)^{-2} - 2e^{2x}(1 + e^x)^{-3} \\ &= \frac{e^x(1 - e^x)}{(1 + e^x)^3} \end{aligned}$$

So $p'' > 0$ for $x < 0$ and $p'' < 0$ for $x > 0$: concave up, then concave down, with an inflection point at $(0, \frac{1}{2})$, exactly where the game is fair.



General Logarithmic and Exponential Functions

1 General logarithmic and exponential functions

Once we understand $\ln(x)$ and e^x , every other base is really just a translated version of those two.

DEFINITION | General Exponential Function

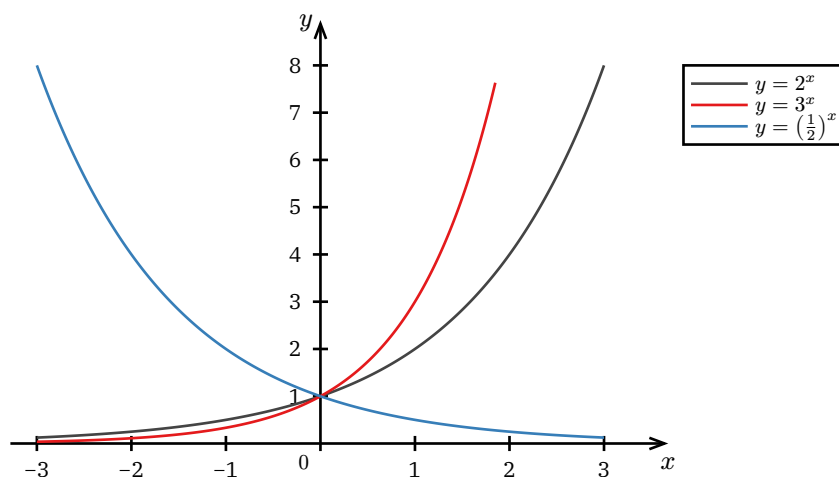
If $a > 0$ and $a \neq 1$, then

$$a^x = e^{x \ln(a)}$$

DEFINITION | General Logarithm

If $a > 0$ and $a \neq 1$, then

$$\log_a(x) = \frac{\ln(x)}{\ln(a)}$$



NOTE | One family, one master key

All these curves pass through $(0, 1)$, and each is just $e^{x \ln(a)}$ stretched horizontally by its own factor $\ln(a)$. Bases $a > 1$ climb, bases $a < 1$ fall, and the case $a = e$ is the one whose tangent at $(0, 1)$ has slope exactly 1.

THEOREM | Derivative Formulae

1. $\frac{d}{dx} a^x = a^x \ln(a)$
2. $\frac{d}{dx} \log_{a(x)} = \frac{1}{x \ln(a)}$

Proof | Reduce everything to e^x and $\ln(x)$

Since

$$a^x = e^{x \ln(a)}$$

the chain rule gives

$$\frac{d}{dx} a^x = \frac{d}{dx} e^{x \ln(a)} = e^{x \ln(a)} \ln(a) = a^x \ln(a)$$

Also,

$$\log_{a(x)} = \frac{\ln(x)}{\ln(a)}$$

and $\ln(a)$ is a constant, so

$$\frac{d}{dx} \log_{a(x)} = \frac{1}{\ln(a)} \cdot \frac{d}{dx} \ln(x) = \frac{1}{x \ln(a)}$$

EXAMPLE | Differentiating a General Exponential with a Composite Exponent

Differentiate

$$y = 5^{x^2 - x}$$

Solution 1 |

Use the chain rule together with the derivative of a^x :

$$y' = 5^{x^2-x} \ln(5) \cdot (2x - 1)$$

EXAMPLE | Differentiating a General Logarithm of a Composite Function

Differentiate

$$y = \log_3(x^2 + 1)$$

Solution 1 |

Using

$$\frac{d}{dx} \log_a(x) = \frac{1}{x \ln(a)}$$

together with the chain rule, we get

$$\begin{aligned} y' &= \frac{1}{(x^2 + 1) \ln(3)} \cdot 2x \\ &= 2 \frac{x}{(x^2 + 1) \ln(3)} \end{aligned}$$

2 Variable base, variable exponent

What about x^x -type functions, where **both** the base and the exponent move? Neither the power rule nor the exponential rule applies alone, but $a^x = e^{x \ln a}$ still saves us.

EXAMPLE | Differentiating $y = x^{\sqrt{x}}$

Differentiate

$$y = x^{\sqrt{x}}$$

for $x > 0$.

Solution 1 |

Rewrite with base e :

$$y = e^{\sqrt{x} \ln(x)}$$

Then by the chain and product rules,

$$\begin{aligned} y' &= x^{\sqrt{x}} \cdot \frac{d}{dx} [\sqrt{x} \ln(x)] \\ &= x^{\sqrt{x}} \left[\frac{\ln(x)}{2\sqrt{x}} + \frac{\sqrt{x}}{x} \right] \\ &= x^{\sqrt{x}} \frac{\ln(x) + 2}{2\sqrt{x}} \end{aligned}$$

Equivalently, we could have taken $\ln$ of both sides and differentiated implicitly : logarithmic differentiation and the $e^{x \ln a}$ rewrite are the same move in different clothes.

THEOREM | The Number e as a Limit

$$e = \lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n$$

Proof | The derivative of $\ln$ at 1 in disguise

Let $f(x) = \ln(x)$, so $f'(1) = 1$. Writing the derivative as a limit :

$$1 = f'(1) = \lim_{x \rightarrow 0} \frac{\ln(1+x) - \ln(1)}{x} = \lim_{x \rightarrow 0} \ln\left((1+x)^{\frac{1}{x}}\right)$$

Since e^x is continuous, we may exponentiate the limit :

$$\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e^1 = e$$

The sequence version follows by setting $x = \frac{1}{n}$.

Exponential Growth and Decay

1 Exponential growth and decay

The most important picture here is feedback : the bigger the quantity is, the faster it changes. That is exactly what the equation $y' = ky$ says.

THEOREM | Growth and Decay Model

If a quantity y satisfies

$$d\frac{y}{dt} = ky$$

then the solution is

$$y = Ce^{kt}$$

If $k > 0$, this is growth. If $k < 0$, this is decay.

Proof | Separate the variables

Assuming $y \neq 0$, we can write

$$\frac{1}{y}dy = kdt$$

Integrate both sides :

$$\int \frac{1}{y}dy = \int kdt$$

$$\ln|y| = kt + C_1$$

Exponentiating gives

$$|y| = e^{C_1}e^{kt}$$

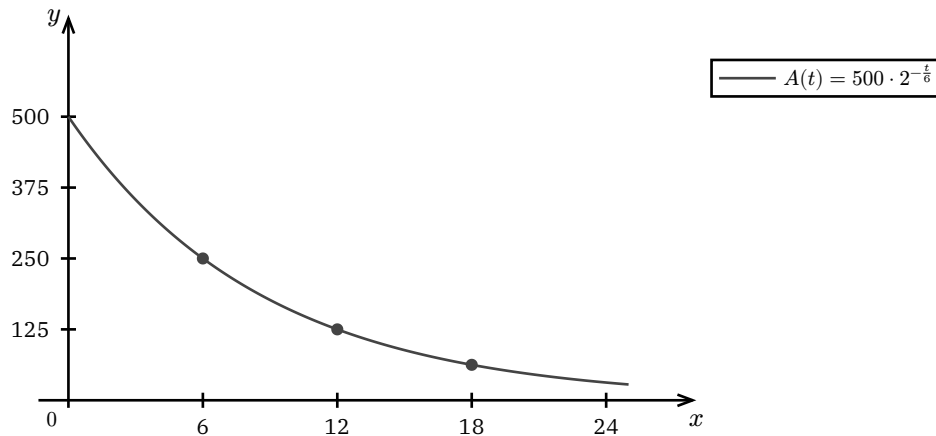
so after absorbing the constant and sign into one symbol, we get

$$y = Ce^{kt}$$

DEFINITION | Doubling Time and Half-Life

For the model $y = Ce^{kt}$:

- The doubling time is $T = \frac{\ln(2)}{k}$ if $k > 0$
- The half-life is $T = \frac{\ln(2)}{|k|}$ if $k < 0$



NOTE | Read the picture

Every 6 hours, the height halves : $500 \rightarrow 250 \rightarrow 125 \rightarrow 62.5$. Equal time steps produce equal **ratios**, not equal differences. That is the signature of an exponential process.

EXAMPLE | Radioactive Decay with a Time Question

A sample begins with 500 mg of a substance and has half-life 6 hours. Find the amount remaining after 14 hours, and find when the amount first drops below 100 mg.

Solution 1 |

The decay model is

$$A(t) = 500e^{kt}$$

Since the half-life is 6 hours,

$$250 = 500e^{6k}$$

so

$$e^{6k} = \frac{1}{2}$$

and

$$k = -\frac{\ln(2)}{6}$$

Thus

$$A(t) = 500e^{-t\frac{\ln(2)}{6}} = 500 \cdot 2^{-\frac{t}{6}}$$

After 14 hours,

$$A(14) = 500 \cdot 2^{-\frac{14}{6}}$$

To find when the amount falls below 100 mg, solve

$$500 \cdot 2^{-\frac{t}{6}} = 100$$

$$2^{-\frac{t}{6}} = \frac{1}{5}$$

Taking natural logarithms,

$$-\frac{t}{6} \ln(2) = -\ln(5)$$

so

$$t = 6 \frac{\ln(5)}{\ln(2)}$$

Therefore the amount first drops below 100 mg after

$$6 \frac{\ln(5)}{\ln(2)}$$

hours, which is about 13.93 hours.

2 Newton's law of cooling

A hot object does not cool at a constant rate : it cools fast while it is much hotter than the room, and slower as it approaches room temperature. The right variable is the temperature **difference**.

THEOREM | Newton's Law of Cooling

If $T(t)$ is the temperature of an object in surroundings of constant temperature T_s , and

$$\frac{dT}{dt} = k(T - T_s)$$

then

$$T(t) = T_s + (T(0) - T_s)e^{kt}$$

Proof | Shift the variable

Let $y = T - T_s$. Since T_s is constant,

$$\frac{dy}{dt} = \frac{dT}{dt} = ky$$

which is exactly the growth-decay equation. Hence

$$y = y(0)e^{kt}$$

and adding T_s back gives the formula.

EXAMPLE | Coffee Cooling in a Room

A cup of coffee at 90°C sits in a 20°C room. After 5 minutes it has cooled to 60°C. When will it reach 40°C?

Solution 1 |

Using $T_s = 20$ and $T(0) = 90$:

$$T(t) = 20 + 70e^{kt}$$

The measurement at $t = 5$ gives

$$60 = 20 + 70e^{5k}$$

so

$$e^{5k} = \frac{4}{7}$$

and

$$k = \frac{1}{5} \ln\left(\frac{4}{7}\right)$$

We want $T(t) = 40$:

$$40 = 20 + 70e^{kt}$$

$$e^{kt} = \frac{2}{7}$$

so

$$t = \frac{\ln(\frac{2}{7})}{k} = 5 \frac{\ln(\frac{2}{7})}{\ln(\frac{4}{7})}$$

Numerically $t \approx 11.2$ minutes. The second 20° drop takes more than twice as long as the first : the shrinking gap slows everything down, exactly as the differential equation predicted.

Inverse Trigonometric Functions

1 Inverse trigonometric functions

The ordinary trig functions repeat too much, so we restrict them first. After that, their inverses behave like normal functions and can be differentiated.

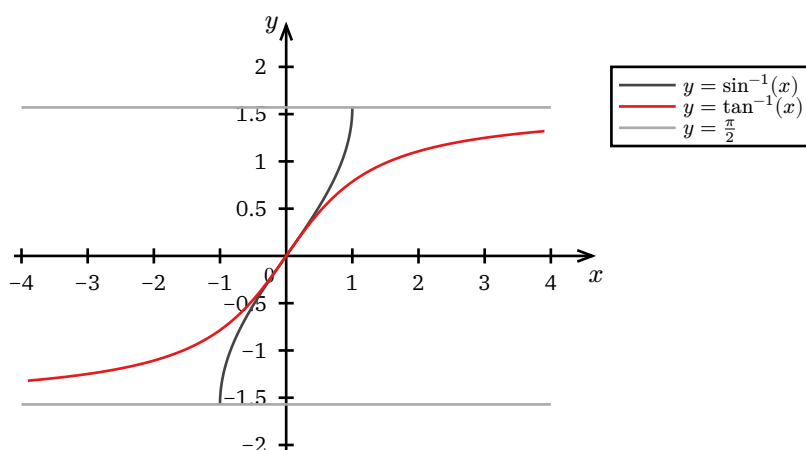
DEFINITION | Inverse Sine and Inverse Tangent

Restrict $\sin$ to $[-\frac{\pi}{2}, \frac{\pi}{2}]$ and $\tan$ to $(-\frac{\pi}{2}, \frac{\pi}{2})$. Then :

$$y = \sin^{-1}(x) \iff \sin(y) = x, \quad y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$y = \tan^{-1}(x) \iff \tan(y) = x, \quad y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Similarly, $\cos^{-1}(x)$ uses the restriction of $\cos$ to $[0, \pi]$.



NOTE | Read the picture

$\sin^{-1}$ lives only on $[-1, 1]$ and exits vertically at the two endpoints, so its derivative must blow up there. $\tan^{-1}$ accepts every real number, but its output is squeezed between the two gray lines $y = \pm \frac{\pi}{2}$: instant horizontal asymptotes.

THEOREM | Derivative Formulae

1. $\frac{d}{dx} \sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}}$
2. $\frac{d}{dx} \cos^{-1}(x) = -\frac{1}{\sqrt{1-x^2}}$
3. $\frac{d}{dx} \tan^{-1}(x) = \frac{1}{1+x^2}$
4. $\frac{d}{dx} \cot^{-1}(x) = -\frac{1}{1+x^2}$
5. $\frac{d}{dx} \sec^{-1}(x) = \frac{1}{x\sqrt{x^2-1}}$

for the standard range choice

6. $\frac{d}{dx} \csc^{-1}(x) = -\frac{1}{|x| \sqrt{x^2-1}}$

with the range $[-\frac{\pi}{2}, \frac{\pi}{2}] \setminus \{0\}$

Proof | Use implicit differentiation

Let

$$y = \sin^{-1}(x)$$

Then

$$\sin(y) = x$$

Differentiate :

$$\cos(y) d\frac{y}{dx} = 1$$

so

$$d\frac{y}{dx} = \frac{1}{\cos(y)}$$

Since $\sin(y) = x$, we have

$$\cos(y) = \sqrt{1-x^2}$$

on the principal interval, so

$$\frac{d}{dx} \sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}}$$

If

$$y = \cos^{-1}(x)$$

then

$$\cos(y) = x$$

and differentiating gives

$$-\sin(y) d\frac{y}{dx} = 1$$

Hence

$$d\frac{y}{dx} = -\frac{1}{\sin(y)} = -\frac{1}{\sqrt{1-x^2}}$$

If

$$y = \tan^{-1}(x)$$

then

$$\tan(y) = x$$

so

$$\sec^2(y) d\frac{y}{dx} = 1$$

Therefore

$$d\frac{y}{dx} = \frac{1}{\sec^2(y)} = \frac{1}{1+\tan^2(y)} = \frac{1}{1+x^2}$$

For the last one, suppose some authors define

$$y = \csc^{-1}(x) \iff \csc(y) = x, \quad y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$$

Differentiating $\csc(y) = x$:

$$-\csc(y) \cot(y) d\frac{y}{dx} = 1$$

so

$$d\frac{y}{dx} = -\frac{1}{\csc(y) \cot(y)}$$

On this range, $\cot(y) = \pm\sqrt{\csc^2(y) - 1}$ takes the sign of y (hence of x), so $\csc(y) \cot(y) = |x| \sqrt{x^2 - 1}$ and

$$\frac{d}{dx} \csc^{-1}(x) = -\frac{1}{|x| \sqrt{x^2 - 1}}$$

valid for $|x| > 1$. The $\sec^{-1}$ and $\cot^{-1}$ formulas come from the same routine.

EXAMPLE | Differentiating a Mixed Inverse-Trig Expression

Differentiate

$$y = x^2 \tan^{-1}(x) + \cos^{-1}(1 - 2x)$$

Solution 1 |

Use the product rule on the first part :

$$\frac{d}{dx}[x^2 \tan^{-1}(x)] = 2x \tan^{-1}(x) + \frac{x^2}{1+x^2}$$

For the second part, use the chain rule :

$$\begin{aligned}\frac{d}{dx} \cos^{-1}(1 - 2x) &= -\frac{1}{\sqrt{1 - (1 - 2x)^2}} \cdot (-2) \\ &= \frac{2}{\sqrt{1 - (1 - 2x)^2}}\end{aligned}$$

Therefore

$$y' = 2x \tan^{-1}(x) + \frac{x^2}{1+x^2} + \frac{2}{\sqrt{1 - (1 - 2x)^2}}$$

THEOREM | The Matching Integrals

1. $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right) + C$
2. $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$

EXAMPLE | An Integral that Secretly Wants $\sin^{-1}$

Evaluate

$$\int_0^{\frac{3\sqrt{3}}{2}} \frac{1}{\sqrt{9 - x^2}} dx$$

Solution 1 |

With $a = 3$,

$$\begin{aligned}\int_0^{\frac{3\sqrt{3}}{2}} \frac{1}{\sqrt{9-x^2}} dx &= \left[\sin^{-1}\left(\frac{x}{3}\right) \right]_0^{\frac{3\sqrt{3}}{2}} \\ &= \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) - \sin^{-1}(0) = \frac{\pi}{3}\end{aligned}$$

2 A full sketch with an inverse trig function

EXAMPLE | Sketching $y = \sin^{-1}(2/x)$

Sketch the curve

$$y = \sin^{-1}\left(\frac{2}{x}\right)$$

domain, asymptotes, monotonicity, concavity and all.

Solution 1 |

Domain. We need $|\frac{2}{x}| \leq 1$, that is $|x| \geq 2$: two separate branches.

Endpoints and asymptotes. At $x = 2$, $y = \sin^{-1}(1) = \frac{\pi}{2}$; at $x = -2$, $y = -\frac{\pi}{2}$. As $x \rightarrow \pm\infty$, $\frac{2}{x} \rightarrow 0$, so $y \rightarrow 0$: the x -axis is a horizontal asymptote on both sides. There is no intercept, since $\frac{2}{x}$ never equals 0.

Monotonicity. By the chain rule, for $|x| > 2$,

$$y' = \frac{1}{\sqrt{1 - (\frac{2}{x})^2}} \cdot \left(-\frac{2}{x^2}\right) = -\frac{2}{|x| \sqrt{x^2 - 4}}$$

which is negative on both branches: each branch falls.

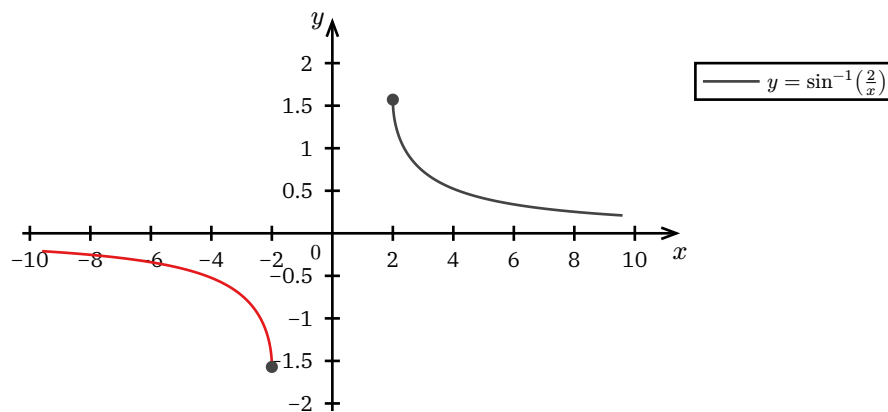
Concavity. On $(2, \infty)$,

$$y' = -\frac{2}{x\sqrt{x^2 - 4}}$$

and differentiating gives

$$y'' = \frac{2(2x^2 - 4)}{x^2(x^2 - 4)^{\frac{3}{2}}} > 0$$

so the right branch is concave up. The function is odd, so the left branch is concave down. At $x = \pm 2$ the derivative blows up : the curve leaves its endpoints vertically.



Hyperbolic Functions

1 Hyperbolic functions

These are exponential functions wearing trigonometric clothes. Their formulas look familiar, but their geometry comes from hyperbolas rather than circles.

DEFINITION | Hyperbolic Functions

The two basic hyperbolic functions are

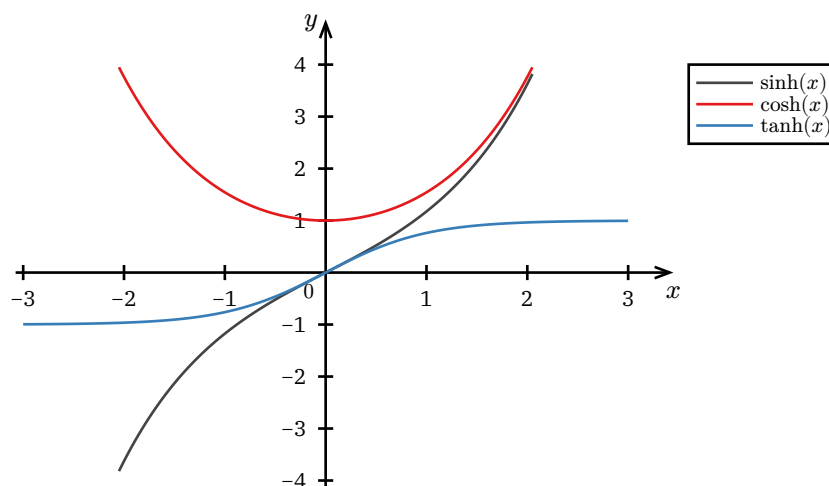
$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

and

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

Also,

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)}$$



NOTE | Three personalities

$\sinh$ is odd and races off like $\frac{e^x}{2}$. $\cosh$ is even, never smaller than 1, and is the exact shape of a hanging chain (a **catenary**). $\tanh$ is the tame one: squeezed between -1 and 1 , it is a smooth switch function.

THEOREM | Basic Facts

1. $\cosh^2(x) - \sinh^2(x) = 1$
2. $\frac{d}{dx} \sinh(x) = \cosh(x)$
3. $\frac{d}{dx} \cosh(x) = \sinh(x)$
4. $\frac{d}{dx} \tanh(x) = \frac{1}{(\cosh(x))^2}$

Proof | Expand the exponential definitions

Start with the identity :

$$\begin{aligned}\cosh^2(x) - \sinh^2(x) &= \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2 \\ &= \frac{e^{2x} + 2 + e^{-2x} - e^{2x} + 2 - e^{-2x}}{4} = 1\end{aligned}$$

For the derivatives,

$$\frac{d}{dx} \sinh(x) = \frac{d}{dx} \frac{e^x - e^{-x}}{2} = \frac{e^x + e^{-x}}{2} = \cosh(x)$$

and

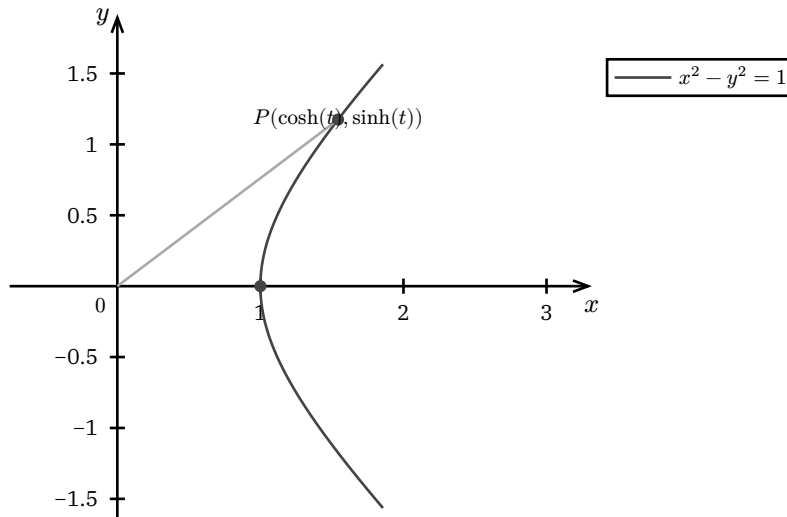
$$\frac{d}{dx} \cosh(x) = \frac{d}{dx} \frac{e^x + e^{-x}}{2} = \frac{e^x - e^{-x}}{2} = \sinh(x)$$

Finally, use the quotient rule :

$$\frac{d}{dx} \tanh(x) = \frac{\cosh^2(x) - \sinh^2(x)}{\cosh^2(x)} = \frac{1}{(\cosh(x))^2}$$

2 Why “hyperbolic”?

The identity $\cosh^2(t) - \sinh^2(t) = 1$ says that the point $(\cosh(t), \sinh(t))$ always sits on the unit hyperbola $x^2 - y^2 = 1$, just as $(\cos(t), \sin(t))$ sits on the unit circle. But t is **not** an angle here. So what is it?



EXAMPLE | The Area of a Hyperbolic Sector

Let $P = (\cosh(t), \sinh(t))$ with $t > 0$ be a point on the hyperbola $x^2 - y^2 = 1$. Show that the area $A(t)$ of the “hyperbolic sector” (the region bounded by the segment OP , the arc of the hyperbola from P down to the vertex $(1, 0)$, and the x -axis) is

$$A(t) = \frac{t}{2}$$

Solution 1 |

Write the sector as a triangle minus the sliver under the hyperbola :

$$A(t) = \frac{1}{2} \cosh(t) \sinh(t) - \int_1^{\cosh(t)} \sqrt{x^2 - 1} dx$$

Instead of computing the ugly integral, differentiate with respect to t . The first term gives

$$\frac{1}{2}(\sinh^2(t) + \cosh^2(t))$$

For the integral, the Fundamental Theorem Part I plus the chain rule (moving bound $\cosh(t)$) gives

$$\sqrt{\cosh^2(t) - 1} \cdot \sinh(t) = \sinh^2(t)$$

Therefore

$$A'(t) = \frac{1}{2}(\sinh^2(t) + \cosh^2(t)) - \sinh^2(t) = \frac{1}{2}(\cosh^2(t) - \sinh^2(t)) = \frac{1}{2}$$

Since $A(0) = 0$ and $A' \equiv \frac{1}{2}$,

$$A(t) = \frac{t}{2}$$

So t measures **twice the sector area**, precisely the same statement that holds for the angle on the unit circle. That is the sense in which $\sinh$ and $\cosh$ deserve their names.

EXAMPLE | Differentiating a Product of Hyperbolic Functions

Differentiate

$$y = \cosh(x^2) \tanh(x)$$

Solution 1 |

Use the product rule :

$$y' = \frac{d}{dx}(\cosh(x^2)) \tanh(x) + \cosh(x^2) \frac{d}{dx}(\tanh(x))$$

Now apply the chain rule and the derivative formulas :

$$\frac{d}{dx}(\cosh(x^2)) = 2x \sinh(x^2)$$

and

$$\frac{d}{dx}(\tanh(x)) = \frac{1}{(\cosh(x))^2}$$

Therefore

$$y' = 2x \sinh(x^2) \tanh(x) + \frac{\cosh(x^2)}{(\cosh(x))^2}$$

3 The reciprocal trio

DEFINITION | Reciprocal Hyperbolic Functions

$$\operatorname{sech}(x) = \frac{1}{\cosh(x)}, \quad \operatorname{csch}(x) = \frac{1}{\sinh(x)}, \quad \operatorname{coth}(x) = \frac{\cosh(x)}{\sinh(x)}$$

EXAMPLE | Sketching $y = \operatorname{sech}(x)$

Sketch the graph of $y = \operatorname{sech}(x)$ by verifying its local extrema, concavity, inflection points, and asymptotes.

Solution 1 |

Domain and symmetry. Since $\cosh(x) \geq 1$ everywhere, sech is defined on all of $\mathbb{R}$, satisfies $0 < \operatorname{sech}(x) \leq 1$, and is even.

Asymptotes. As $x \rightarrow \pm\infty$, $\cosh(x) \rightarrow \infty$, so $\operatorname{sech}(x) \rightarrow 0$: the x -axis is a two-sided horizontal asymptote.

Extrema. Differentiate :

$$\frac{d}{dx} \operatorname{sech}(x) = -\frac{\sinh(x)}{\cosh^2(x)} = -\operatorname{sech}(x) \tanh(x)$$

This is positive for $x < 0$ and negative for $x > 0$, so the only critical point $x = 0$ is a global maximum, with value

$$\operatorname{sech}(0) = 1$$

Concavity. Differentiating once more (product rule on $-\operatorname{sech} \tanh$) :

$$\frac{d^2}{dx^2} \operatorname{sech}(x) = \operatorname{sech}(x) [\tanh^2(x) - \operatorname{sech}^2(x)]$$

The sign is decided by $\tanh^2(x) - \operatorname{sech}^2(x)$, i.e. by $\sinh^2(x) - 1$ after multiplying through by $\cosh^2(x)$. So the inflection points solve

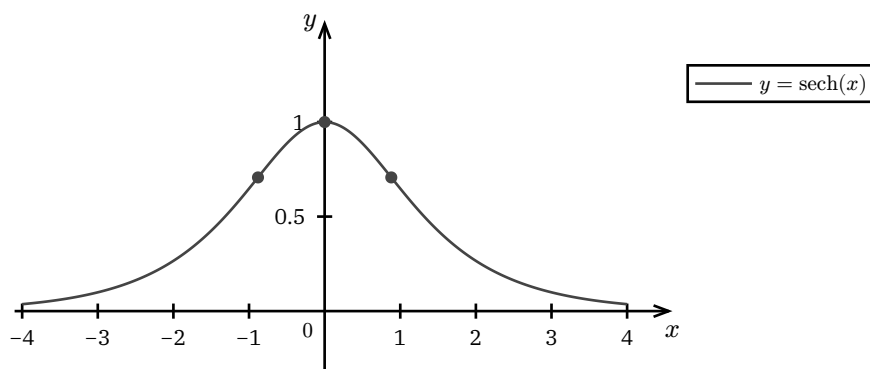
$$\sinh(x) = \pm 1$$

that is

$$x = \pm \sinh^{-1}(1) = \pm \ln(1 + \sqrt{2})$$

using the logarithm formula proven below. At these points $\cosh^2(x) = 1 + \sinh^2(x) = 2$, so the height is exactly $\frac{1}{\sqrt{2}}$.

The curve is concave down on the middle interval and concave up on the two tails : a smooth, symmetric bump often mistaken for the bell curve e^{-x^2} , though its tails die exponentially rather than super-exponentially.



4 Inverse hyperbolic functions

Since $\sinh$ is increasing on all of $\mathbb{R}$, it has a global inverse ; $\cosh$ needs the usual restriction to $[0, \infty)$. Because the originals are built from e^x , the inverses can be written with logarithms, so no new functions are required.

THEOREM | Logarithm Formulas for the Inverses

$$1. \quad \sinh^{-1}(x) = \ln(x + \sqrt{x^2 + 1})$$

for all x

$$2. \quad \cosh^{-1}(x) = \ln(x + \sqrt{x^2 - 1})$$

for $x \geq 1$

$$3. \quad \tanh^{-1}(x) = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$$

for $|x| < 1$

Proof | Solve a quadratic in e^y

Let $y = \sinh^{-1}(x)$, so

$$x = \frac{e^y - e^{-y}}{2}$$

Multiply by $2e^y$:

$$e^{2y} - 2xe^y - 1 = 0$$

This is a quadratic in e^y , so

$$e^y = x \pm \sqrt{x^2 + 1}$$

Since $e^y > 0$ we must take the + sign, and taking logarithms gives

$$y = \ln\left(x + \sqrt{x^2 + 1}\right)$$

The other two formulas come from the same trick applied to their definitions.

THEOREM | Derivatives of the Inverses

1. $\frac{d}{dx} \sinh^{-1}(x) = \frac{1}{\sqrt{x^2 + 1}}$
2. $\frac{d}{dx} \cosh^{-1}(x) = \frac{1}{\sqrt{x^2 - 1}}$
3. $\frac{d}{dx} \tanh^{-1}(x) = \frac{1}{1 - x^2}$

Proof | Differentiate the logarithm formula

For the first,

$$\begin{aligned} \frac{d}{dx} \ln\left(x + \sqrt{x^2 + 1}\right) &= \frac{1 + \frac{x}{\sqrt{x^2 + 1}}}{x + \sqrt{x^2 + 1}} \\ &= \frac{\frac{\sqrt{x^2 + 1} + x}{\sqrt{x^2 + 1}}}{x + \sqrt{x^2 + 1}} = \frac{1}{\sqrt{x^2 + 1}} \end{aligned}$$

The others are identical exercises.

Indeterminate Forms and l'Hospital's Rule

1 Indeterminate forms and l'Hospital's Rule

When direct substitution gives something like $\frac{0}{0}$ or $\frac{\infty}{\infty}$, the graph is telling us that numerator and denominator are racing each other. l'Hospital's Rule compares their rates instead.

DEFINITION | Indeterminate Forms

The two indeterminate **quotient** forms are

$$\frac{0}{0} \quad \text{and} \quad \frac{\infty}{\infty}$$

Five more indeterminate forms reduce to these :

$$0 \cdot \infty, \quad \infty - \infty, \quad 0^0, \quad \infty^0, \quad 1^\infty$$

THEOREM | l'Hospital's Rule

Suppose f and g are differentiable near a , with $g'(x) \neq 0$ nearby. If

$$\lim_{x \rightarrow a} f(x) = 0$$

and

$$\lim_{x \rightarrow a} g(x) = 0$$

or if both limits are $\pm\infty$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} f' \frac{x}{g'}(x)$$

provided the limit on the right exists.

Proof | Sketch using the Mean Value Theorem

The visual idea is that if both functions are heading to the same dangerous place, then the quotient should be controlled by their slopes.

In the $\frac{0}{0}$ case, for $x \neq a$, Cauchy's Mean Value Theorem gives a point c between x and a such that

$$\frac{f(x) - f(a)}{g(x) - g(a)} = f' \frac{c}{g'}(c)$$

Since $f(a) = g(a) = 0$, this becomes

$$\frac{f(x)}{g(x)} = f' \frac{c}{g'}(c)$$

As $x \rightarrow a$, the point $c \rightarrow a$ as well. So if

$$\lim_{x \rightarrow a} f' \frac{c}{g'}(c) = L$$

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L$$

The $\frac{\infty}{\infty}$ case is handled in the same spirit by applying the argument on a tail interval where both functions are defined and large.

NOTE | Two warnings before we compute anything

1. Check the form **every single time**. Applying the rule to a limit that is not $\frac{0}{0}$ or $\frac{\infty}{\infty}$ produces confident nonsense.
2. The rule differentiates the top and the bottom **separately**. It is not the quotient rule.

EXAMPLE | A $\frac{0}{0}$ Form that Needs Two Differentiations

Evaluate

$$\lim_{x \rightarrow 0} \frac{\ln(1+x) - \sin(x)}{x^2}$$

Solution 1 |

Direct substitution gives $\frac{0}{0}$, so apply l'Hospital's Rule :

$$\lim_{x \rightarrow 0} \frac{\ln(1+x) - \sin(x)}{x^2} = \lim_{x \rightarrow 0} \frac{\frac{1}{1+x} - \cos(x)}{2x}$$

This is still $\frac{0}{0}$, so apply l'Hospital's Rule again :

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{-\frac{1}{(1+x)^2} + \sin(x)}{2} \\ &= \frac{-1 + 0}{2} = -\frac{1}{2} \end{aligned}$$

EXAMPLE | An $\frac{\infty}{\infty}$ Form

Evaluate

$$\lim_{x \rightarrow \infty} \frac{x^2 + 3x}{e^x}$$

Solution 1 |

Direct substitution gives $\frac{\infty}{\infty}$, so apply l'Hospital's Rule :

$$\lim_{x \rightarrow \infty} \frac{x^2 + 3x}{e^x} = \lim_{x \rightarrow \infty} \frac{2x + 3}{e^x}$$

This is still $\frac{\infty}{\infty}$, so apply it again :

$$= \lim_{x \rightarrow \infty} \frac{2}{e^x} = 0$$

2 Indeterminate products and differences

NOTE | The conversion trick

For a $0 \cdot \infty$ product, push one factor into the denominator :

$$fg = \frac{f}{1/g}$$

For an $\infty - \infty$ difference, combine into a single fraction first (common denominators are usually enough).

EXAMPLE | A $0 \cdot \infty$ Product

Evaluate

$$\lim_{x \rightarrow 0^+} \sin(3x) \ln(\sin(3x))$$

Solution 1 |

As $x \rightarrow 0^+$, the factor $\sin(3x) \rightarrow 0^+$ while $\ln(\sin(3x)) \rightarrow -\infty$: a $0 \cdot \infty$ tug-of-war.

Substituting $t = \sin(3x)$, which slides to 0^+ , the limit becomes

$$\lim_{t \rightarrow 0^+} t \ln(t) = \lim_{t \rightarrow 0^+} \frac{\ln(t)}{\frac{1}{t}}$$

which is $\frac{-\infty}{\infty}$. l'Hospital's Rule gives

$$\lim_{t \rightarrow 0^+} \frac{\frac{1}{t}}{-\frac{1}{t^2}} = \lim_{t \rightarrow 0^+} (-t) = 0$$

The polynomial-ish factor wins : $0 \cdot \infty = 0$ here, but only because we **proved** it.

EXAMPLE | An $\infty - \infty$ Difference

Evaluate

$$\lim_{x \rightarrow 0} \left[\frac{1}{\ln(1+2x)} - \frac{1}{2x} \right]$$

Solution 1 |

Both terms blow up near 0, with the same sign on each side. Combine them :

$$\frac{1}{\ln(1+2x)} - \frac{1}{2x} = \frac{2x - \ln(1+2x)}{2x \ln(1+2x)}$$

Write $u = 2x$ to reduce clutter ; as $x \rightarrow 0$, $u \rightarrow 0$ and the limit is

$$\lim_{u \rightarrow 0} \frac{u - \ln(1+u)}{u \ln(1+u)}$$

which is $\frac{0}{0}$. Apply l'Hospital :

$$\lim_{u \rightarrow 0} \frac{1 - \frac{1}{1+u}}{\ln(1+u) + \frac{u}{1+u}} = \lim_{u \rightarrow 0} \frac{u/(1+u)}{\ln(1+u) + \frac{u}{1+u}}$$

Multiply top and bottom by $(1+u)$:

$$= \lim_{u \rightarrow 0} \frac{u}{(1+u)\ln(1+u) + u}$$

Still $\frac{0}{0}$; one more round :

$$= \lim_{u \rightarrow 0} \frac{1}{\ln(1+u) + 1 + 1} = \frac{1}{2}$$

3 Indeterminate powers

For 0^0 , ∞^0 and 1^∞ , take logarithms : the exponent comes down, the power becomes a product, and we are back on familiar ground. At the very end, exponentiate.

EXAMPLE | A 1^∞ Power

Evaluate

$$\lim_{x \rightarrow 0} (\cos(2x))^{\frac{1}{x^2}}$$

Solution 1 |

The base tends to 1 and the exponent to ∞ : a 1^∞ form. Let L be the limit and take logarithms :

$$\ln(L) = \lim_{x \rightarrow 0} \frac{\ln(\cos(2x))}{x^2}$$

which is $\frac{0}{0}$. l'Hospital's Rule :

$$\ln(L) = \lim_{x \rightarrow 0} \frac{-2 \tan(2x)}{2x} = - \lim_{x \rightarrow 0} \frac{\tan(2x)}{x}$$

This is $\frac{0}{0}$ once more, and a second application gives

$$\ln(L) = - \lim_{x \rightarrow 0} \frac{2 \sec^2(2x)}{1} = -2$$

Therefore

$$L = e^{-2}$$

The graph of $(\cos 2x)^{\frac{1}{x^2}}$ looks like it should collapse to 1, yet instead it hovers near 0.135. Powers of nearly-1 numbers are dangerous.

EXAMPLE | An ∞^0 Power

Evaluate

$$\lim_{x \rightarrow \infty} x^{\frac{1}{x}}$$

Solution 1 |

The base runs to ∞ and the exponent to 0. Take logarithms :

$$\ln(L) = \lim_{x \rightarrow \infty} \frac{\ln(x)}{x}$$

which is $\frac{\infty}{\infty}$. l'Hospital's Rule gives

$$\ln(L) = \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = 0$$

so

$$L = e^0 = 1$$

Huge numbers raised to dying exponents settle at 1 : logarithmic growth is simply too slow to fight the $\frac{1}{x}$.

4 A capstone : every chapter at once

EXAMPLE | A Limit of an Accumulated Exponential

Let

$$f(x) = \exp\left(\int_2^x \tan^{-1}(\sinh(2-t))dt\right)$$

Find

$$\lim_{x \rightarrow 2} \frac{f'(x)}{x^2 - 4}$$

Solution 1 |

First differentiate f with the chain rule and the Fundamental Theorem Part I:

$$f'(x) = f(x) \cdot \tan^{-1}(\sinh(2-x))$$

At $x = 2$ the integral is empty, so $f(2) = e^0 = 1$, while $\tan^{-1}(\sinh(0)) = 0$. The requested limit is therefore $\frac{0}{0}$ in disguise:

$$\lim_{x \rightarrow 2} \frac{f'(x)}{x^2 - 4} = \lim_{x \rightarrow 2} f(x) \cdot \frac{\tan^{-1}(\sinh(2-x))}{x^2 - 4}$$

The first factor tends to 1. For the second, apply l'Hospital's Rule:

$$\lim_{x \rightarrow 2} \frac{\tan^{-1}(\sinh(2-x))}{x^2 - 4} = \lim_{x \rightarrow 2} \frac{-\cosh(2-x)/(1 + \sinh^2(2-x))}{2x}$$

Since $1 + \sinh^2(u) = \cosh^2(u)$, the numerator simplifies to $-\frac{1}{\cosh(2-x)}$, which tends to -1 . The denominator tends to 4.

Therefore

$$\lim_{x \rightarrow 2} \frac{f'(x)}{x^2 - 4} = 1 \cdot \left(-\frac{1}{4}\right) = -\frac{1}{4}$$

One limit, and it quietly used the FTC, hyperbolic identities, inverse trig, and l'Hospital: the whole chapter in one line.

Chapter 07

Techniques of Integration

Differentiation is mechanical; integration is an art of recognition. This chapter assembles the working integrator's toolbox: integration by parts for products, identity games for trigonometric powers, trigonometric substitution for roots of quadratics, partial fractions for rational functions, and an overall strategy for choosing among them. It then confronts integrals that resist exact methods, with tables, technology, and the Trapezoidal and Simpson's Rules, and finishes with improper integrals, where areas over infinite regions are tamed by limits, p -tests, and comparison.

Chapter 07.01

Integration by Parts

1 Integration by parts

This method is what happens when the product rule is run backward. The picture is simple : move the derivative from one factor to the other if that makes the integral easier.

THEOREM | Integration by Parts

$$\int u dv = uv - \int v du$$

Proof | Start with the product rule

The product rule says

$$\frac{d}{dx}[uv] = u \frac{d}{dx}v + v \frac{d}{dx}u$$

Integrate both sides :

$$\int \frac{d}{dx}[uv] dx = \int u \frac{d}{dx}v dx + \int v \frac{d}{dx}u dx$$

so

$$uv = \int u dv + \int v du$$

Rearranging gives

$$\int u dv = uv - \int v du$$

NOTE | What should we look for?

Try to choose u so that differentiating it makes the expression simpler. Then choose dv so that integrating it is harmless. A useful priority order for choosing u is **L-I-A-T-E** : Logarithm, Inverse trig, Algebraic, Trig, Exponential. The earlier in the list, the more it wants to be u .

EXAMPLE | A Repeated By-Parts Integral

Evaluate

$$\int x^2 e^x dx$$

Solution 1 |

Let

$$u = x^2 \quad \text{and} \quad dv = e^x dx$$

Then

$$du = 2x dx \quad \text{and} \quad v = e^x$$

So

$$\int x^2 e^x dx = x^2 e^x - 2 \int x e^x dx$$

Apply integration by parts again to

$$\int x e^x dx$$

with

$$u = x$$

and

$$dv = e^x dx$$

Then

$$\int x e^x dx = x e^x - e^x$$

Therefore

$$\begin{aligned} \int x^2 e^x dx &= x^2 e^x - 2(x e^x - e^x) + C \\ &= e^x (x^2 - 2x + 2) + C \end{aligned}$$

EXAMPLE | By Parts with a Logarithm

Evaluate

$$\int x \ln(x) dx$$

Solution 1 |

Let

$$u = \ln(x) \quad \text{and} \quad dv = x dx$$

Then

$$du = \frac{1}{x} dx \quad \text{and} \quad v = \frac{x^2}{2}$$

Therefore

$$\begin{aligned} \int x \ln(x) dx &= \frac{x^2}{2} \ln(x) - \int \frac{x^2}{2} \cdot \frac{1}{x} dx \\ &= \frac{x^2}{2} \ln(x) - \frac{1}{2} \int x dx \\ &= \frac{x^2}{2} \ln(x) - \frac{x^2}{4} + C \end{aligned}$$

EXAMPLE | The Invisible Second Factor

Evaluate

$$\int \ln(x) dx$$

Solution 1 |

There seems to be no product here, until we write one in :

$$u = \ln(x) \quad \text{and} \quad dv = 1 \cdot dx$$

Then

$$du = \frac{1}{x} dx \quad \text{and} \quad v = x$$

So

$$\int \ln(x)dx = x \ln(x) - \int x \cdot \frac{1}{x}dx = x \ln(x) - x + C$$

The same trick integrates $\tan^{-1}(x)$ and $\sin^{-1}(x)$: any function whose **derivative** is friendly can be integrated by parts against $dv = dx$.

EXAMPLE | A Cyclic Integral

Evaluate

$$\int e^x \sin(x)dx$$

Solution 1 |

Let I denote the integral. Take

$$u = \sin(x), \quad dv = e^x dx$$

so that

$$I = e^x \sin(x) - \int e^x \cos(x)dx$$

Apply by parts again to the new integral, with $u = \cos(x)$:

$$\int e^x \cos(x)dx = e^x \cos(x) + \int e^x \sin(x)dx = e^x \cos(x) + I$$

Substituting back :

$$I = e^x \sin(x) - e^x \cos(x) - I$$

so

$$2I = e^x (\sin(x) - \cos(x))$$

and

$$I = \frac{1}{2}e^x (\sin(x) - \cos(x)) + C$$

Nothing ever got simpler. Instead, the original integral came back around, and we **solved for it** like an unknown in an equation.

EXAMPLE | By Parts Meets an Inverse Tangent

Evaluate

$$\int_{-\ln(\sqrt{3})}^{\ln(\sqrt{3})} e^{-x} \tan^{-1}(e^x) dx$$

Solution 1 |

L-I-A-T-E says the inverse trig factor should be u :

$$u = \tan^{-1}(e^x), \quad dv = e^{-x} dx$$

Then

$$du = \frac{e^x}{1 + e^{2x}} dx, \quad v = -e^{-x}$$

So the integral becomes

$$\left[-e^{-x} \tan^{-1}(e^x) \right]_{-\ln(\sqrt{3})}^{\ln(\sqrt{3})} + \int_{-\ln(\sqrt{3})}^{\ln(\sqrt{3})} e^{-x} \cdot \frac{e^x}{1 + e^{2x}} dx$$

The leftover integral is

$$\int \frac{1}{1 + e^{2x}} dx$$

Multiply top and bottom by e^{-2x} :

$$\int \frac{e^{-2x}}{e^{-2x} + 1} dx = -\frac{1}{2} \ln(e^{-2x} + 1)$$

Now evaluate everything. At $x = \ln(\sqrt{3})$: $e^x = \sqrt{3}$, $e^{-x} = \frac{1}{\sqrt{3}}$, and $\tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$. At $x = -\ln(\sqrt{3})$: $e^x = \frac{1}{\sqrt{3}}$, $e^{-x} = \sqrt{3}$, and $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$.

The boundary term :

$$-\frac{1}{\sqrt{3}} \cdot \frac{\pi}{3} + \sqrt{3} \cdot \frac{\pi}{6} = \frac{\sqrt{3}\pi}{18}$$

The logarithm term :

$$-\frac{1}{2} \left[\ln\left(\frac{4}{3}\right) - \ln(4) \right] = \frac{1}{2} \ln(3)$$

Therefore

$$\int_{-\ln(\sqrt{3})}^{\ln(\sqrt{3})} e^{-x} \tan^{-1}(e^x) dx = \frac{\sqrt{3}\pi}{18} + \frac{1}{2} \ln(3)$$

Chapter 07.02

Trigonometric Integrals

1 Trigonometric integrals

The main idea is to rewrite the trigonometric powers until one factor becomes the derivative of the inside of the other.

DEFINITION | Useful identities

1. $\sin^2(x) + \cos^2(x) = 1$
2. $1 + \tan^2(x) = \sec^2(x)$
3. $1 + \cot^2(x) = \csc^2(x)$
4. $\sin^2(x) = \frac{1 - \cos(2x)}{2}$
5. $\cos^2(x) = \frac{1 + \cos(2x)}{2}$

NOTE | The visual pattern

Save one factor so that it can become du , and convert everything else with identities until the integral is in one variable.

- Odd power of $\sin$: save one $\sin(x)$, convert the rest to $\cos$, substitute $u = \cos(x)$.
- Odd power of $\cos$: save one $\cos(x)$, convert the rest to $\sin$, substitute $u = \sin(x)$.
- Both powers even : no factor can be saved, so use the half-angle identities (4) and (5) instead.
- For $\tan$ - $\sec$ (or $\cot$ - $\csc$) integrals, the derivative pairs are $d(\tan) = \sec^2 dx$ and $d(\sec) = \sec \tan dx$.

EXAMPLE | A Higher-Power Trigonometric Integral

Evaluate

$$\int \sin^5(x) \cos^4(x) dx$$

Solution 1 |

Since the power of $\sin(x)$ is odd, save one factor of $\sin(x)$:

$$\sin^5(x) = \sin(x)(\sin^2(x))^2 = \sin(x)(1 - \cos^2(x))^2$$

So

$$\int \sin^5(x) \cos^4(x) dx = \int \sin(x)(1 - \cos^2(x))^2 \cos^4(x) dx$$

Let

$$u = \cos(x)$$

Then

$$du = -\sin(x) dx$$

Therefore

$$\begin{aligned} \int \sin^5(x) \cos^4(x) dx &= - \int (1 - u^2)^2 u^4 du \\ &= - \int (u^4 - 2u^6 + u^8) du \\ &= -\frac{u^5}{5} + 2\frac{u^7}{7} - \frac{u^9}{9} + C \\ &= -\frac{\cos^5(x)}{5} + \frac{2\cos^7(x)}{7} - \frac{\cos^9(x)}{9} + C \end{aligned}$$

EXAMPLE | Both Powers Even

Evaluate

$$\int_0^{\pi} \sin^4(x) dx$$

Solution 1 |

No odd factor to save, so use the half-angle identity twice :

$$\sin^4(x) = \left(\frac{1 - \cos(2x)}{2} \right)^2 = \frac{1}{4} [1 - 2\cos(2x) + \cos^2(2x)]$$

and

$$\cos^2(2x) = \frac{1 + \cos(4x)}{2}$$

So

$$\sin^4(x) = \frac{3}{8} - \frac{1}{2} \cos(2x) + \frac{1}{8} \cos(4x)$$

The cosine terms integrate to sines, which vanish at both 0 and π . Hence

$$\int_0^\pi \sin^4(x) dx = \frac{3}{8} \cdot \pi = \frac{3\pi}{8}$$

2 Powers of tangent and secant

THEOREM | Two Integrals Everyone Needs

1. $\int \tan(x) dx = \ln|\sec(x)| + C$
2. $\int \sec(x) dx = \ln|\sec(x) + \tan(x)| + C$

Proof | One substitution each

For (1), write $\tan(x) = \frac{\sin(x)}{\cos(x)}$ and substitute $u = \cos(x)$:

$$\int \frac{\sin(x)}{\cos(x)} dx = - \int \frac{du}{u} = -\ln|u| + C = \ln|\sec(x)| + C$$

For (2), multiply by a clever form of 1:

$$\int \sec(x) \cdot \frac{\sec(x) + \tan(x)}{\sec(x) + \tan(x)} dx = \int \frac{\sec^2(x) + \sec(x) \tan(x)}{\sec(x) + \tan(x)} dx$$

The numerator is exactly the derivative of the denominator, so the integral is

$$\ln|\sec(x) + \tan(x)| + C$$

EXAMPLE | A Pure Power of Tangent

Evaluate

$$\int_0^{\frac{\pi}{4}} \tan^4(x) dx$$

Solution 1 |

Peel off $\tan^2(x)$ and convert it :

$$\tan^4(x) = \tan^2(x)(\sec^2(x) - 1) = \tan^2(x) \sec^2(x) - \tan^2(x)$$

and convert the remaining $\tan^2(x) = \sec^2(x) - 1$ as well :

$$\tan^4(x) = \tan^2(x) \sec^2(x) - \sec^2(x) + 1$$

Each piece is now integrable on sight :

$$\int \tan^4(x) dx = \frac{\tan^3(x)}{3} - \tan(x) + x + C$$

Therefore

$$\int_0^{\frac{\pi}{4}} \tan^4(x) dx = \frac{1}{3} - 1 + \frac{\pi}{4} = \frac{\pi}{4} - \frac{2}{3}$$

The same peeling reduces $\tan^6$, $\tan^8$, ... down the ladder two rungs at a time.

EXAMPLE | Cosecants and Cotangents Play the Same Game

Evaluate

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \csc^4(\theta) \cot^2(\theta) d\theta$$

Solution 1 |

The derivative of $\cot(\theta)$ is $-\csc^2(\theta)$, so save one $\csc^2$ and convert the other :

$$\csc^4(\theta) \cot^2(\theta) = (1 + \cot^2(\theta)) \cot^2(\theta) \csc^2(\theta)$$

Let

$$u = \cot(\theta)$$

Then $du = -\csc^2(\theta) d\theta$, and the bounds $\theta = \frac{\pi}{4}, \frac{\pi}{2}$ become $u = 1, 0$.

Therefore

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \csc^4(\theta) \cot^2(\theta) d\theta = - \int_1^0 (u^2 + u^4) du = \int_0^1 (u^2 + u^4) du$$

$$= \frac{1}{3} + \frac{1}{5} = \frac{8}{15}$$

NOTE | A summary table to carry with you

For $\int \tan^m(x) \sec^n(x) dx$:

- n even : save $\sec^2(x)$, convert the rest to $\tan$, use $u = \tan(x)$.
- m odd : save $\sec(x) \tan(x)$, convert the rest to $\sec$, use $u = \sec(x)$.
- m even and n odd : the hard case. Convert everything to $\sec$ and integrate powers of $\sec$ by parts.

Trigonometric Substitution

1 Trigonometric substitution

When a square root contains a quadratic expression, trigonometric substitution turns that root into something we can actually see and simplify.

DEFINITION | Standard substitutions

1. For $\sqrt{a^2 - x^2}$, use $x = a \sin(\theta)$
2. For $\sqrt{a^2 + x^2}$, use $x = a \tan(\theta)$
3. For $\sqrt{x^2 - a^2}$, use $x = a \sec(\theta)$

NOTE | Why these choices work

They are built to trigger the identities

$$1 - \sin^2(\theta) = \cos^2(\theta)$$

and

$$1 + \tan^2(\theta) = \sec^2(\theta)$$

so the square roots collapse nicely.

EXAMPLE | A Trig Substitution with Extra Algebra

Evaluate

$$\int \frac{x^2}{\sqrt{9 - x^2}} dx$$

Solution 1 |

Since the root has the form $\sqrt{9 - x^2}$, let

$$x = 3 \sin(\theta)$$

Then

$$dx = 3 \cos(\theta) d\theta$$

and

$$\sqrt{9 - x^2} = 3 \cos(\theta)$$

Therefore

$$\begin{aligned} \int \frac{x^2}{\sqrt{9 - x^2}} dx &= \int \frac{9 \sin^2(\theta)}{3 \cos(\theta)} \cdot 3 \cos(\theta) d\theta \\ &= 9 \int \sin^2(\theta) d\theta \end{aligned}$$

Use

$$\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$$

so

$$\begin{aligned} 9 \int \sin^2(\theta) d\theta &= \frac{9}{2} \int (1 - \cos(2\theta)) d\theta \\ &= \frac{9}{2} \theta - \frac{9}{4} \sin(2\theta) + C \end{aligned}$$

Now

$$\theta = \sin^{-1}\left(\frac{x}{3}\right)$$

and

$$\sin(2\theta) = 2 \sin(\theta) \cos(\theta) = 2 \cdot \frac{x}{3} \cdot \frac{\sqrt{9 - x^2}}{3}$$

Thus

$$\int \frac{x^2}{\sqrt{9 - x^2}} dx = \frac{9}{2} \sin^{-1}\left(\frac{x}{3}\right) - x \frac{\sqrt{9 - x^2}}{2} + C$$

NOTE | Getting back to x : draw the triangle

After substituting $x = a \sin(\theta)$, draw a right triangle with hypotenuse a , opposite side x , and adjacent side $\sqrt{a^2 - x^2}$. Every trig expression of θ can then be read straight off the triangle. This is how $\sin(2\theta)$ became $2x \frac{\sqrt{9 - x^2}}{9}$ above without any guessing.

EXAMPLE | A Definite Integral with a Squared Quadratic

Evaluate

$$\int_0^1 \frac{1}{(1+x^2)^2} dx$$

Solution 1 |

No square root in sight, but $(1+x^2)^2$ still begs for

$$x = \tan(\theta)$$

Then

$$dx = \sec^2(\theta) d\theta$$

and

$$(1+x^2)^2 = \sec^4(\theta)$$

The bounds $x = 0, 1$ become $\theta = 0, \frac{\pi}{4}$.

Therefore

$$\begin{aligned} \int_0^1 \frac{1}{(1+x^2)^2} dx &= \int_0^{\frac{\pi}{4}} \frac{\sec^2(\theta)}{\sec^4(\theta)} d\theta = \int_0^{\frac{\pi}{4}} \cos^2(\theta) d\theta \\ &= \left[\frac{\theta}{2} + \frac{\sin(2\theta)}{4} \right]_0^{\frac{\pi}{4}} = \frac{\pi}{8} + \frac{1}{4} \end{aligned}$$

EXAMPLE | A Root of a Quotient

Evaluate

$$\int_0^{\frac{1}{2}} \sqrt{\frac{1-x}{1+x}} dx$$

Solution 1 |

Multiply inside the root by the conjugate $\frac{1-x}{1-x}$:

$$\sqrt{\frac{1-x}{1+x}} = \sqrt{\frac{(1-x)^2}{1-x^2}} = \frac{1-x}{\sqrt{1-x^2}}$$

for $0 \leq x < 1$. Now split :

$$\int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}} dx - \int_0^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}} dx$$

The first integral is $\sin^{-1}(x)$ on sight. The second falls to $u = 1 - x^2$ and gives $-\sqrt{1-x^2}$. So the antiderivative is

$$\sin^{-1}(x) + \sqrt{1-x^2}$$

Evaluating :

$$\begin{aligned} \left[\sin^{-1}(x) + \sqrt{1-x^2} \right]_0^{\frac{1}{2}} &= \left(\frac{\pi}{6} + \frac{\sqrt{3}}{2} \right) - (0 + 1) \\ &= \frac{\pi}{6} + \frac{\sqrt{3}}{2} - 1 \end{aligned}$$

The “trig substitution” here was invisible : conjugation pre-simplified the root so thoroughly that only its two easiest fragments survived.

EXAMPLE | Completing the Square First

Evaluate

$$\int \frac{1}{\sqrt{x^2 + 4x + 13}} dx$$

Solution 1 |

The quadratic is not centered, so complete the square :

$$x^2 + 4x + 13 = (x + 2)^2 + 9$$

Now it has the $a^2 + u^2$ shape with $u = x + 2$ and $a = 3$, so let

$$x + 2 = 3 \tan(\theta)$$

Then $dx = 3 \sec^2(\theta) d\theta$ and $\sqrt{(x+2)^2 + 9} = 3 \sec(\theta)$:

$$\int \frac{3 \sec^2(\theta)}{3 \sec(\theta)} d\theta = \int \sec(\theta) d\theta = \ln |\sec(\theta) + \tan(\theta)| + C$$

From the triangle, $\tan(\theta) = \frac{x+2}{3}$ and $\sec(\theta) = \frac{\sqrt{x^2+4x+13}}{3}$, so

$$\int \frac{1}{\sqrt{x^2 + 4x + 13}} dx = \ln \left| x + 2 + \sqrt{x^2 + 4x + 13} \right| + C$$

after absorbing the constant $-\ln(3)$ into C .

Integration of Rational Functions by Partial Fractions

1 Partial fractions

Rational functions often hide simpler pieces inside them. Partial fractions is the method of pulling those pieces back out.

DEFINITION | Partial Fraction Decomposition

Let $\frac{P(x)}{Q(x)}$ be a **proper** rational function (degree of P less than degree of Q). Factor Q over the reals ; each factor donates terms as follows :

1. A distinct linear factor $(ax + b)$ donates

$$\frac{A}{ax + b}$$

2. A repeated linear factor $(ax + b)^r$ donates

$$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \dots + \frac{A_r}{(ax + b)^r}$$

3. A distinct irreducible quadratic $(ax^2 + bx + c)$ donates

$$\frac{Ax + B}{ax^2 + bx + c}$$

4. A repeated irreducible quadratic donates a chain of such terms with rising powers, just like case 2.

NOTE | If the fraction is not proper

Divide first. Long division rewrites the function as

$$\text{polynomial} + \text{proper fraction}$$

and only the proper part needs decomposing. Forgetting this step is the single most common error in the whole method.

EXAMPLE | Long Division Before Anything Else

Evaluate

$$\int \frac{x^2 + 1}{x^2 - 3x + 2} dx$$

Solution 1 |

The degrees match, so divide :

$$\frac{x^2 + 1}{x^2 - 3x + 2} = 1 + \frac{3x - 1}{x^2 - 3x + 2}$$

Factor the denominator and decompose the proper part :

$$\frac{3x - 1}{(x - 1)(x - 2)} = \frac{A}{x - 1} + \frac{B}{x - 2}$$

Multiplying out and substituting the roots : at $x = 1$, $A = \frac{3-1}{1-2} = -2$; at $x = 2$, $B = \frac{6-1}{2-1} = 5$.

Therefore

$$\int \frac{x^2 + 1}{x^2 - 3x + 2} dx = x - 2 \ln|x - 1| + 5 \ln|x - 2| + C$$

EXAMPLE | A Repeated Linear Factor

Evaluate

$$\int \frac{4x}{(x - 1)^2(x + 1)} dx$$

Solution 1 |

Case 2 says the squared factor donates **two** terms :

$$\frac{4x}{(x - 1)^2(x + 1)} = \frac{A}{x - 1} + \frac{B}{(x - 1)^2} + \frac{C}{x + 1}$$

Multiply through by the denominator :

$$4x = A(x - 1)(x + 1) + B(x + 1) + C(x - 1)^2$$

Substituting $x = 1$ gives $4 = 2B$, so $B = 2$. Substituting $x = -1$ gives $-4 = 4C$, so $C = -1$. Comparing x^2 coefficients gives $0 = A + C$, so $A = 1$.

Therefore

$$\int \frac{4x}{(x-1)^2(x+1)} dx = \ln|x-1| - \frac{2}{x-1} - \ln|x+1| + C$$

Note the middle term : the $\frac{B}{(x-1)^2}$ piece integrates to a **rational** function, not a logarithm. Repeated factors are exactly where non-log pieces sneak in.

EXAMPLE | A Rational Function with Linear and Irreducible Factors

Evaluate

$$\int \frac{2x^2 + 3x + 5}{x(x^2 + 1)} dx$$

Solution 1 |

Decompose it as

$$\frac{2x^2 + 3x + 5}{x(x^2 + 1)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1}$$

Multiply through by $x(x^2 + 1)$:

$$\begin{aligned} 2x^2 + 3x + 5 &= A(x^2 + 1) + (Bx + C)x \\ &= (A + B)x^2 + Cx + A \end{aligned}$$

Match coefficients :

$$A = 5, C = 3, A + B = 2$$

so

$$B = -3$$

Therefore

$$\begin{aligned} \int \frac{2x^2 + 3x + 5}{x(x^2 + 1)} dx &= \int \left[\frac{5}{x} + \frac{-3x + 3}{x^2 + 1} \right] dx \\ &= 5 \int \frac{1}{x} dx - 3 \int \frac{x}{x^2 + 1} dx + 3 \int \frac{1}{x^2 + 1} dx \end{aligned}$$

$$= 5 \ln|x| - \frac{3}{2} \ln(x^2 + 1) + 3 \tan^{-1}(x) + C$$

EXAMPLE | Two Irreducible Quadratics at Once

Evaluate

$$\int_0^1 \frac{2x^2 + x + 3}{(x^2 + 1)(x^2 + 3)} dx$$

Solution 1 |

Both factors are irreducible quadratics, so each donates a linear-over-quadratic term :

$$\frac{2x^2 + x + 3}{(x^2 + 1)(x^2 + 3)} = \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{x^2 + 3}$$

Multiplying out :

$$2x^2 + x + 3 = (Ax + B)(x^2 + 3) + (Cx + D)(x^2 + 1)$$

Compare coefficients degree by degree :

- x^3 : $A + C = 0$
- x^2 : $B + D = 2$
- x^1 : $3A + C = 1$
- x^0 : $3B + D = 3$

From the first and third, $A = \frac{1}{2}$ and $C = -\frac{1}{2}$. From the second and fourth, $B = \frac{1}{2}$ and $D = \frac{3}{2}$.

Now integrate the four pieces over $[0, 1]$:

$$\frac{1}{2} \int_0^1 \frac{x}{x^2 + 1} dx = \frac{1}{4} \ln(2)$$

$$\frac{1}{2} \int_0^1 \frac{1}{x^2 + 1} dx = \frac{\pi}{8}$$

$$-\frac{1}{2} \int_0^1 \frac{x}{x^2 + 3} dx = -\frac{1}{4} \ln\left(\frac{4}{3}\right)$$

$$\frac{3}{2} \int_0^1 \frac{1}{x^2 + 3} dx = \frac{3}{2} \cdot \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) = \frac{\sqrt{3}\pi}{12}$$

Adding the logarithms : $\frac{1}{4}(\ln(2) - \ln(\frac{4}{3})) = \frac{1}{4} \ln(\frac{3}{2})$. Therefore

$$\int_0^1 \frac{2x^2 + x + 3}{(x^2 + 1)(x^2 + 3)} dx = \frac{1}{4} \ln\left(\frac{3}{2}\right) + \frac{\pi}{8} + \frac{\sqrt{3}\pi}{12}$$

NOTE | What can possibly come out?

Integrating any rational function only ever produces three species : rational functions, logarithms, and arctangents. If your answer contains anything else, something went wrong.

Strategy for Integration

1 Strategy for integration

There is no universal button for integration. The real skill is learning what the integrand is trying to become after one good rewrite.

DEFINITION | A rough checklist

1. Simplify the integrand first if possible (algebra, identities, splitting fractions)
2. Look for a composition whose inner derivative is present : substitution
3. Look for a product of unrelated species : integration by parts
4. For trigonometric powers, use identities and the save-one-factor patterns
5. For rational functions, try partial fractions (divide first if improper)
6. For roots of quadratics, complete the square and use trigonometric substitution
7. For roots of other things, try **rationalizing** : make the whole root the new variable

NOTE | The reader should ask one question

What change makes this look **more familiar**? That is usually the right first move. And if the first attempt stalls, do not push. Return to the list and pick a different door.

EXAMPLE | Choosing Simplification and Substitution Together

Evaluate

$$\int x^3 \sqrt{x^2 + 1} dx$$

Solution 1 |

The square root suggests substitution, but first notice that

$$x^3 dx = x^2 \cdot x dx$$

so if

$$u = x^2 + 1$$

then

$$x^2 = u - 1$$

and

$$du = 2x dx$$

Thus

$$\begin{aligned} \int x^3 \sqrt{x^2 + 1} dx &= \frac{1}{2} \int (u - 1) u^{\frac{1}{2}} du \\ &= \frac{1}{2} \int \left(u^{\frac{3}{2}} - u^{\frac{1}{2}} \right) du \\ &= \frac{1}{2} \left[\frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} \right] + C \\ &= \frac{1}{5} u^{\frac{5}{2}} - \frac{1}{3} u^{\frac{3}{2}} + C \end{aligned}$$

Substitute back :

$$\int x^3 \sqrt{x^2 + 1} dx = \frac{1}{5} (x^2 + 1)^{\frac{5}{2}} - \frac{1}{3} (x^2 + 1)^{\frac{3}{2}} + C$$

EXAMPLE | An Identity Opens the Door

Evaluate

$$\int \tan^3(x) dx$$

Solution 1 |

Rule 1 : simplify with an identity before reaching for machinery.

$$\tan^3(x) = \tan(x)(\sec^2(x) - 1) = \tan(x) \sec^2(x) - \tan(x)$$

The first piece is $u = \tan(x)$ on sight ; the second is a known integral :

$$\int \tan^3(x) dx = \frac{\tan^2(x)}{2} - \ln|\sec(x)| + C$$

EXAMPLE | Rationalizing a Stubborn Root

Evaluate

$$\int \frac{1}{x\sqrt{x+1}} dx$$

Solution 1 |

The root is glued to the variable, so make the **whole root** the new variable :

$$u = \sqrt{x+1}$$

Then

$$x = u^2 - 1$$

and

$$dx = 2u du$$

Therefore

$$\int \frac{1}{x\sqrt{x+1}} dx = \int \frac{2u}{(u^2-1)u} du = 2 \int \frac{1}{u^2-1} du$$

A tiny partial fraction finishes it :

$$2 \int \frac{1}{u^2-1} du = \int \left[\frac{1}{u-1} - \frac{1}{u+1} \right] du = \ln \left| \frac{u-1}{u+1} \right| + C$$

Substituting back :

$$\int \frac{1}{x\sqrt{x+1}} dx = \ln \left| \frac{\sqrt{x+1}-1}{\sqrt{x+1}+1} \right| + C$$

One integral, three doors from the checklist : rationalize, then substitute, then partial fractions. Chains like this are the norm, not the exception.

Integration Using Tables and Technology

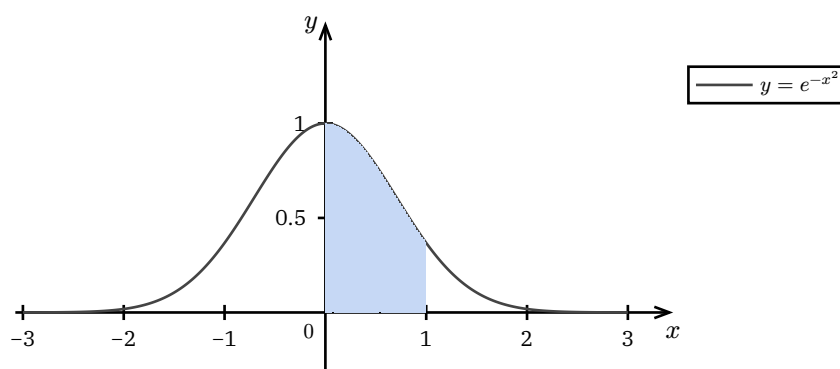
1 Integration using tables and technology

Some integrals are not meant to end in elementary functions. In those cases, technology is not a shortcut. It is the correct tool.

DEFINITION | Nonelementary antiderivatives

Some integrals cannot be expressed with the elementary functions we normally use in calculus. Then we either :

- leave the answer in integral form,
- define a new special function,
- or approximate the value numerically.



EXAMPLE | A Classical Nonelementary Integral

Consider

$$\int e^{-x^2} dx$$

Solution 1 |

This integral has no elementary antiderivative : the bell curve above encloses perfectly good areas, but no formula made of powers,

exponentials, logs and trig functions describes them. So a definite value is usually found with technology or by defining a special function.

For instance, the shaded region is

$$\int_0^1 e^{-x^2} dx \approx 0.7468$$

Another well-known example is

$$\int_0^1 \sin(x^2) dx \approx 0.3103$$

which also has no elementary antiderivative.

2 Using a table of integrals

A table is a museum of solved shapes. The skill is not reading the table ; it is **deforming your integral** until it matches an exhibit.

EXAMPLE | Matching a Table Entry

A standard table lists the entry

$$\int \frac{1}{u\sqrt{a^2 + u^2}} du = -\frac{1}{a} \ln \left| \frac{a + \sqrt{a^2 + u^2}}{u} \right| + C$$

Use it to evaluate

$$\int \frac{1}{x\sqrt{4 + x^2}} dx$$

Solution 1 |

Pattern-match : $u = x$ and $a^2 = 4$, so $a = 2$. Substituting directly,

$$\int \frac{1}{x\sqrt{4 + x^2}} dx = -\frac{1}{2} \ln \left| \frac{2 + \sqrt{4 + x^2}}{x} \right| + C$$

No cleverness required, but note that we could also have earned this result ourselves with the substitution $x = 2 \tan(\theta)$ from the previous sections.

Tables are frozen trig substitutions.

NOTE | When the computer disagrees with you

A computer algebra system may answer in a different costume : it might return $\sinh^{-1}(\frac{x}{2})$ where your table said $\ln(x + \sqrt{x^2 + 4})$, or produce answers differing by a constant. Both are correct : antiderivatives are only unique **up to constants and identities**. Differentiate the machine's answer if you want to referee.

NOTE | What the student should learn here

The goal is not to force every integral into hand computation. The goal is to recognize when the hand computation is over.

Chapter 07.07

Approximate Integration

1 Approximate integration

If an exact antiderivative is unavailable, we can still estimate the area very well by replacing the curve with shapes we understand.

DEFINITION | Midpoint Rule

If $[a, b]$ is split into n equal pieces of width Δx , and $\bar{x}_i$ denotes the midpoint of the i th piece, then

$$M_n = \Delta x [f(\bar{x}_1) + f(\bar{x}_2) + \dots + f(\bar{x}_n)]$$

DEFINITION | Trapezoidal Rule

If $[a, b]$ is split into n equal pieces of width Δx , then

$$T_n = \Delta x \left[\frac{f(x_0) + f(x_n)}{2} + f(x_1) + f(x_2) + \dots + f(x_{n-1}) \right]$$

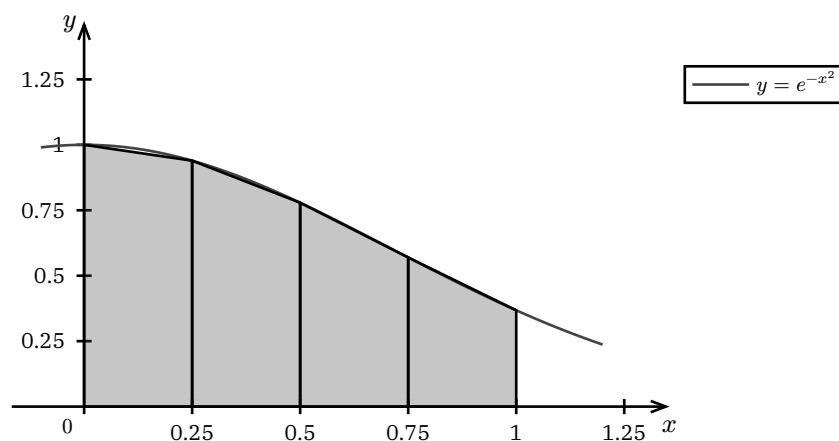
DEFINITION | Simpson's Rule

If n is even, then

$$S_n = \Delta x \left[\frac{f(x_0) + f(x_n)}{3} + 4f(x_1) + 2f(x_2) + 4f(x_3) + \dots + 2f(x_{n-2}) + 4f(x_{n-1}) \right]$$

NOTE | How should we picture them?

The Midpoint Rule uses flat-topped rectangles balanced at the middle. The Trapezoidal Rule connects nearby points by line segments. Simpson's Rule goes one step better and uses parabolic arcs. That is why its weights dance 1, 4, 2, 4, ..., 4, 1.



EXAMPLE | Approximating a Nonelementary Integral

Use $n = 4$ to approximate

$$\int_0^1 e^{-x^2} dx$$

by the Trapezoidal Rule and Simpson's Rule.

Solution 1 |

Here

$$\Delta x = \frac{1}{4}$$

and

$$f(x) = e^{-x^2}$$

The needed values are

$$f(0) = 1$$

$$f\left(\frac{1}{4}\right) \approx 0.9394$$

$$f\left(\frac{1}{2}\right) \approx 0.7788$$

$$f\left(\frac{3}{4}\right) \approx 0.5698$$

$$f(1) \approx 0.3679$$

Therefore

$$T_4 = \frac{1}{8}[1 + 2(0.9394) + 2(0.7788) + 2(0.5698) + 0.3679]$$

$$\approx 0.7430$$

And

$$S_4 = \frac{1}{12}[1 + 4(0.9394) + 2(0.7788) + 4(0.5698) + 0.3679]$$

$$\approx 0.7469$$

So Simpson's Rule is already extremely close to the actual value (≈ 0.74682). The trapezoids undershoot because the curve is mostly concave down there : each chord hangs below the arc, exactly as the picture shows.

EXAMPLE | Simpson's Rule against an Exact Answer

Approximate

$$\int_0^2 3^{-x} dx$$

with Simpson's Rule and $n = 4$, then compare with the exact value.

Solution 1 |

Here $\Delta x = \frac{1}{2}$ and the sample values are

$$f(0) = 1, \quad f\left(\frac{1}{2}\right) \approx 0.5774, \quad f(1) \approx 0.3333, \quad f\left(\frac{3}{2}\right) \approx 0.1925, \quad f(2) \approx 0.1111$$

So

$$S_4 = \frac{1}{6}[1 + 4(0.5774) + 2(0.3333) + 4(0.1925) + 0.1111] \approx 0.8095$$

This time we **can** check. Since

$$\int 3^{-x} dx = -\frac{3^{-x}}{\ln(3)} + C$$

the exact value is

$$\int_0^2 3^{-x} dx = \frac{1 - \frac{1}{9}}{\ln(3)} = \frac{8}{9 \ln(3)} \approx 0.8091$$

Four subintervals, and Simpson's Rule is already correct to three decimal places.

2 How wrong can we be?

An approximation without an error estimate is a rumor. The error of each rule is controlled by how **curved** the function is.

THEOREM | Error Bounds

Suppose $|f''(x)| \leq K$ on $[a, b]$. Then the errors satisfy

$$|E_T| \leq \frac{K(b-a)^3}{12n^2} \quad \text{and} \quad |E_M| \leq \frac{K(b-a)^3}{24n^2}$$

If moreover $|f^{(4)}(x)| \leq K_4$ on $[a, b]$, then Simpson's error satisfies

$$|E_S| \leq \frac{K_4(b-a)^5}{180n^4}$$

NOTE | Read the exponents

Doubling n divides the trapezoid error by about 4, but divides Simpson's error by about 16. And the midpoint rule is (perhaps surprisingly) about **twice as accurate** as the trapezoid rule, with the opposite sign of error.

EXAMPLE | Choosing n in Advance

How large must n be so that the Trapezoidal Rule approximates

$$\int_0^1 e^{-x^2} dx$$

with error at most 10^{-4} ?

Solution 1 |

We need a bound on the second derivative. From

$$f''(x) = (4x^2 - 2)e^{-x^2}$$

the largest magnitude on $[0, 1]$ occurs at $x = 0$, giving

$$|f''(x)| \leq 2$$

So take $K = 2$, $b - a = 1$:

$$|E_T| \leq \frac{2}{12n^2} = \frac{1}{6n^2}$$

Demanding $\frac{1}{6n^2} \leq 10^{-4}$ gives

$$n^2 \geq \frac{10^4}{6} \approx 1667$$

so

$$n \geq 41$$

41 trapezoids for four decimal places, while Simpson's Rule earned three decimals with $n = 4$. This is why Simpson's Rule is the default workhorse of numerical integration.

Chapter 07.08

Improper Integrals

1 Improper integrals

Improper integrals appear when the region never ends or when the graph shoots off vertically. In both cases, the right way to read the picture is through a limit.

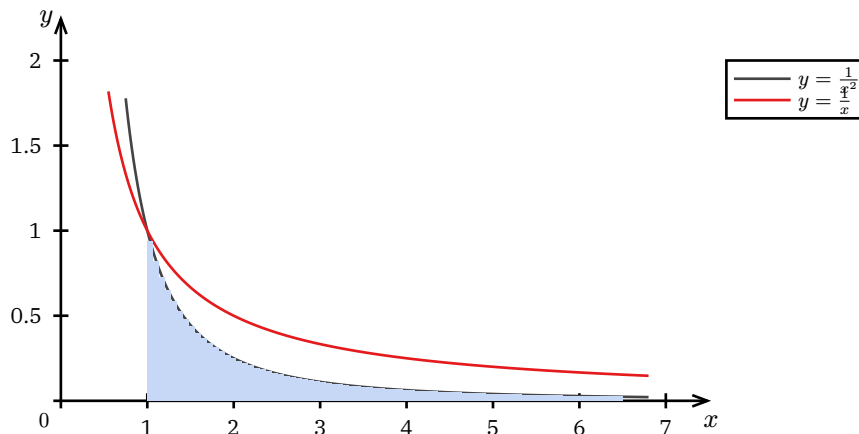
DEFINITION | Improper Integral

An integral is called **improper** if :

- one or both bounds are infinite, or
- the integrand becomes unbounded on the interval

NOTE | What are we really doing?

We are replacing a dangerous endpoint by a moving cutoff, computing an ordinary integral first, and only then taking a limit.



- Both curves above sink to 0, but their infinite tails behave completely differently : the shaded region under $\frac{1}{x^2}$ has **finite** total area, while the region under $\frac{1}{x}$ (which looks barely bigger) has infinite area. Speed of decay is everything.

EXAMPLE | A Convergent Improper Integral on an Infinite Interval

Evaluate

$$\int_1^{\infty} \frac{1}{x^2 + 4x + 3} dx$$

Solution 1 |

First factor the denominator :

$$x^2 + 4x + 3 = (x + 1)(x + 3)$$

Use partial fractions :

$$\frac{1}{(x + 1)(x + 3)} = \frac{1}{2} \cdot \frac{1}{x + 1} - \frac{1}{2} \cdot \frac{1}{x + 3}$$

Therefore

$$\begin{aligned} \int_1^\infty \frac{1}{x^2 + 4x + 3} dx &= \lim_{b \rightarrow \infty} \frac{1}{2} \int_1^b \left[\frac{1}{x + 1} - \frac{1}{x + 3} \right] dx \\ &= \lim_{b \rightarrow \infty} \frac{1}{2} [\ln(x + 1) - \ln(x + 3)]_1^b \\ &= \frac{1}{2} \lim_{b \rightarrow \infty} \left[\ln\left(\frac{b + 1}{b + 3}\right) - \ln\left(\frac{2}{4}\right) \right] \\ &= \frac{1}{2} \left[0 - \ln\left(\frac{1}{2}\right) \right] = \frac{1}{2} \ln(2) \end{aligned}$$

So the integral converges to $\frac{1}{2} \ln(2)$.

EXAMPLE | A Divergent Improper Integral with a Vertical Asymptote

Evaluate

$$\int_0^1 \frac{1}{x^2} dx$$

Solution 1 |

Because the integrand blows up at $x = 0$, we define

$$\begin{aligned} \int_0^1 \frac{1}{x^2} dx &= \lim_{a \rightarrow 0^+} \int_a^1 \frac{1}{x^2} dx \\ &= \lim_{a \rightarrow 0^+} \left[-\frac{1}{x} \right]_a^1 \end{aligned}$$

$$= \lim_{a \rightarrow 0+} \left(-1 + \frac{1}{a} \right)$$

Since $\frac{1}{a} \rightarrow \infty$ as $a \rightarrow 0+$, the integral diverges.

2 The p-integrals : the measuring sticks

THEOREM | The p-Test at Infinity

$$\int_1^{\infty} \frac{1}{x^p} dx \quad \text{converges if and only if } p > 1$$

Proof | Just integrate

For $p \neq 1$,

$$\int_1^b x^{-p} dx = \frac{b^{1-p} - 1}{1-p}$$

As $b \rightarrow \infty$: if $p > 1$, then $b^{1-p} \rightarrow 0$ and the limit is $\frac{1}{p-1}$. If $p < 1$, then $b^{1-p} \rightarrow \infty$ and the integral diverges.

For $p = 1$,

$$\int_1^b \frac{1}{x} dx = \ln(b) \rightarrow \infty$$

so it diverges too.

THEOREM | The p-Test at Zero

$$\int_0^1 \frac{1}{x^p} dx \quad \text{converges if and only if } p < 1$$

Proof | Same computation, other endpoint

For $p \neq 1$,

$$\int_a^1 x^{-p} dx = \frac{1 - a^{1-p}}{1-p}$$

As $a \rightarrow 0^+$: if $p < 1$, then $a^{1-p} \rightarrow 0$ and the limit is $\frac{1}{1-p}$. If $p > 1$, then $a^{1-p} \rightarrow \infty$ and the integral diverges. The case $p = 1$ gives $-\ln(a) \rightarrow \infty$.

Notice the perfect duality with the previous theorem : the two tests split the world of powers at $p = 1$, in opposite directions.

3 Comparison : convergence without computation

THEOREM | Comparison Theorem

Suppose f and g are continuous with $0 \leq g(x) \leq f(x)$ for $x \geq a$.

1. If $\int_a^\infty f(x)dx$ converges, then $\int_a^\infty g(x)dx$ converges.
2. If $\int_a^\infty g(x)dx$ diverges, then $\int_a^\infty f(x)dx$ diverges.

EXAMPLE | Convergence Decided by Comparison

For which values of $a > 0$ does

$$\int_1^\infty \frac{3 + \cos(x)}{x^a} dx$$

converge?

Solution 1 |

The numerator oscillates, but harmlessly : for all x ,

$$2 \leq 3 + \cos(x) \leq 4$$

So the integrand is trapped :

$$\frac{2}{x^a} \leq \frac{3 + \cos(x)}{x^a} \leq \frac{4}{x^a}$$

If $a > 1$, the upper bound $\frac{4}{x^a}$ has a convergent p-integral, so ours converges by comparison. If $a \leq 1$, the lower bound $\frac{2}{x^a}$ has a divergent p-integral, so ours diverges.

Therefore the integral converges **exactly** when $a > 1$. The cosine never had a vote : only the power of x mattered.

4 Two exam-grade computations

EXAMPLE | An Infinite Integral Needing By Parts

Evaluate

$$\int_0^{\infty} x^2 e^{-3x} dx$$

Solution 1 |

Integrate by parts twice (or recycle the pattern from the by-parts section) :

$$\int x^2 e^{-3x} dx = e^{-3x} \left[-\frac{x^2}{3} - \frac{2x}{9} - \frac{2}{27} \right] + C$$

Now take the limit. For any polynomial P , $P(b)e^{-3b} \rightarrow 0$ as $b \rightarrow \infty$, because exponential decay beats polynomial growth (l'Hospital says so). Hence

$$\int_0^{\infty} x^2 e^{-3x} dx = 0 - \left[-\frac{2}{27} \right] = \frac{2}{27}$$

EXAMPLE | An Endpoint Singularity Cured by Substitution

Evaluate

$$\int_0^1 \frac{e^{-\frac{2}{x}}}{x^2} dx$$

Solution 1 |

The integrand is undefined at $x = 0$, so this is improper :

$$\int_0^1 \frac{e^{-\frac{2}{x}}}{x^2} dx = \lim_{a \rightarrow 0^+} \int_a^1 \frac{e^{-\frac{2}{x}}}{x^2} dx$$

Spot the derivative pair : since

$$\frac{d}{dx} e^{-\frac{2}{x}} = e^{-\frac{2}{x}} \cdot \frac{2}{x^2}$$

we get

$$\int_a^1 \frac{e^{-\frac{2}{x}}}{x^2} dx = \frac{1}{2} \left[e^{-\frac{2}{x}} \right]_a^1 = \frac{1}{2} \left[e^{-2} - e^{-\frac{2}{a}} \right]$$

As $a \rightarrow 0^+$, the exponent $-\frac{2}{a} \rightarrow -\infty$, so $e^{-\frac{2}{a}} \rightarrow 0$. Therefore

$$\int_0^1 \frac{e^{-\frac{2}{x}}}{x^2} dx = \frac{1}{2e^2}$$

The “singularity” was a bluff : the function actually dives to 0 near the origin faster than any power climbs.